SYPT 2027 · Problem B8 “Cold drink”

The B8 reports, unified

All four B8 reports in one volume, with a check of where they disagree. The reports are reproduced unchanged after the discrepancy review. Each keeps its own section, equation and page numbering; the page numbers below are pages of this combined PDF.

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The nine single figures in figs/ are already embedded in Part A, so they are not repeated.

Discrepancy review

The four reports were written days apart, for different models, and they say so: Part A is the 24 Oct equilibrium model plus the simulator, Part B derives that same model, Part C is the second edition of the compendium, and Part D (29 Oct) is a separate closed-form model with no convection. Most differences are therefore different **assumptions**, not errors. Below, each one is classified: **[Conflict]** the reports give different answers to the same question; **[Constant]** a handbook value differs; **[Setup]** the rig is described differently; **[Internal]** a report disagrees with itself.

1 · Results that conflict

Item	A · salted ice bath	B · derivation	C · compendium	D · strong case	Assessment
[Conflict] Rig-note charge 3.0 kg ice / 2.6 kg water / 0.95 kg salt: where the bath settles	−17.78 °C, all salt dissolved	−17.8 °C (used for L_{eff} = 411 500 J/kg)	—	B4: −14.7 °C, regime U (needs 1.409 kg salt for the eutectic)	All agree it never reaches the eutectic. The 3 K gap comes mainly from the start temperatures: A puts the water in at 4 °C, D at room temperature 25 °C. The extra 2.6 kg × 4190 × 21 K ≈ 230 kJ melts about 0.7 kg more ice and dilutes the brine. Quote the temperature together with the water start temperature, and measure it (Part A, experiment E1).
[Conflict] Buffered charge 4.0 ice / 1.6 water / 1.0 salt: salt left solid	193 g at −21.2 °C	—	193 g (measured diagram)	B1: 36 g at −21.2 °C	Same cause: water at 4 °C (A, C) against 25 °C (D). With 25 °C water, 1.0 kg is only just enough (D: 0.963 kg needed). To keep a real buffer, chill the water or add salt.
[Conflict] Time for the drink to reach 5 °C, Can 2	222 s simulator (can lying, 28→5 °C); 169 s equilibrium model (Newton); 184 s well-mixed	closed forms only	165–246 s depending on k_m, orientation, floating	151–223 s depending on where the ice sits (25→5 °C, upright, no convection); lower bound 813 s with no convection inside the can	Not comparable as they stand: different start temperature (28 vs 25 °C), orientation (lying vs upright) and heat-transfer physics (fitted U vs pure conduction). The spread, ~150–250 s, is a fair model uncertainty until U is measured on the rig.

2 · Rig and can described differently

Item	Values in the reports	Assessment
[Setup] Room and drink start temperature	A, C: 28 °C. D: 25 °C (also water and salt at 25 °C). B: symbolic.	Pick one for the presentation; 28 °C matches the rig note.
[Setup] Water start temperature	A, C: 4 °C. D: room temperature.	The largest single cause of the conflicts above.
[Setup] Can orientation	A, C: lying (as in the rig). D: upright, base 1 cm off the floor, held by the lid board.	Both are possible; state which one each number is for.
[Setup] Headspace in Can 2	A, C: measured 1.2 cm upright (41 mL), drink 10.46 cm. D: 66.5 mL, 19.6 mm, drink 97.0 mm (ideal cylinder minus 330 mL).	Use the measured 1.2 cm / 41 mL: D's ideal cylinder ignores the neck and domed base, as C notes.
[Setup] Drink mass and heat capacity	A, C: 330 g, c = 4000 → C_d = 1332 J/K, can + drink 343 g. D: 330 mL × 1040 kg/m³ = 343 g, c = 4050 → 1402 J/K, can + drink 356 g. B: water properties (ρ = 999.1, k = 0.589).	Weigh the can full and empty. The 5 % difference in C_d moves t_5 by about 5 %. The draft in eutectic brine (C: 8.48 cm, D: 8.85 cm) follows from the mass.

Item	Values in the reports	Assessment
[Setup] Box conductance $(UA)_{box}$	0.46 W/K (foam alone, rig note) and 0.34 W/K (with films and lid gaps) both appear in A and C; the A table uses 0.46, the simulator 0.34.	Not a contradiction, but say which. Measure it with the box test (box_test.toml).

3 · Handbook constants that differ

Constant	A	B	C	D	Note
L_f (J/kg)	333 550	333 550	333 500	333 550	333.55 kJ/kg is the standard value.
c_w (J/kg K)	4180	4180	4180	4190	Both within the 0–30 °C range.
c_i (J/kg K)	2000 (sim: IAPWS)	2000	2100	2060	Ice c falls from 2097 at 0 °C to 1950 at −20 °C; so $c_w - c_i$ is 2180 (A, B), 2080 (C), 2130 (D).
c_s NaCl (J/kg K)	864	864	864	860	Negligible.
Δh_s	3.88 kJ/mol = 66.4 kJ/kg	66 400 J/kg	3900 J/mol = 66 735 J/kg	3.88 kJ/mol	3.88 kJ/mol is the handbook value at 25 °C, infinite dilution.
k_b brine (W/m K)	0.54; sim 0.5725 − 0.149 w	0.54	—	0.52	Within the uncertainty.
ρ_b eutectic (kg/m³)	1180 table, 1182 text; sim 1000 + 786 w	1182	1182	1000 + 765 w (≈1177)	Within 0.5 %.
k of can aluminium (W/m K)	—	160	160	205	The wall resistance is negligible either way.
Freezing curve	Bodnar fit, eutectic −21.2 °C, w = 0.2318	same	dilute law $2K_fb$ (floor −22.7 °C); measured diagram as a correction	bound-water fit K_s = 3.173, ω = 0.044	C's dilute law is ~2 K too warm near the eutectic, which C states. Use Bodnar.

4 · Inside one report

[Internal] Part A gives the SINTEF can time constant as τ ≈ 265 s in Table 1 and τ = 268 s in §7 (U_0 = 160 W/m²K, A = 310 cm²). The difference is rounding and does not matter. No other internal inconsistency was found in the constants and headline numbers checked.

5 · What agrees

The eutectic (−21.2 °C, 23.2 wt %, 5.163 mol/kg, 0.302 kg salt per kg water); K_f = 1.86 K kg/mol; Kirchhoff's law for the latent heat below 0 °C; Can 2's size (66.0 × 116.6 mm, 13 g aluminium, wall area 310 cm²); the tank (14.2 × 29.0 cm floor, 412 cm²); the film coefficients h_i ≈ 400 and h_o ≈ 250–300 W/m²K where they are used; that a 330 mL can floats in brine and is unstable upright, so it must be held; and the main conclusion that **reaching the eutectic is an energy condition, not a mass condition**. A, C and D all show it, and all show that the rig-note charge has no thermostat.

6 · Recommendations

• Fix one set of rig inputs (room, drink, water and salt start temperatures; can orientation; measured headspace; drink mass) and rerun every model on it. Then the headline tables can be compared.

• Use 333.55 kJ/kg, 3.88 kJ/mol and Bodnar's freezing curve everywhere. Choose one c_i, or use the IAPWS curve.

• Run experiment E1 (Part A) first. It decides the rig-note bath temperature and shows directly whether the water start temperature explains the A–D gap.

• Measure U on the rig (cool a can in a bath held at a steady temperature). That shrinks the 150–250 s spread in t_5 more than anything else.

Method: text extracted from all four PDFs. Constant tables, headline results and the rig/can descriptions were compared line by line. The causes given in section 1 are energy-balance estimates; they were not checked by rerunning the models.

SYPT 2027 · PROBLEM B8 · COLD DRINK

The Salted Ice Bath

The physics of the bath and the drink, from first principles

Handover report — October 2026 · built from the Drive file `sypt code`

A can of drink is cooled in a bath of crushed ice, water and salt. This report derives everything that decides how cold the bath gets and how fast the drink cools, starting from conservation of energy and of mass, the thermodynamics of salt water, and the laws of conduction, convection and diffusion. Every equation is derived where it is first used, its symbols are named with their units, and the conditions it needs are stated. Where physics gives only the form of a law and experiments supply its constant (the freezing curve of strong brine, free-convection correlations, the permeability of a heap of ice), the report says so and gives the source.

Two programs put the equations to work. `b8_sim.py` treats the bath as one body at equilibrium and answers how cold (§12). `b8_model.py`, behind the interactive simulator, follows every process at its own rate and answers how fast (§13). The colour code of the reporter deck is kept: **FLOOR** for equilibrium thermodynamics (how cold) and **RATE** for heat transfer (how fast).

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1 Summary and status

What the physics says. The temperature of the bath is not fixed by its recipe alone but by its energy: the brine composition depends on how much ice has melted, and that depends on the temperature (§2). Closing that loop with an energy balance gives four results that change the rig. **The charge in the rig note does not reach the eutectic**: it settles near $-17.8\,^{\circ}$C with no salt left, so it has no thermostat (§12.7). **4.0 kg of ice, 1.6 kg of water and 1.0 kg of salt** reaches $-21.2\,^{\circ}$C and keeps 193 g of salt as a buffer. Reaching the eutectic is an energy condition, not a mass condition (§5.5). And the drink's heat does not reach the whole bath at once: it spreads through the ice bed no faster than the brine can carry it (§8). With the can lying on its side as in the rig, the simulator puts Can 2 at 5 °C **222 s** after it goes in (§13).

Solid. The enthalpy formulation and the equilibrium solve it implies (§5); the reproduction of every bath number in the earlier B8 notes once their starting temperatures are matched (Table 2); the result that the bath temperature cannot be found from composition alone; the $1/$size scaling of bath drift; the condition for reaching the eutectic; the numerical methods, which match closed-form solutions and conserve energy to 10^{-9} J.

Computed but not measured. The adiabatic temperature of any charge; the buffer map (Fig. 5); the drink cooling curves for Can 2; the blank-run warming rate; how far the drink's heat spreads; the box's heat leak.

Uncertain. The heat-transfer coefficients across the can, taken from a bottle study and a SINTEF time constant rather than measured on this rig: every **RATE** number scales with them. The effective conductivity of the ice bed, an estimate good to a factor of two or three. The rates at which ice melts and salt dissolves. The **FLOOR** results move by at most 0.3 K over the plausible range of the textbook constants (§16).

Not attempted. Layering of the bath from top to bottom; the metastable eutectic and hydrohalite nucleation; flow inside the drink (the COMSOL model covers that); and any dissolution law that could explain the bench stall at $-7.5\,^{\circ}$C (§14).

2 Starting point: what `sypt code` did

The script defines the right ingredients — 1 kg ice, 2 kg water, 0.5 kg salt, $L_f = 333.55\,\mathrm{kJ\,kg^{-1}}$, $K_f = 1.86\,\mathrm{K\,kg\,mol^{-1}}$, $i = 2$, and a linear solubility fit from the Sigma-Aldrich table — and stops. Read bluntly:

Line	Problem, and what it costs
`bathT = 25.0 ## K`	It is 25 °C. 25 K is $-248\,^\circ$C. Every temperature in the model is Celsius; the label should say so.
`solubility()`	Compares `paramsaltMass` in kg with a fit in *grams per 100 g of water*. 0.5 is always less than 35.2, so the cap can never bind, for any input. Correct form: $s_{\max} = 10^{-2}\,(0.0338\,T + 35.24)\,m_w$ kg. The fit is also the *halite* branch, valid above $+0.1\,^\circ$C; below that the solid in equilibrium is hydrohalite and the solubility falls to 30.2 g/100 g at the eutectic, not the 34.5 the fit extrapolates to.
`temp()`	Computes ΔT_f and returns nothing, so the function evaluates to `None`.
molality in `temp()`	Divides by the initial 2 kg of water. The liquid water at equilibrium is 2 kg *plus whatever ice melted*, and that is the unknown.
Blagden, $i = 2$	Over-predicts the depression by 11 % at 5 wt %, under-predicts by 9 % near the eutectic (§4.2), and has no floor: at 30 wt % it gives $-27.3\,^\circ$C for a brine that cannot exist.
`1.201 Density ... as a %`	Density is in g mL^{-1}, not a percentage. Unused, but `waterVol` is used as a mass: fine for water, 20 % wrong for brine.
`import matplotlib, time`	`pyplot` is never imported; `main()` times nothing. Harmless.

The one that matters is structural, not a bug. **The temperature depends on the brine composition, the composition depends on how much ice has melted, and how much ice melts depends on the temperature.** Composition alone cannot close that loop; an energy balance can. For the script's own charge, `temp()` would have reported $-15.9\,^\circ$C (20 wt %, all salt in the original 2 kg). The equilibrium is $-13.7\,^\circ$C with 647 g of ice left if the water starts at 0 °C, and $-10.5\,^\circ$C with only 51 g of ice left if it starts at 25 °C — a bath one cold drink away from running out of ice. Everything below is the loop closed.

Part I · The physics

3 The system, and what is conserved

3.1 The parts

The rig is a can of drink standing (or lying) in a glass tank, inside a styrofoam box with a lid, in a room at $T_{\text{room}} = 28\,^\circ$C. The tank holds the bath: liquid *brine* (water with salt dissolved in it), crushed *ice*, and undissolved *salt crystals* on the floor. Each part holds heat and passes it to its neighbours:

$$\text{room} \to \text{foam} \to \text{glass} \to \text{brine}, \qquad \text{room} \to \text{foam lid} \to \text{air} \to \text{brine},$$
$$\text{drink} \to \text{can wall} \to \text{brine} \to \text{ice}.$$

Two quantities are conserved in every part and every exchange: energy and mass. Everything that follows is these two conservation laws, written for each part, together with the laws that say how

fast energy and mass move between parts.

3.2 Energy: the first law at constant pressure

For any body that does no work other than by changing its volume against the surrounding pressure p, the first law of thermodynamics reads

$$\mathrm{d}U = \delta Q - p\,\mathrm{d}V, \tag{1}$$

with U the internal energy (J), δQ the heat received (J) and V the volume (m^3). The tank is open to the room, so p stays at one atmosphere. Defining the *enthalpy* $H = U + pV$,

$$\mathrm{d}H = \mathrm{d}U + p\,\mathrm{d}V + V\,\mathrm{d}p = \delta Q \qquad (\mathrm{d}p = 0). \tag{2}$$

Where it comes from. Equation (1) is energy conservation: what a body gains as heat it keeps as internal energy, less the work it does pushing the atmosphere back as it expands. Adding pV folds that work into the bookkeeping, so at constant pressure every joule of heat that arrives shows up as enthalpy. Ice shrinks by 8 % as it melts and brine expands slightly as it warms; with H these volume changes need no separate treatment.

When it holds. At constant pressure, with no stirring work counted as heat (a stirrer's power P_{stir} is added to the heat it stirs into).

For a part k receiving heat flows $\dot{Q}_{jk}$ (W) from its neighbours j,

$$\frac{\mathrm{d}H_k}{\mathrm{d}t} = \sum_j \dot{Q}_{jk}, \qquad \dot{Q}_{jk} = -\dot{Q}_{kj}. \tag{3}$$

Because every flow between two parts appears once with each sign, adding (3) over all the parts inside the box cancels them: the total enthalpy changes only by the heat that crosses the box from the room, plus the stirrer. The simulator checks exactly this after every run (the *energy bookkeeping*, §11).

3.3 Mass: water and salt

Water is present as ice (mass m_i) and as liquid in the brine (m_l); salt is present dissolved (s) and as crystals ($S - s$). Melting moves water from one form to the other and dissolving moves salt, but neither is created:

$$m_i + m_l = W, \qquad s + (S - s) = S, \qquad w = \frac{s}{m_l + s}, \tag{4}$$

with W and S the totals put in (kg) and w the mass fraction of salt in the brine (kg of salt per kg of brine). A brine of $w = 0.232$ is 23.2 wt % salt.

3.4 The enthalpy of each phase

Enthalpy is defined up to a constant, so a reference state is chosen: **liquid water and solid salt at 0 °C have zero enthalpy.** Then, for each kilogram, from (2) with $\delta Q = c\,\mathrm{d}T$:

Liquid water. Warming from 0 °C to T at constant pressure takes $c_w T$, with c_w the specific heat capacity (J kg^{-1}K^{-1}): $h_w(T) = c_w T$.

Ice. Start from the reference, liquid water at 0 °C. Freezing it releases the latent heat, $L_0 = 333.55$ kJ per kg; cooling the ice to T then takes out $\int c_i \,\mathrm{d}T$:

$$h_i(T) = -L_0 + \int_0^T c_i(T')\,\mathrm{d}T' \;\approx\; c_i T - L_0. \tag{5}$$

The heat that melts 1 kg of ice at a temperature T below zero is then the difference of the two enthalpies at that temperature,

$$L(T) = h_w(T) - h_i(T) = L_0 + (c_w - c_i)\,T, \tag{6}$$

Kirchhoff's law. Because $c_w > c_i$, melting below 0 °C takes *less* heat: 287 kJ kg^{-1} at $-21.2\,^\circ$C instead of 334. The simulator uses the IAPWS heat capacity of ice, which falls from 2097 J kg^{-1}K^{-1} at 0 °C to 1950 at $-20\,^\circ$C ($c_i = c_{i0} + c_i' T$, so the integral in (5) gives $c_{i0}T + \frac{1}{2}c_i' T^2$); `b8_sim.py` holds c_i constant.

Salt. Solid salt warms like any solid: $h_s = c_s T$, $c_s = 864$ J kg^{-1}K^{-1} (50.5 J mol^{-1}K^{-1} divided by 58.44 g mol^{-1}).

Dissolved salt. Dissolving NaCl in water absorbs heat: breaking the crystal lattice costs more than hydrating the ions returns. The heat of solution is $\Delta h_s = 3.88$ kJ mol^{-1}, or 66.4 kJ per kg of salt (measured at 25 °C and infinite dilution; it is held constant). A kilogram of salt that dissolves therefore carries Δh_s more enthalpy than when it was a crystal, and the brine, which has to supply it, cools.

The whole bath. Adding the parts, with the brine (liquid water m_l and dissolved salt s) taken to warm with the heat capacity $c_b(w)$ of brine of its composition:

$$H = \underbrace{(m_l + s)\,c_b(w)\,T}_{\text{brine, sensible}} + \underbrace{s\,\Delta h_s}_{\text{dissolution}} + \underbrace{m_i\,(c_i T - L_0)}_{\text{ice}} + \underbrace{(S - s)\,c_s T}_{\text{undissolved salt}}. \tag{7}$$

Where it comes from. Equation (7) is the sum of the per-kilogram enthalpies above, each multiplied by its mass, with every phase at the same temperature T. It is the single equation that connects the heat put into the bath with its temperature and phase amounts.

When it holds. When the bath is at one temperature (relaxed in the simulator, which gives every ring of brine and its ice their own temperatures, §7), when Δh_s does not depend on concentration (±20 % moves the bath by less than 0.05 K), and when mixing water with salt changes the heat capacity but not the enthalpy otherwise.

The heat capacity of brine falls as salt is added. `b8_sim.py` uses $c_b = 4180 - 4400\,w$; the simulator uses a fit to Melinder (2010), $c_b = 4189 - 5794\,w + 8579\,w^2$ J kg^{-1}K^{-1}, good to ± 33 between -20 and $+10\,^\circ$C. Table 1 lists every constant.

4 Salt water: freezing, solubility and the eutectic

4.1 Why salt lowers the freezing point

Ice and liquid water coexist when a molecule of water has no reason to move from one to the other: when the *chemical potential* of water, the Gibbs energy per mole, is the same in both. For pure water

Table 1. Constants. The first block is used by `b8_sim.py`; the simulator reads the same quantities, with the fits shown where they differ, from `inputs.toml`.

Constant	Value	Unit	Source
L_0 (ice, 0 °C)	333.55	kJ kg^{-1}	handbook
c_w, c_i, c_s	4.18, 2.00, 0.864	kJ kg^{-1}K^{-1}	c_i: simulator uses IAPWS R10-06
$c_b(w)$	$4.18 - 4.4\,w$	kJ kg^{-1}K^{-1}	simulator: Melinder fit
ρ_b	1180	kg m^{-3}	eutectic brine; simulator: $1000 + 786\,w$ (Melinder)
Δh_s	$+66.4$	kJ kg^{-1}	3.88 kJ mol^{-1}, infinite dilution, 25 °C
T_E, w_E	-21.2, 0.2318	°C, –	end of the Bodnar fit
T_P, w_P	$+0.1$, 0.26285	°C, –	Swenne (1983) peritectic
$k_b(w)$	$0.5725 - 0.149\,w$	W m^{-1}K^{-1}	Melinder fit, simulator only
C_d	1.332	kJ K^{-1}	330 g drink at $c = 4.0$ + 13 g Al
U_0, A	160, 0.03102	W m^{-2}K^{-1}, m^2	SINTEF $\tau \approx 265$ s; Can 2
$(UA)_{\text{box}}$, T_{room}	0.46, 28	W K^{-1}, °C	rig note, 30 mm EPS; simulator 0.34

this happens at $T_0 = 273.15$ K. In brine, dissolved ions dilute the water, which lowers its chemical potential by $RT \ln a_w$, with a_w the activity of water ($a_w = 1$ for pure water, $a_w < 1$ in brine). Ice contains no salt, so its chemical potential is unchanged. Equilibrium at a new temperature T needs

$$\mu_{\text{ice}}(T) = \mu_w^*(T) + RT \ln a_w, \tag{8}$$

with μ_w^* the chemical potential of pure liquid water and $R = 8.314$ J mol^{-1}K^{-1}. Dividing by RT and using the Gibbs–Helmholtz relation $\mathrm{d}(\mu/T)/\mathrm{d}T = -h_m/T^2$ for each phase (h_m the molar enthalpy), the temperature derivative of $\ln a_w$ is the molar latent heat $L_m = 6.01$ kJ mol^{-1} over RT^2; integrating from T_0, where $a_w = 1$,

$$\ln a_w = -\frac{L_m}{R}\left(\frac{1}{T} - \frac{1}{T_0}\right). \tag{9}$$

For a dilute, ideal solution the activity is the mole fraction of water, so $\ln a_w = \ln(1 - x) \approx -x$, and $x \approx M_w\, i\, b$, with b the molality (moles of NaCl per kg of water), $i = 2$ the number of ions each formula unit gives, and $M_w = 0.018015$ kg mol^{-1}. For a small depression $\Delta T = T_0 - T$, $1/T - 1/T_0 \approx \Delta T/T_0^2$. Together:

$$\Delta T = K_f\, i\, b, \qquad K_f = \frac{R\,T_0^2\,M_w}{L_m} = 1.86 \text{ K kg mol}^{-1}, \qquad b = \frac{w}{M_{\text{NaCl}}(1 - w)}. \tag{10}$$

This is Blagden's law, the one `sypt code` used, and its constant follows from the latent heat alone.

When it holds. Only when the brine is dilute (a few per cent), ideal (no ion–ion interactions), and fully dissociated, and when $\Delta T \ll T_0$. None of these is true near the eutectic, where $b = 5.2$ mol kg^{-1} and one particle in six is an ion.

4.2 The measured freezing curve

Strong brine is not ideal: water molecules bound in the ions' hydration shells are not free to freeze, and the ions screen each other. Thermodynamics still gives (9) exactly; what fails is the step $a_w \approx 1 - M_w i b$. Rather than model the activity, the model uses the measured freezing curve, the *liquidus*, in the form fitted by Bodnar (1993) to fluid-inclusion data, with $\theta = -T$ the depression below 0 °C:

$$w_L(T) = 0.0178\,\theta - 4.42 \times 10^{-4}\,\theta^2 + 5.572 \times 10^{-6}\,\theta^3, \qquad 0 \le \theta \le 21.2. \tag{11}$$

Where it comes from. Experiment, with a check from (10). For dilute brine, (10) gives $w \approx 0.0157\,\theta$, since $M_{\mathrm{NaCl}}/(iK_f) = 0.0157\,\mathrm{K}^{-1}$: close to the leading term of the fit. The higher terms carry the non-ideality. The ratio of Blagden's depression to the measured one is 1.106, 1.069, 1.019, 0.952 and 0.909 at 5, 10, 15, 20 and 23 wt % (Fig. 1). The fit is explicit in T, which makes the equilibrium solve fast; the simulator inverts it by Newton's method when it needs T from w.

4.3 Solubility, the eutectic and the phase rule

How many of the bath's variables are free is set by Gibbs' phase rule. At fixed pressure, a mixture of C components in P phases at equilibrium has

$$F = C - P + 1 \tag{12}$$

free intensive variables: each phase has $C - 1$ independent mass fractions, the temperature adds one, and each of the $C(P-1)$ equalities of chemical potential between phases removes one. With water and salt ($C = 2$):

- brine alone ($P = 1$, $F = 2$): temperature and composition are both free;
- brine with ice, or brine with salt crystals ($P = 2$, $F = 1$): composition follows from temperature — the liquidus (11), or the solubility line $w_{\mathrm{sat}}(T)$;
- brine with ice *and* salt ($P = 3$, $F = 0$): nothing is free. This is the *eutectic* point, $T_E = -21.2\,^{\circ}\mathrm{C}$, $w_E = w_L(T_E) = 0.2318$.

On the eutectic the bath is a thermostat: heat put in cannot change the temperature, only the amounts of the three phases. The solubility of halite (rock salt) is $35.65 + 0.0147\,T$ g per 100 g of water between 0 and 30 °C, $w_{\mathrm{sat}} = g/(100+g)$. Below the peritectic, $T_P = +0.1\,^{\circ}\mathrm{C}$, the stable solid is hydrohalite NaCl·2H_2O, whose solubility falls from $w_P = 0.2628$ to w_E at the eutectic (taken as a straight line). Brine on the liquidus is below both solubility lines everywhere above T_E, so **crystals and ice can coexist only at the eutectic**: with ice present, salt dissolves until it is all gone or the brine reaches w_E.

5 The bath at equilibrium: how cold

5.1 One idea: describe the bath by its enthalpy

Given the totals W and S and the enthalpy H, the bath has exactly one equilibrium state. The temperature is then an output, not an input: heat that crosses the bath boundary changes H (3), and temperature, ice, brine composition and salt left on the floor are all read off H. This removes the circularity of §2, and five behaviours that are usually bolted on as separate rules come out of one function: the adiabatic mixing temperature, the drift along the liquidus as ice melts, the eutectic thermostat, the exhaustion of the ice, and the temperature dependence of the latent heat.

5.2 The equilibrium solve $T_b = \mathcal{T}(H;\, W, S)$

Combining the mass balances (4), the phase rule (§4.3) and the enthalpy (7) gives four regimes:

Regime	Phase amounts (from mass and phase equilibrium)	T_b
`liquid`	no ice, $m_l = W$; $s = \min(S,\ W\, r_{\text{sat}}(T))$	from (7)
`liquidus`	$w = w_L(T)$, $s = S$, $m_l = S\,(1 - w_L)/w_L$, $m_i = W - m_l$	from (7)
`eutectic`	$w = w_E$, $0 \le m_l \le \min(W, S/r_E)$, $s = r_E\, m_l$	T_E, fixed
`solid`	no liquid	below T_E

Here $r = w/(1-w)$ is kilograms of salt per kilogram of water, and $r_E = 0.3018$ at the eutectic.

Where it comes from. `liquidus`: with ice present the brine sits on the freezing curve ($F = 1$); all salt is dissolved because the liquidus lies below saturation (§4.3); the salt balance $s = w\,(m_l + s)$ then gives m_l, and the water balance gives m_i. `eutectic`: three phases, $F = 0$, so T and w are fixed and only the amount of brine is unknown; (7) is linear in m_l there, so the solve is a single division.

Uniqueness. On the `liquidus` branch, raising T melts ice (which takes $L(T)$ per kilogram) and warms everything, so $H(T)$ rises strictly; on the `liquid` branch it rises with the heat capacity alone. H is continuous where the branches meet, so $H(T)$ is strictly increasing: **for each H there is exactly one temperature**, found by bracketing on $(T_E, 200\,^\circ\text{C})$. (7) is evaluated at both ends of the eutectic plateau with the same w_E — computed from (11), not typed — or the branch and the plateau fail to meet by 10^{-3} kJ and the bracket breaks.

5.3 The latent heat that the bath actually charges

On the liquidus with S fixed, $m_l = S(1-w)/w$, so melting $\mathrm{d}m$ of ice changes the composition by $\mathrm{d}w/\mathrm{d}m = -w^2/S$, and the temperature follows the liquidus, $\mathrm{d}T_b = \mathrm{d}w/w_L'(T_b)$. The heat taken per kilogram melted is the derivative of (7):

$$\frac{\mathrm{d}T_b}{\mathrm{d}m_{\text{melt}}} = \frac{w^2}{S\,|w_L'(T_b)|}, \qquad L_{\text{eff}} \equiv \frac{\mathrm{d}H}{\mathrm{d}m_{\text{melt}}} = \underbrace{L_0 + (c_b - c_i)\,T_b}_{L(T_b)} + \underbrace{C_{\text{bath}}\,\frac{\mathrm{d}T_b}{\mathrm{d}m_{\text{melt}}}}_{\text{warming the bath}}. \tag{13}$$

Where it comes from. The first term is Kirchhoff's latent heat (6) at the bath temperature, with the meltwater joining brine of heat capacity c_b; the second is the heat spent warming the whole bath, of heat capacity C_{bath}, along the liquidus while that kilogram melts (the dilution raises the freezing point). (7) contains both automatically; the code evaluates L_{eff} by finite difference in H. Both factors in $\mathrm{d}T_b/\mathrm{d}m$ carry $1/S$: **double every mass and the drift per kilogram melted halves**, the $1/\text{size}$ law of the rig note.

5.4 Where the cold comes from

Integrating (7) between the unmixed ingredients and the mixed bath at constant H, the heat that cools the bath is paid by only two terms: $L(T)\,m_{\text{melt}}$, the ice melting, and $s\,\Delta h_s$, the salt dissolving. Their ratio can be read off any equilibrium; for the charges in the earlier notes the salt supplies about a tenth of the cold and the ice the rest (§12).

5.5 Reaching the eutectic is an energy condition

Adiabatic mixing conserves H. A charge reaches the thermostat only if its starting enthalpy lies below the top of the eutectic plateau, and keeps solid salt only if it lies further below:

$$H_0 \;\le\; H(T_E;\ m_l = \min(W,\ S/r_E)), \tag{14}$$

with H_0 the enthalpy of the ingredients at their own starting temperatures.

Where it comes from. The plateau's top is the state with all the water that can be brined at w_E already liquid; any more enthalpy and the bath leaves the plateau. Free water is expensive: every kilogram of it must be cooled to $-21.2\,^{\circ}\mathrm{C}$ and salted to w_E, and the only payment is melting ice, which brings more water. Checking that the salt can saturate the free water — a mass condition — is necessary but not sufficient.

6 How heat moves

6.1 Conduction

In a solid or a still fluid heat flows down the temperature gradient. Fourier's law,

$$\mathbf{q} = -k\,\nabla T, \tag{15}$$

gives the heat flux $\mathbf{q}$ ($\mathrm{W\,m^{-2}}$), with k the thermal conductivity ($\mathrm{W\,m^{-1}K^{-1}}$). Energy conservation (3) for a small volume, whose enthalpy changes at $\rho c\,\partial T/\partial t$ per unit volume, turns it into the heat equation

$$\rho c\,\frac{\partial T}{\partial t} = \nabla\cdot(k\nabla T), \qquad \frac{\partial T}{\partial t} = \alpha\,\nabla^2 T, \qquad \alpha = \frac{k}{\rho c}. \tag{16}$$

Where it comes from. The net heat flowing into a small volume through its faces is $-\nabla\cdot\mathbf{q}$ per unit volume; it raises the temperature of the mass ρ per unit volume of heat capacity c. The thermal diffusivity α ($\mathrm{m^2\,s^{-1}}$) is the only combination of the three properties that appears; in time t a change of temperature diffuses a distance

$$\delta(t) \sim \sqrt{\alpha t}, \tag{17}$$

because $\partial/\partial t \sim 1/t$ and $\nabla^2 \sim 1/\delta^2$ must balance (the exact solution for a suddenly heated wall is an error function of $x/2\sqrt{\alpha t}$). For eutectic brine $k_b = 0.54\,\mathrm{W\,m^{-1}K^{-1}}$, $\rho_b = 1182\,\mathrm{kg\,m^{-3}}$, $c_b = 3307\,\mathrm{J\,kg^{-1}K^{-1}}$, so $\alpha_b = 0.14\,\mathrm{mm^2\,s^{-1}}$ and in three minutes conduction alone carries heat **5 mm**; for ice $\alpha = 1.2\,\mathrm{mm^2\,s^{-1}}$.

6.2 Steady conduction through layers

When nothing stores heat, the same heat flow $\dot{Q}$ passes through every layer in a stack. Integrating (15) across a flat slab of thickness δ and area A, and across a cylindrical shell of length λ between radii r_1 and r_2 (where the flux falls as $1/r$ because the same $\dot{Q}$ crosses every circle $2\pi r\lambda$):

$$\dot{Q} = \frac{kA}{\delta}\,\Delta T, \qquad \dot{Q} = \frac{2\pi k\lambda}{\ln(r_2/r_1)}\,\Delta T. \tag{18}$$

The coefficient of ΔT is a conductance G ($\mathrm{W\,K^{-1}}$); its inverse is a resistance. Since the temperature drops of layers in series add while the heat flow is common, **resistances in series add**; side by side, the heat flows add, so **conductances in parallel add**. Every heat path in the rig is built from these two rules.

6.3 Convection and the film coefficient

At a wall in a moving fluid, the fluid touching the wall is at rest and passes heat only by conduction, through a thin *thermal boundary layer* of thickness δ_T in which the temperature changes from the wall's T_w to the fluid's T_∞. Outside it the flow mixes the fluid. So

$$q = h\,(T_w - T_\infty), \qquad h \approx \frac{k}{\delta_T}, \qquad \mathrm{Nu} = \frac{h\mathcal{L}}{k} \approx \frac{\mathcal{L}}{\delta_T}. \tag{19}$$

Where it comes from. The first equation, Newton's law of cooling, *defines* the film coefficient h ($\mathrm{W\,m^{-2}K^{-1}}$); the physics is in how δ_T depends on the flow, and the Nusselt number Nu measures it against a length $\mathcal{L}$ of the body.

6.4 Free convection

Nothing stirs the bath or the drink: fluid moves only because warm fluid is lighter. Next to a wall at T_w the fluid's density differs by $\Delta\rho = \rho\beta\,\Delta T$ (β the thermal expansion coefficient), so the boundary layer feels a buoyancy force $\rho g\beta\Delta T$ per unit volume. In a viscous liquid ($\mathrm{Pr} = \nu/\alpha \gg 1$: 7–13 for the drink, 33 for cold brine) friction balances it, $\rho g\beta\Delta T \sim \mu u/\delta_T^2$, which sets the speed u of the rising (or sinking) layer. Heat carried along the wall over a length $\mathcal{L}$ balances heat conducted across the layer, $u\,\Delta T/\mathcal{L} \sim \alpha\,\Delta T/\delta_T^2$. Eliminating u:

$$\delta_T \sim \mathcal{L}\,\mathrm{Ra}^{-1/4}, \qquad \mathrm{Nu} \sim \mathrm{Ra}^{1/4}, \qquad \mathrm{Ra} = \frac{g\beta\,\Delta T\,\mathcal{L}^3}{\nu\alpha}, \qquad \Longrightarrow \qquad h \propto \left(\frac{\Delta T}{\mathcal{L}}\right)^{1/4}. \tag{20}$$

Where it comes from. Two balances, buoyancy against friction and convection against conduction, give the thickness of the layer; the Rayleigh number Ra is the ratio of the two. Two consequences are used directly. **(i)** The film coefficient grows as $\Delta T^{1/4}$: the drink-side film in the simulator follows $h = h_{\mathrm{ref}}(\Delta T/\Delta T_0)^{1/4}$ with $\Delta T_0 = 25\,\mathrm{K}$. **(ii)** It shrinks as $\mathcal{L}^{-1/4}$: a can lying on its side, for which $\mathcal{L}$ is the 6.6 cm diameter rather than the 11.7 cm height, has films $(11.66/6.6)^{1/4} = 1.15$ times stronger from the scaling alone.

When it holds. For laminar boundary layers, $\mathrm{Ra} \lesssim 10^9$: here Ra is 10^7–10^8.

The scaling fixes the exponents; the constants come from experiment. Churchill & Chu (1975) fitted, for a vertical wall of height $\mathcal{L}$ and a horizontal cylinder of diameter D,

$$\mathrm{Nu}_{\mathcal{L}} = 0.68 + \frac{0.670\,\mathrm{Ra}^{1/4}}{[1+(0.492/\mathrm{Pr})^{9/16}]^{4/9}}, \qquad \mathrm{Nu}_D = \left\{0.60 + \frac{0.387\,\mathrm{Ra}_D^{1/6}}{[1+(0.559/\mathrm{Pr})^{9/16}]^{8/27}}\right\}^2.$$

For brine at $-15\,^\circ$C on Can 2 these give $h = 240$–$358\,\mathrm{W\,m^{-2}K^{-1}}$ upright and 280–457 lying down, for a 5–25 K difference: a film 1.5–2.2 mm thick, and a ratio of 1.17–1.28 between the two orientations (1.15–1.33 for the drink inside). The simulator takes the brine-side $h_o = 270\,\mathrm{W\,m^{-2}K^{-1}}$ and the drink-side $h_i = 400$ for an upright can (a bottle study, §17), and multiplies both by `on_side_factor` $= 1.2$ for a can on its side.

6.5 Radiation

Two facing grey surfaces at absolute temperatures T_1 and T_2 exchange $q = \varepsilon_{\mathrm{eff}}\sigma(T_1^4 - T_2^4)$, with $\sigma = 5.67\times10^{-8}\,\mathrm{W\,m^{-2}K^{-4}}$ and $1/\varepsilon_{\mathrm{eff}} = 1/\varepsilon_1 + 1/\varepsilon_2 - 1$. Factoring $T_1^4 - T_2^4 = (T_1 - T_2)(T_1 + T_2)(T_1^2 + T_2^2) \approx 4\bar{T}^3(T_1 - T_2)$ for close temperatures,

$$q \approx h_{\mathrm{rad}}\,(T_1 - T_2), \qquad h_{\mathrm{rad}} = 4\,\varepsilon_{\mathrm{eff}}\,\sigma\,\bar{T}^3, \tag{21}$$

5.0 $\mathrm{W\,m^{-2}K^{-1}}$ at 290 K for $\varepsilon = 0.9$ and 3.7 at 270 K between two such surfaces. Radiation therefore adds to convection like a parallel film. The box's air films (§10) are built this way: room air on the foam, $h \approx 5 + 4 = 9$; the 1 cm gap between foam and glass, conduction $k_{\mathrm{air}}/\delta = 0.024/0.01 = 2.4$ plus radiation 3.5, $h \approx 6$; the still air under the lid, warm above cold so that it cannot convect, $h \approx 4$.

6.6 The drink as one temperature

The drink is lumped: one temperature T_d for the whole can. By conduction alone it would not be uniform — the Biot number $\mathrm{Bi} = h_i\,(V/A)/k_w = 400 \times 0.0129/0.58 = 8.9$ compares the film outside

with conduction inside and is well above 0.1 — but free convection inside the can (Ra $\sim 10^8$) mixes it; the lumped temperature is the volume average of the full flow problem solved in COMSOL. Energy conservation (3) for the drink and can, of total heat capacity $C_d = m_d c_d + m_{\text{can}} c_{\text{Al}} = 1332\,\text{J K}^{-1}$, losing heat through the can wall of area A:

$$C_d \frac{\mathrm{d}T_d}{\mathrm{d}t} = -UA\,(T_d - T_b), \qquad \frac{1}{U} = \frac{1}{h_i} + \frac{\delta_{\text{Al}}}{k_{\text{Al}}} + \frac{1}{h_o}. \tag{22}$$

Where it comes from. The heat flows from the drink, across the inner film, the aluminium and the outer film in series (§6.2); the wall's resistance, $10^{-4}/237 = 4 \times 10^{-7}\,\text{m}^2\text{K W}^{-1}$, is negligible beside $1/h \approx 3 \times 10^{-3}$. For a constant bath temperature and constant U, (22) is solved by an exponential, and with $h \propto \Delta T^{1/4}$ on both sides ($U = U_0\,(\theta/\Delta T_0)^{1/4}$, $\theta = T_d - T_b$) by separation of variables, $\theta^{-5/4}\mathrm{d}\theta = -\mathrm{d}t/(\tau \Delta T_0^{1/4})$:

$$\tau = \frac{C_d}{UA}, \qquad t_{\text{target}} = \tau \ln \frac{T_{d0} - T_b}{T_{\text{target}} - T_b}, \qquad \theta(t) = \theta_0 \Big(1 + \frac{t}{t^*}\Big)^{-4}, \quad t^* = 4\tau \Big(\frac{\Delta T_0}{\theta_0}\Big)^{1/4}. \tag{23}$$

With $U_0 = 160\,\text{W m}^{-2}\text{K}^{-1}$ and $A = 310\,\text{cm}^2$, $\tau = 268\,\text{s}$. Both closed forms are used to check the integrators, not as the model.

7 Melting, dissolving and freezing: the rates

7.1 The Stefan condition at the ice surface

Ice melts at its surface, and the surface can store no heat. Applying (3) to a thin layer straddling it: heat arrives from the brine across the brine-side film, $h_{bi} A_i (T_b - T_s)$, and from the ice's own interior, $G_c (T_{\text{ice}} - T_s)$; it is all spent turning ice at T_s into water at T_s:

$$\dot{m}_{\text{melt}}\,[h_w(T_s) - h_i(T_s)] = h_{bi} A_i\,(T_b - T_s) + G_c\,(T_{\text{ice}} - T_s). \tag{24}$$

Where it comes from. Energy conservation at a moving phase boundary (the Stefan condition). A_i is the wetted ice area, $h_{bi} = 100\,\text{W m}^{-2}\text{K}^{-1}$ the brine-side film coefficient for the slow free convection among the pieces, T_s the surface temperature, T_{ice} the interior temperature of the pieces and G_c the conductance from interior to surface. A negative right-hand side freezes brine onto the ice. The interior cools as it gives heat to the surface: $C_{\text{ice}}\,\mathrm{d}T_{\text{ice}}/\mathrm{d}t = -G_c(T_{\text{ice}} - T_s)$.

Area and interior conductance. The heap is n cubes of side a, so $a = (m_i/n\rho_i)^{1/3}$ and the wetted area is $A_i = 6na^2\,\phi_{\text{wet}}$, with ϕ_{wet} the fraction of the heap below the liquid surface (§9). As the pieces shrink, $A_i \propto m_i^{2/3}$. Heat inside a piece crosses a mean distance comparable with its volume-to-area ratio $a/6$, so $G_c = k_i A_i/(a/6)$ (for a sphere the exact quasi-steady value is a fifth of the radius, 0.6 times this distance, so G_c is if anything an underestimate). The interior's time constant C_{ice}/G_c is about 9 s for 2 cm pieces.

7.2 The temperature of the ice surface

Bulk (default): the surface sits at the freezing point of the brine around it, $T_s = T_f(w)$ from (11). This assumes salt reaches the surface as fast as melting dilutes it there.

Salt-limited: meltwater is fresh, so it dilutes the brine at the surface to $w_s < w$, whose freezing point is higher. Three conditions fix T_s, w_s and the melt rate: the Stefan condition (24); the liquidus at the

surface, $T_s = T_f(w_s)$; and a salt balance at the surface,

$$\rho\,\gamma_S A_i\,(w - w_s) = \dot{m}_{\text{melt}}\,w_s, \qquad \gamma_S = \frac{\gamma_T}{\mathcal{R}}, \quad \gamma_T = \frac{h_{bi}}{\rho c_b}, \tag{25}$$

Where it comes from. Salt diffuses to the surface across the boundary layer with a transfer velocity γ_S (m s^{-1}), the mass-transfer analogue of $h/\rho c$, and must arrive fast enough to bring the meltwater, $\dot{m}$ per second, up to w_s. This is the three-equation ice–ocean interface of Holland & Jenkins (1999). Salt diffuses about a hundred times more slowly than heat ($\alpha/D_{\text{salt}} \approx 0.14\times10^{-6}/1.5\times10^{-9}$), and across a boundary layer the transfer velocities scale with the diffusivities to the power $2/3$, so $\mathcal{R} = \gamma_T/\gamma_S \approx 100^{2/3} \approx 20$; the literature gives 10–200, the default is 30.

7.3 Salt dissolving

A salt grain dissolves as salt diffuses away from its surface, where the brine is saturated, into the bulk brine. Fick's law, the diffusion analogue of (15), across a concentration boundary layer of thickness δ_c, gives a mass flux $j = D\rho_b(w_{\text{sat}} - w)/\delta_c$, so

$$\dot{s} = k_d\,A_s\,\rho_b\,(w_{\text{sat}}(T) - w), \qquad k_d = \frac{D}{\delta_c}, \qquad A_s = A_{s0}\Big(\frac{S-s}{S}\Big)^{2/3}. \tag{26}$$

Where it comes from. k_d (m s^{-1}) is the mass-transfer coefficient: with $D = 1.5\times10^{-9}$ m^2 s^{-1} and $\delta_c \approx 0.1$ mm in stirred brine, $k_d \approx 1.5\times10^{-5}$ (default 10^{-5}); an unstirred layer ten times thicker gives 10^{-6}. The grains are cubes of side 0.5 mm, whose total area $A_{s0} = 6S/(\rho_s a_s) = 5.5\,\text{m}^2$ for 1 kg shrinks as their mass to the $2/3$. Above saturation the grains grow back. While there is ice, dissolving stops at w_E: beyond it salt would only come out again as hydrohalite (§4.3). Each kilogram that dissolves takes Δh_s from the brine, as (7) records.

7.4 The drink freezing

The drink is mostly water with sugar (≈ 10 %) and dissolved CO$_2$. By (10) with $i = 1$, 10 % sucrose ($b = 0.33$ mol kg^{-1}) lowers its freezing point by 0.6 K; the CO$_2$ adds about 0.3 K: $T_{f0} = -0.9\,^\circ$C. When a fraction φ of the drink's water has frozen, the solutes are all in the remaining liquid, so their molality is $1/(1-\varphi)$ times larger and, by the same law,

$$T_f(\varphi) = \frac{T_{f0}}{1-\varphi}, \qquad H_d = C_d T - \varphi\,m_{dw}\,L(T) \quad \text{on} \quad T = T_f(\varphi). \tag{27}$$

Where it comes from. Freezing concentrates the solution, so the drink freezes over a range of temperatures, not at one: 50 % of its water is frozen at $-1.8\,^\circ$C, 90 % at $-9\,^\circ$C. Its enthalpy is that of the liquid less the latent heat (6) of the ice formed, with $m_{dw} = 0.89\,m_d$ the drink's water; given H_d from (3), the model solves for T and φ together. A still drink can supercool: it stays liquid down to $T_{f0} - \Delta T_{\text{sc}}$, and when ice nucleates the latent heat released warms it back onto the curve.

8 Where the drink's heat goes: spreading through the bath

8.1 Why the heat cannot reach the whole bath at once

Once Can 2 is in, about 217 W leaves it ($T_d = 26.4\,^\circ$C, with the brine next to it already at $-17.9\,^\circ$C). A perfectly mixed bath would share that at once among 2.55 kg of ice and 3.97 kg of brine. But by (17) conduction carries heat only 5 mm through still brine in the three minutes the drink takes to cool. The brine next to the can warms, its ice melts first, and the drink faces warmer brine than the bulk.

8.2 How far the heat gets before the ice takes it

Brine warmer than the ice surface loses heat to the ice at $h_{bi}a_v(T - T_s)$ per unit volume, with a_v the wetted ice area per unit volume of bath (135 m^{-1} at drink-in). In a steady state near the can, heat conducted outwards at an effective conductivity k_{eff} is absorbed by the ice on the way. Energy conservation (16) with this sink, in one dimension x out from the can:

$$k_{\text{eff}}\,\frac{\mathrm{d}^2T}{\mathrm{d}x^2} = h_{bi}a_v\,(T - T_s) \implies T - T_s \propto e^{-x/\ell}, \qquad \ell = \sqrt{\frac{k_{\text{eff}}}{h_{bi}\,a_v}}. \tag{28}$$

Where it comes from. Conduction in, absorption by melting out: the temperature excess decays exponentially over the *penetration length* ℓ, 0.6 cm for still brine and 2.8 cm for $k_{\text{eff}} = 20\,k_b$. Most of the drink's heat is absorbed within a few centimetres of the can.

8.3 The ice bed conducts and convects

Still brine is only a lower bound for k_{eff}. Two things raise it. First, ice conducts four times better than brine. A bed of porosity $\varepsilon = 0.4$ conducts between the series and parallel limits of its two phases,

$$\left(\frac{\varepsilon}{k_b} + \frac{1-\varepsilon}{k_i}\right)^{-1} \le k_{\text{bed}} \le \varepsilon k_b + (1-\varepsilon)k_i, \tag{29}$$

0.98–1.54 W m^{-1}K^{-1}, or 2–3 times k_b: the two limits are layers of brine and ice across and along the heat flow, in series and in parallel (§6.2). Second, the brine moves through the bed. Flow through a porous bed obeys Darcy's law, speed proportional to the driving pressure gradient, here buoyancy: $u \sim Kg\beta\Delta T/\nu$, with K the permeability (m^2). Modelling the gaps as channels of width $\propto \varepsilon a/(1-\varepsilon)$ with Poiseuille flow gives the Kozeny–Carman form $K = a^2\varepsilon^3/180(1-\varepsilon)^2 = 4\times10^{-7}$ m^2 (the 180 is empirical). Repeating the boundary-layer balance of §6.4 with Darcy's law in place of viscous friction, $u\,\Delta T/\mathcal{L} \sim \alpha_m \Delta T/\delta^2$ with u now independent of δ, gives

$$\mathrm{Nu} \sim \mathrm{Ra}_D^{1/2}, \qquad \mathrm{Ra}_D = \frac{g\beta\,\Delta T\,K\,\mathcal{L}}{\nu\,\alpha_m}, \tag{30}$$

with mean $\mathrm{Nu} = 0.89\,\mathrm{Ra}_D^{1/2}$ for a heated vertical wall (Cheng & Minkowycz 1977) and α_m the bed's diffusivity. For a 14.5 cm bed and $\Delta T = 1$–6 K, $\mathrm{Ra}_D \approx 10^2$–$10^3$ and Nu $\approx$ 9–27. Together, the *convection factor* $f = k_{\text{eff}}/k_b \approx (k_{\text{bed}}/k_b)\,\mathrm{Nu} \approx 20$–60. The estimate is rough — the gaps (about 1 cm) are not small next to the 2 mm film on the can, and warm brine rising past the can spreads along the top of the bath — so f is an input; the default 20 is the low end. $f = 1$ is still brine, $f = \infty$ a perfectly mixed bath.

8.4 Rings round the can

The bath is divided into N rings (default 8): shells round the can's axis from its wall, $r_c = 3.30$ cm, out to $R = \sqrt{A_f/\pi} = 11.45$ cm, the radius of a circle with the floor area A_f of the tank. Each ring is a finite volume obeying (3), with its own brine temperature T_j, ice, ice-interior temperature and crystals; the salt fraction w is shared, since the brine that carries heat also mixes salt. The edges are spaced evenly in $\ln r$,

$$r_j = r_c\,(R/r_c)^{j/N}, \qquad j = 0, \dots, N, \tag{31}$$

so the rings are thin by the can (0.56 cm) and wide at the walls (1.65 cm), where little happens. Ring j holds the share $\psi_j = (r_{j+1}^2 - r_j^2)/(R^2 - r_c^2)$ of the brine, ice and crystals. The drink heats only the

inner ring; the glass, the lid, the gaps in the box and the stirrer heat only the outer one:

$$C_j\,\dot{T}_j = G\,(T_{j-1}-T_j) + G\,(T_{j+1}-T_j) - Q_{i,j} - \Delta h_s\,\dot{s}_j \\ + \delta_{j0}\,UA\,(T_d - T_0) + \delta_{j,N-1}\,(Q_{\text{glass}} + Q_{\text{lid}} + Q_{\text{gaps}} + P_{\text{stir}}), \tag{32}$$

$$G = \frac{2\pi\,k_{\text{eff}}\,\lambda}{\ln(r_{j+1}/r_j)} = \frac{2\pi N k_{\text{eff}}\,\lambda}{\ln(R/r_c)}, \qquad k_{\text{eff}} = f\,k_b(w). \tag{33}$$

Where it comes from. (32) is (3) for ring j, of heat capacity C_j: heat conducted in from each neighbour, heat lost to its ice ($Q_{i,j}$, (24)) and to dissolving its crystals, and the drink or the room where they touch it (δ is 1 for the named ring and 0 otherwise). G is the cylindrical-shell conductance (18) between the middles of neighbouring rings, which are one step of $\ln r$ apart; with even spacing in $\ln r$ it is the same for every pair. λ is the length of the shells: the can's own length (11.66 cm) for a can on its side, the depth of brine for an upright one. At drink-in, $G = 51\,\mathrm{W\,K^{-1}}$; from can to walls the bath adds $\ln(R/r_c)/(2\pi k_{\text{eff}}\lambda) = 0.16\,\mathrm{K\,W^{-1}}$ in series with the can's own $1/UA = 0.20\,\mathrm{K\,W^{-1}}$, although by (28) most of the heat is absorbed before the walls. The rings must resolve ℓ: 8 rings are within 4 % of 50 at the default, but still brine needs 50 (§13).

9 The liquid level and the areas that follow it

Every heat path into or out of the brine crosses a wetted surface, and most of those surfaces depend on how high the liquid stands. At every step the level is found from the volumes, and each area from the level.

9.1 Volumes

The liquid occupies $V_l = (m_l + s)/\rho_b(w) + (S - s)/\rho_s$: brine of density $\rho_b = 1000 + 786\,w\,\mathrm{kg\,m^{-3}}$ (Melinder), and the crystals on the floor, $\rho_s = 2165$. Dissolving a kilogram of salt changes the brine volume by $[\rho_b - 786(1-w)]/\rho_b^2 = 0.41\,\mathrm{L}$, less than the 0.46 L of the crystal, and makes the brine denser.

9.2 A heap resting on the floor, or floating

The crushed ice is a heap of porosity ε: a fraction $1-\varepsilon$ of its volume is ice. Its thickness is $t_h = m_i/[\rho_i(1-\varepsilon)A_f]$, 17.7 cm for 4 kg in a 412 $\mathrm{cm^2}$ tank. While it rests on the floor the liquid has only the gaps; once there is enough of it, the heap floats. By Archimedes, a floating heap displaces its own weight of brine, $V_{\text{sub}} = m_i/\rho_b$, wherever its ice is. The level Z is where the volume below it, $A_f Z$, holds the liquid and the submerged ice:

$$Z = \frac{V_l}{\varepsilon A_f}\ \text{(resting)}, \qquad Z = \frac{V_l + m_i/\rho_b}{A_f}\ \text{(floating)}, \tag{34}$$

and the part of the can under the surface adds its own volume over A_f.

Where it comes from. Volume conservation and Archimedes. The heap lifts off when the liquid fills its gaps to the depth at which it would float, $V_l = \varepsilon A_f\,m_i/[\rho_b(1-\varepsilon)A_f]$; at that point both expressions give the same Z, so the level is continuous. The wetted fraction of the ice is Z/t_h while resting and ρ_i/ρ_b while floating, again continuous.

With the defaults the level rises from 12.5 to 13.7 cm in the first half-minute: the heap still rests on the floor, and water melting off the ice fills its gaps, raising the level 2.5 times faster per litre than open liquid would. The brine then becomes dense enough to float the heap, and as the rest of the

salt dissolves, the denser brine lifts the floating ice, so the level eases back to 13.4 cm. That is Archimedes for a bath mixed all at once at $t = 0$; a real bath, made up over a minute, rises over that minute. As floating ice melts later, its meltwater takes a little more room than the brine it displaced, and the level creeps up about 2 mm.

9.3 The can in the bath

In the rig the can lies on its side, along the tank. The part of its circular end below a depth y measured from its lowest point is a circular segment, a sector of the circle less the triangle over the chord:

$$S(y) = r_c^2 \arccos\!\left(\frac{r_c - y}{r_c}\right) - (r_c - y)\sqrt{2r_c y - y^2}, \qquad V_{\text{under}}(y) = H_c\, S(y), \tag{35}$$

with $H_c = 11.66$ cm the can's length ($V_{\text{under}} = \pi r_c^2 y$ for an upright can). With the can's lowest point at z_c, the depth under the surface y satisfies $y = Z_0 - z_c + V_{\text{under}}(y)/A_f$, because the displaced volume raises the level (Z_0 is the level without the can); this has one root, since H_c times the chord is less than A_f. A can held on its holder comes to rest at $z_c = 1$ cm; a free can floats where $\rho_b V_{\text{under}} = m_{\text{can}}$ (68 % of its height under, lying down). It is pushed down from the surface over 10 s, so its 0.40 L raises the level by $0.40/412 = 0.97$ cm smoothly, within the first 5 s: lying down it is only 6.6 cm high.

9.4 The can's heat-passing area

Heat crosses the can wall only where there is liquid on both sides: brine outside, up to the brine's surface, and drink inside, up to the drink's own surface. Above the drink is gas, a headspace of $h_s = 1.2$ cm measured upright, which passes little heat. Lying down, the same 41 cm^3 of gas makes a strip of depth h_g along the top, found from $H_c\, S(h_g) = \pi r_c^2 h_s$: $h_g = 1.05$ cm. The can then passes heat through both ends and the curved wall below the strip:

$$A_c = 2\, S(y) + 2r_c \arccos\!\left(\frac{r_c - y}{r_c}\right) H_c, \qquad y = \min(d,\ 2r_c - h_g), \tag{36}$$

with d the depth of can under the brine.

Where it comes from. The ends contribute two segments of depth y and the curved wall an arc of length $2r_c \arccos[(r_c - y)/r_c]$ along the whole length. Submerged, Can 2 passes heat through 240 cm^2 of its 310 cm^2. Standing upright it would be the bottom and the side up to the drink, $\pi r_c^2 + 2\pi r_c \min(d, H_c - h_s) = 251$ cm^2. Either way the area grows smoothly with the level.

Glass. The brine wets the floor and the walls up to Z: $A_g = A_f + 2(W_t + \ell_t)Z$, with W_t and ℓ_t the tank's width and length; it passes heat across the brine film and half the glass thickness, $1/U_g = 1/h_{bg} + \delta_g/2k_g$, with $h_{bg} = 270$ W m^{-2}K^{-1}.

10 The box: heat from the room

The room is 48 K warmer than the bath, so heat leaks in through the styrofoam all the time.

10.1 A layer of foam that stores heat

Each path through the box is a series of layers (§6.2): the room's air film, the foam, and whatever is inside. The foam also holds heat, $C_f = \rho_f c_f A t$ (280 J K^{-1} for the whole box). Lumping that heat at the middle of the foam splits the path in two:

$$G_{\text{out}} = \frac{A}{1/h_{\text{out}} + t/2k_f}, \quad G_{\text{in}} = \frac{A}{t/2k_f + 1/h_{\text{in}}}, \quad C_f \frac{\mathrm{d}T_f}{\mathrm{d}t} = G_{\text{out}}(T_{\text{room}} - T_f) - G_{\text{in}}(T_f - T_{\text{inside}}). \tag{37}$$

Where it comes from. (3) for the foam node, with each half of the foam ($t/2$ of conductivity $k_f = 0.035\,\mathrm{W\,m^{-1}K^{-1}}$) in series with its film. In steady state the node passes on what it receives and the two halves add back to the full series resistance, $UA = A/(1/h_{\mathrm{out}} + t/k_f + 1/h_{\mathrm{in}})$. Areas are taken at the middle of the foam.

10.2 Four paths

Path	Leak
foam walls → air gap → tank glass → brine	$0.20\,\mathrm{W\,K^{-1}}$
foam floor → tank glass → brine (the tank stands on the foam: no air gap)	$0.07\,\mathrm{W\,K^{-1}}$
foam lid → still air → brine	$0.05\,\mathrm{W\,K^{-1}}$
gaps → brine: lid seam, the lid's thin lip, holes for probe leads	$0.02\,\mathrm{W\,K^{-1}}$
all of them	$0.34\,\mathrm{W\,K^{-1}}$

The gaps bypass the foam. Air in the box is cold and dense, so through a slot of height δ_s and width W_s in the lid seam it flows out at the bottom of the slot and room air flows in at the top. The pressure difference driving it grows linearly with height from the middle of the slot, $\Delta p = \Delta\rho\, g\, z$, and Bernoulli's equation turns it into a speed $u = \sqrt{2\Delta p/\rho} = \sqrt{2g'z}$, with $g' = g\Delta T/T$; integrating over the half-height that flows each way,

$$Q_{\mathrm{air}} = \frac{C_D}{3}\, W_s\, \delta_s \sqrt{g'\delta_s}, \tag{38}$$

Where it comes from. Hydrostatics and Bernoulli, with a discharge coefficient $C_D \approx 0.6$ for the contraction of the jet (Linden 1999). For a 1 mm seam round the lid ($W_s = 1.1$ m) it carries $\rho c_p Q_{\mathrm{air}} \Delta T \approx 0.2$ W, 0.005–0.01 $\mathrm{W\,K^{-1}}$ of leak; a 5 mm gap carries $5^{3/2} = 11$ times as much. With the lid's thin lip ($\approx 0.01\,\mathrm{W\,K^{-1}}$) the gaps are estimated at $0.02\,\mathrm{W\,K^{-1}}$.

10.3 Measuring the leak: the box test

Fill the tank with cold water only, close the box, and log the water. After the first half-hour, while the room-temperature glass gives up its heat, the water and glass (heat capacity C) warm only by the leak, $C\,\mathrm{d}T/\mathrm{d}t = (UA)_{\mathrm{box}}(T_{\mathrm{room}} - T)$, whose solution approaches the room exponentially. Taking logarithms at two times,

$$(UA)_{\mathrm{box}} = \frac{C}{t_2 - t_1}\,\ln\frac{T_{\mathrm{room}} - T(t_1)}{T_{\mathrm{room}} - T(t_2)}, \qquad C = m_w c_w + C_{\mathrm{glass}} = m_w \times 4190 + 2020\ \mathrm{J\,K^{-1}}. \tag{39}$$

The model predicts 6.7 °C at 30 min and 9.4 °C at 3 h for 5 kg of water from 4 °C, and (39) applied to its own curve gives back its leak to 1 %. `box_test.toml` sets the run up; a measured value goes into `[box] measured_UA`, which scales every path to add up to it.

11 Putting it together: two models and how they are solved

11.1 The equilibrium model, `b8_sim.py`

The bath is one body at equilibrium, described by its enthalpy H_b; the drink is a lump. Energy conservation (3) for each, with the drink's loss (22) as the bath's gain:

$$\begin{aligned} C_d\,\dot{T}_d &= -\,U A\,(T_d - T_b), \\ \dot{H}_b &= \underbrace{U A\,(T_d - T_b)}_{\text{from the drink}} + \underbrace{(UA)_{\text{box}}\,(T_{\text{room}} - T_b)}_{\text{leak}} + \underbrace{P_{\text{stir}}}_{\text{stirrer}}, \qquad T_b = \mathcal{T}(H_b), \end{aligned} \tag{40}$$

with $U = U_0$ (Newton) or $U = U_0\,(|T_d - T_b|/\Delta T_0)^{1/4}$ (free convection). Salt added during a run enters as ΔS with $\Delta H = \Delta S\, c_s T_{\text{room}}$. The pair is integrated by Heun's method (a predictor step and a trapezoidal corrector, second-order accurate) at $\Delta t = 0.25$ s; against the closed forms (23) it reproduces the time to 5 °C to 0.01 s.

11.2 The time-dependent model, `b8_model.py`

The simulator follows, as functions of time: the drink's enthalpy (and whether ice has nucleated in it); the liquid water and dissolved salt; for each ring, its brine enthalpy, its crystals, its ice and the ice's interior temperature; the glass temperature; and the temperatures of the foam walls and lid. Each obeys (3) or a mass balance, with the flows of §6–10: the drink (22) and (27); the rings (32); the ice (24); the crystals (26); the glass $C_g\dot{T}_g = Q_{\text{walls}} - U_g A_g (T_g - T_b)$; the foam (37). After each melt or dissolution the brine is shared out again between the rings in proportion to their volume, carrying its heat with it.

Time stepping. Each quantity is stepped forward by Euler's method, $y_{n+1} = y_n + \Delta t\, \dot{y}(y_n)$, with $\Delta t = 0.5$ s. Euler's method is stable for a process of time constant τ only if $\Delta t < 2\tau$, and accurate if $\Delta t \ll \tau$. The fastest processes are the interior of the ice pieces (9 s), the brine next to the can (60 s) and the glass (75 s); the program checks these at the start and warns if Δt is too long. The spreading between rings can be much faster when the inner ring is thin, so it alone is stepped backward,

$$C_j(T_j^{n+1} - T_j^{*}) = G\,\Delta t\,(T_{j-1}^{n+1} - 2T_j^{n+1} + T_{j+1}^{n+1}),$$

a tridiagonal system solved exactly in one sweep: stable for any G, and since each flux appears with both signs, it conserves energy exactly. With $f = \infty$ all rings take one temperature, which reproduces the first version of the simulation (`inputs_original.toml`) to 10^{-13} K.

Energy bookkeeping. After every run the change in total enthalpy is compared with the heat let in from the room and the stirrer (3). It closes to 10^{-9} J of the 43 kJ let in. That proves the bookkeeping, not the physics: a wrong heat-transfer coefficient conserves energy just as well. Water and salt are conserved to 10^{-9} kg.

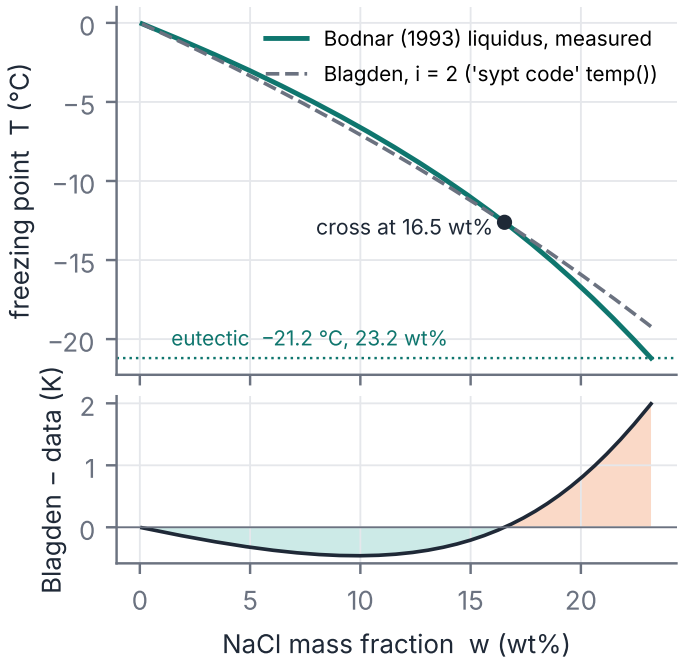


Figure 1. FLOOR. Freezing point against composition: Bodnar's measured liquidus and Blagden's law as coded in `temp()`. Lower panel: Blagden minus data. The residual is small for the dilute brines and grows to 2.0 K at the eutectic, where Blagden also loses the floor.

Part II · What the models say

12 How cold: results of the equilibrium model

12.1 Blagden's law against the data

FLOOR. Blagden's law (10), as coded in `temp()`, crosses Bodnar's liquidus (11) at **16.5 wt %** and **−12.61** °C, the dossier's value to rounding. The ratio of ideal to measured depression is 1.106, 1.069, 1.019, 0.952, 0.909 at 5, 10, 15, 20, 23 wt % (Fig. 1). The crossing belongs to the linear approximation: keeping $\ln(1-x)$ in (9) instead of $-x$ moves it near 12 wt %.

12.2 Pre-dissolving has a floor near $-13\,^{\circ}\mathrm{C}$

Salt dissolved in water at room temperature spends its Δh_s cooling water that the room then rewarms, and the brine arrives at the ice carrying sensible heat that can only be paid in meltwater, which dilutes it. For the five pre-dissolved recipes in the 10 Sep notes the model gives -12.0 to $-12.5\,^{\circ}\mathrm{C}$ (notes: -12.1 to -12.6); **saturating the brine at 25 °C moves the floor only to -12.9 to $-13.1\,^{\circ}\mathrm{C}$**. The 10 Sep conclusion — the eutectic needs staged addition onto ice — follows, and the staged protocol 20 + 6 + 6 + 6 g reproduces as -12.4, -15.8, -19.1, $-21.2\,^{\circ}\mathrm{C}$ (notes: -12.4, -15.7, -18.9, -21.2).

12.3 How much ice the eutectic needs

By (14), for 5.6 kg of water in total, 1.0 kg of salt and water at 4 °C, the ice fraction must exceed **0.612** to reach T_E and **0.691** to keep 150 g of salt solid (ice at 0 °C); 0.555 and 0.627 with ice from a $-18\,^{\circ}\mathrm{C}$ freezer (Fig. 5).

Table 2. Reproduction of earlier B8 numbers. Each earlier number assumed a starting temperature it did not state; the column says which one reproduces it.

Case	Notes	This model	Start
75 g ice + 25 g water + 18 g salt	eutectic, ∼2 g solid	$-21.2\,^{\circ}$C, 2.7 g solid	25 °C water
stall at $-7.5\,^{\circ}$C forces w	11.1 wt %	11.1 wt %	–
150 + 60 + 20 pre-dissolved	$-12.4\,^{\circ}$C	$-12.42\,^{\circ}$C	25 °C brine
staged 20 + 6 + 6 + 6 g	$-12.4/-15.7/-18.9/-21.2$	$-12.4/-15.8/-19.1/-21.2$	25 °C
1200 ice + 400 water, 20 g salt	$-2.07\,^{\circ}$C	$-2.08\,^{\circ}$C	25 °C water
… 90 g / 200 g	-8.61 / $-17.58\,^{\circ}$C	-8.67 / $-17.92\,^{\circ}$C	25 °C water
Blagden crossover	16.4–16.5 wt %	16.5 wt %	–
drift, four bath sizes	$+5.91/1.86/0.97/0.49$ K	$+5.91/1.85/0.97/0.49$ K	on liquidus
L_{eff}/L_f	1.15–1.70	1.15 at $-11.2\,^{\circ}$C	–
latent share of the cold	∼92 %	87 % (0 °C) to 90–92 % (25 °C)	see note

12.4 Drift and the buffer

On the liquidus, 300 g of salt at 20 wt % gives $+17.3$ K per kg melted by (13) (equation bank: $+17$). Rebuilding the rig note's four baths on their own liquidus and melting 150 g gives $+5.91$, $+1.85$, $+0.97$, $+0.49$ K, its table to two decimals, the $1/$size law. On the plateau, forming 1 kg of eutectic brine absorbs **241 kJ** and consumes 0.960 g of solid salt per kJ. The rig note's 272 kJ kg^{-1} used L_0 at 0 °C; at $-21.2\,^{\circ}$C the latent heat is lower (6), so the buffer is used up **13 % faster** than that note assumed.

12.5 Rate and floor optimise at different loadings

RATE and **FLOOR** part company (Fig. 3b). For 4 kg ice and 1.6 kg water, the time to 5 °C falls with salt until the starting bath reaches T_E at salt : ice ≈ 0.20 (2.82 min) and is flat beyond. The drink temperature after 40 min keeps falling until ≈ 0.23, where enough solid salt survives the drink and 40 min of leak to keep the bath pinned. Beyond that, more salt buys only hold time.

12.6 The earlier notes, reproduced

Where a number does not reproduce, the starting temperature is the reason. With the water at 0 °C instead of 25 °C the Session-1 sweep runs 0.6–3.3 K colder at every step, and the latent share drops to 87 %: warm free water must be paid for by melting ice, which raises the latent share. The earlier 92 % therefore belongs to a warm-water charge. The conclusion is unchanged — the salt supplies about a tenth of the cold and the ice the rest — but the earlier notes should state their starting temperatures.

12.7 Numbers for the rig

Tank 7.21 L, Can 2 (∅ 66.0 mm, 116.6 mm, $A = 310.2$ cm^2), 330 g of drink, EPS box at 0.46 W K^{-1}, room 28 °C, drink from 28 °C. Ice at 0 °C, water at 4 °C, salt at room temperature.

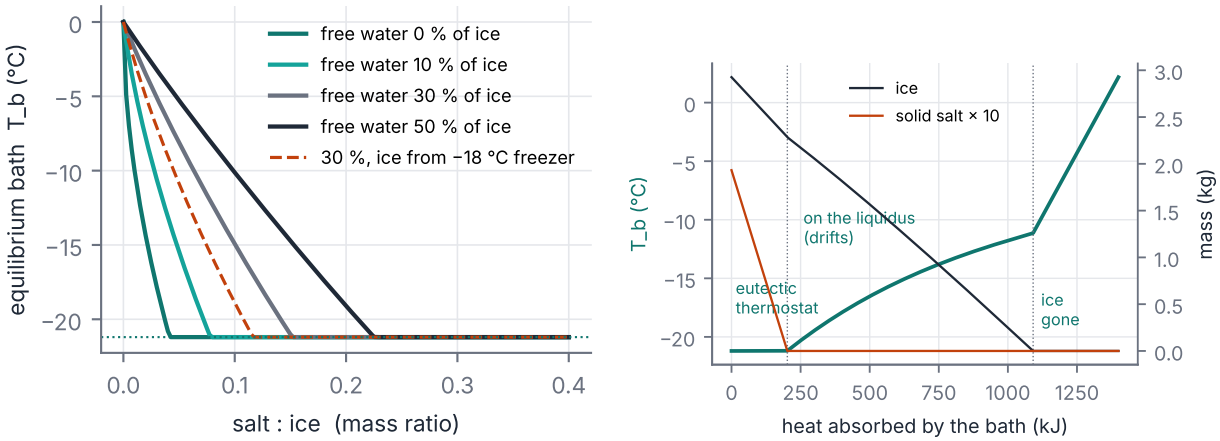


Figure 2. FLOOR. *Left:* adiabatic bath temperature against salt : ice, ice at 0 °C. More free water needs more salt to reach the floor; at 30 % free water, $-18\,^{\circ}$C freezer ice cuts the salt needed from 0.15 to 0.12. *Right:* the buffered charge (§12.7) as heat is added. Three regimes in order: the plateau, where solid salt is consumed at fixed T_E; the liquidus, where the bath drifts as ice melts; ice exhausted, where it warms freely.

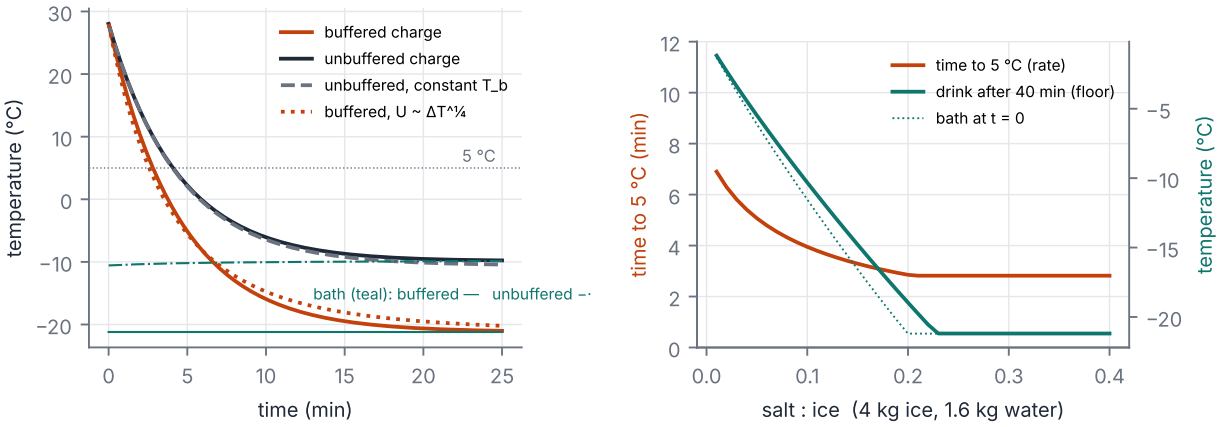


Figure 3. RATE. *Left:* Can 2 from 28 °C in three charges. The buffered bath holds $-21.2\,^{\circ}$C throughout; the unbuffered one warms by 0.7 K, and the constant-T_b model misses that by 2.2 s in time to 5 °C. The dotted curve is the $\Delta T^{1/4}$ law. *Right:* time to 5 °C (rate) and drink temperature after 40 min (floor) against salt; they saturate at 0.20 and 0.23.

	rig note 3.0 : 2.6 : 0.95 kg	**buffered** **4.0 : 1.6 : 1.00 kg**	unbuffered 3.0 : 2.6 : 0.55 kg
starting T_b	$-17.78\,^{\circ}$C	$-$**21.20** °C	$-10.57\,^{\circ}$C
regime	liquidus	eutectic	liquidus
solid salt at start	0 g	193 g	0 g
time to 5 °C (Newton)	189 s	169 s	246 s
$\Delta T^{1/4}$ law	–	155 s	–
T_b after 25 min	$-16.50\,^{\circ}$C	$-21.20\,^{\circ}$C	$-9.86\,^{\circ}$C
ice melted / salt used in 25 min	222 g / –	316 g / 95 g	204 g / –
blank-run warming	1.13 K h^{-1}	0 for 2.5 h	0.62 K h^{-1}

The rig-note charge has no thermostat. It saturates its free water, which is what the rig note checked, but adiabatically it lands at $-17.8\,^{\circ}$C with every gram of salt dissolved, and even with $-18\,^{\circ}$C freezer ice only at $-19.5\,^{\circ}$C. The "≥ 150 g stays solid" line was a mass balance; the energy balance (14) says otherwise. **Use 4.0 kg ice + 1.6 kg water + 1.0 kg salt**: $-21.2\,^{\circ}$C with 193 g solid,

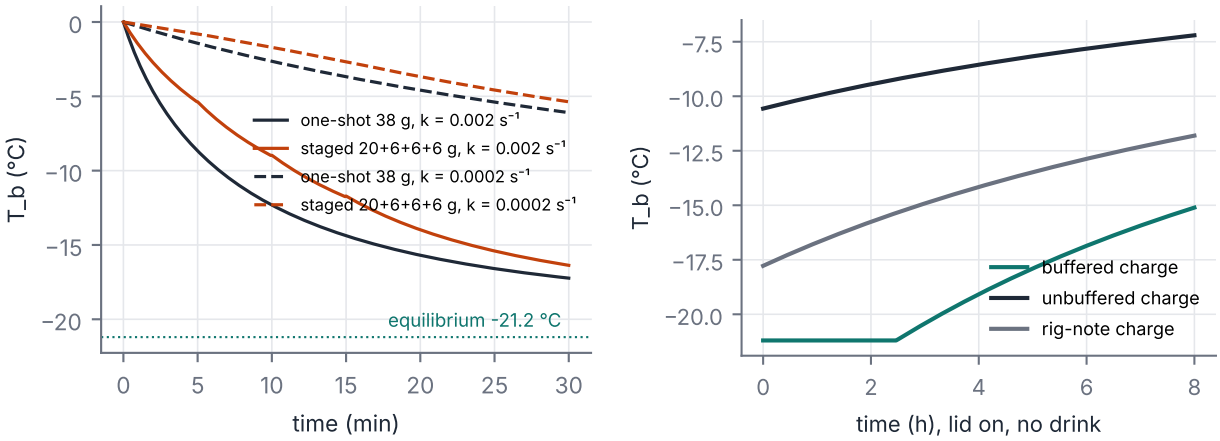


Figure 4. *Left:* well-mixed kinetic dissolution (41), 150 g ice + 60 g water, the same 38 g of salt added at once or in four tranches, for two rate constants. One-shot is *faster* in this model — the opposite of the bench (§14). *Right:* blank runs, lid on, no drink. The buffered charge holds the plateau for 2.5 h on the leak alone.

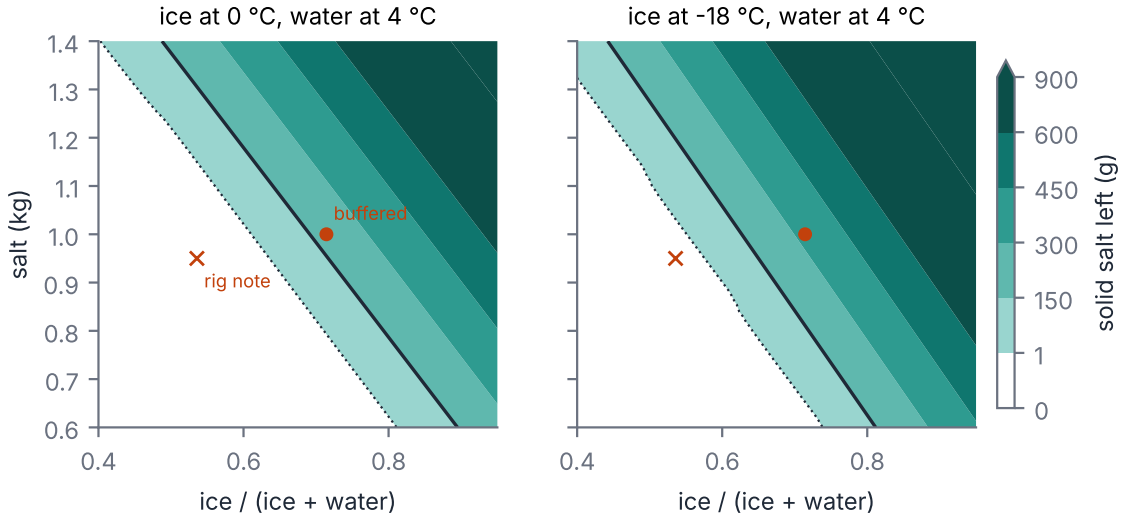


Figure 5. FLOOR. Solid salt left at adiabatic equilibrium, 5.6 kg of water in total, water at 4 °C, salt at 28 °C. White: never reaches T_E. Dotted: just reaches it. Solid line: 150 g of salt stays solid. Cross: the rig-note charge. Dot: the buffered charge of §12.7.

an equilibrium volume of about 6.2 L, plus 0.40 L for the can, inside the 6.8 L to the 16.5 cm line. Add the ice in two lots; 4 kg of crushed ice heaps above the brim until the first lot melts back. One drink plus 25 min of leak uses 95 g of the 193 g, so top up the salt between runs and watch for crystals on the floor.

The blank-run inversion needs L_{eff}. Inverting the warming rate as $\dot{Q} = L\,\dot{T}_b/(\mathrm{d}T_b/\mathrm{d}m)$ recovers the true 17.7 W leak of the unbuffered charge as 15.5 W with L_f ($-13\,\%$) and 17.6 W with L_{eff} (13) ($-1\,\%$); for the rig-note charge, 16.9 against 21.1 W ($-20\,\%$) and 20.8 W ($-1\,\%$). Fit the slope over the first ten minutes: $\mathrm{d}T_b/\mathrm{d}m$ itself falls as the bath dilutes, so a one-hour average biases the answer by another 5 %. And the buffered bath cannot be used for this at all — it reads zero slope for 2.5 h — which is why the rig note asked for the blank to be run unbuffered.

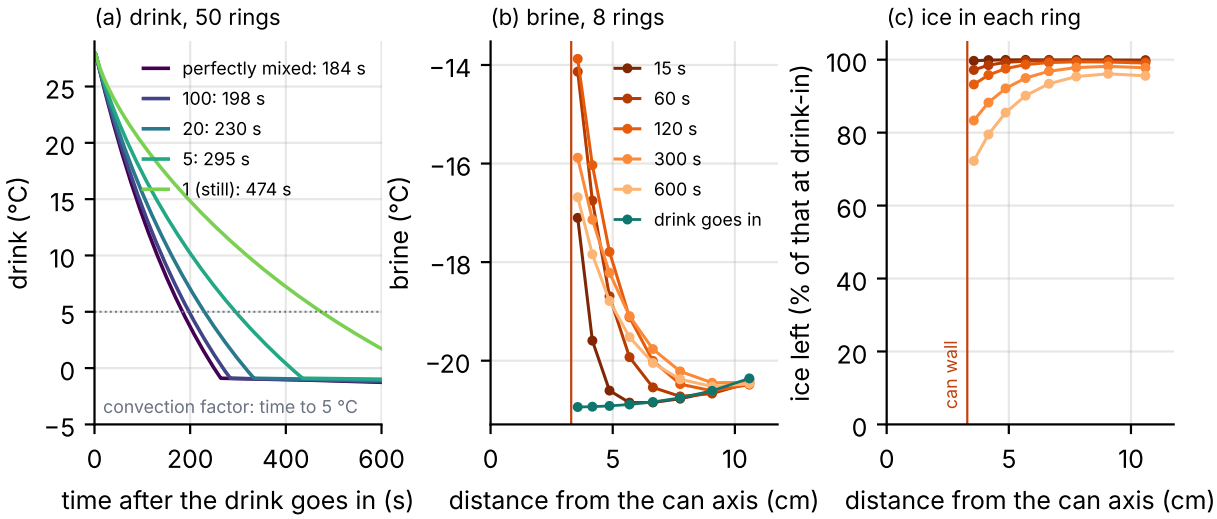


Figure 6. Heat spreading through the bath. (a) The drink for convection factors from perfectly mixed to still brine, run with 50 rings. (b) Brine temperature in each of the 8 rings, and (c) the share of each ring's ice still there, at times after the drink goes in; the default inputs. `python3 b8_simple.py --report-figure` redraws it.

13 How fast: results of the simulator

The simulator's numbers are for the buffered charge on the rig of 20 October, with every input in `inputs.toml`, the can lying on its side, and are printed by `python3 b8_simple.py`. With the bath perfectly mixed it gives 184 s to 5 °C, against 169 s in the table of §12.7, which had the whole outside of an upright can passing heat (§9.4).

With the defaults the drink reaches 5 °C in **222 s instead of 184 s** for a perfectly mixed bath. The inner ring warms 6 K above the rest within a minute and peaks at $-13.8\,^{\circ}\mathrm{C}$, while the bulk stays at $-20.4\,^{\circ}\mathrm{C}$; the ice next to the can goes first (Fig. 6c: more than a quarter of it in 10 minutes, against 4 % at the walls). Even before the drink, the bath is not uniform: the glass warms the outer ring, 0.3 K above a perfectly mixed bath. Over the run, 138.2 kJ comes from the drink and 43.2 kJ from the room through the box.

f	time to 5 °C		brine by the can, peak	bath at drink-in
	8 rings	50 rings	8 rings	8 rings
∞ (perfectly mixed)	184 s	184 s	$-19.6\,^{\circ}\mathrm{C}$	$-20.70\,^{\circ}\mathrm{C}$
100 (stirred)	196.5 s	198 s	$-17.6\,^{\circ}\mathrm{C}$	$-20.62\,^{\circ}\mathrm{C}$
20 (default)	**222 s**	**230 s**	$\mathbf{-13.8}\,^{\circ}\mathrm{C}$	$\mathbf{-20.36}\,^{\circ}\mathrm{C}$
5	266 s	295 s	$-9.0\,^{\circ}\mathrm{C}$	$-19.88\,^{\circ}\mathrm{C}$
1 (still brine)	345.5 s	474 s	$-3.5\,^{\circ}\mathrm{C}$	$-19.30\,^{\circ}\mathrm{C}$

Rings. The rings need to resolve ℓ (28). At the default, 4, 8, 16, 32 and 50 rings give 214, 222, 226.5, 229 and 230 s, so 8 are within 4 %. In still brine ℓ is 0.6 cm and shrinks as the ice by the can melts away, so 8 rings (345.5 s) are far from the 50-ring 474 s; use 50 for f below about 5. Between $f = 20$ and 60 the time to 5 °C moves only from 222 to 202.5 s, because the can's own films then dominate; below 10 it rises quickly.

What matters most. Over plausible ranges the inputs that move the time to 5 °C most are h_o ($\div 3$ to $\times 3$: 442.5 to 147.5 s), a measured U (120–200 W m^{-2}K^{-1}: 339.5 to 218.5 s), `on_side_factor` (1.0–

1.4: 259 to 195.5 s), the drink's mass or heat capacity (± 20 %: 268 to 176.5 s) and f (5–100: 266 to 196.5 s); then the can's orientation (on its side or upright: 222 or 246 s) and the headspace (0.6–2.0 cm: 210 to 237 s). The **FLOOR** numbers hardly move. A fit of U from a drink run (E3) depends on where the brine probe is: **within about 5 mm of the can it measures the can alone**; out in the bath it measures the can and the spreading together, and should be used with $f = \infty$. Two brine probes, one at the can and one at least 5 cm away, measure f: the default predicts a peak difference of about 6.5 K, still brine 16 K, a stirred bath 3 K.

The box. A box leaking anything from 0.25 to 1.0 W K^{-1} gives 222 to 223 s: next to the can's 4.9 W K^{-1} the leak hardly touches the drink. It shows in the bath instead: by the end of the 50-minute run the default bath is at $-19.3\,^{\circ}$C with 2.09 kg of ice left, against $-17.5\,^{\circ}$C and 1.93 kg at 1.0 W K^{-1}.

Still left out. Each ring has one temperature from floor to surface, so the vertical layering of §14 is not in it. Convection is a fixed multiple of k_b, not a flow driven by the temperatures it carries. The tank is taken as a cylinder of the same floor area, and the drink as one temperature. Crushed ice pressed against the can or the glass would conduct directly; only brine is counted.

The simulator. Every number it uses is in `inputs.toml`, with its unit and a tag saying where it comes from; a short file with only the lines that change overrides it. `b8_app.py` and the web page (`https://darkpyxlz.github.io/sypt/`, the same Python files run in the browser) put the main inputs on sliders, each with a box for typing an exact value, and draw the tank to scale with the brine ring by ring. A run takes about 0.5 s in Python and 1 s in the browser.

Part III · What to do next

14 Open problem: why did the one-shot bath stall?

On 10 Sep, 75 g ice + 25 g water + 18 g salt added at once stalled at $-7.5\,^{\circ}$C, against an equilibrium of $-21.2\,^{\circ}$C. The notes attribute this to a crust on the floor in cold, near-saturated brine. The simulation can test the simplest version of that idea, a well-mixed bath with the dissolution law (26) written as

$$\dot{s} = k\,(S - s)^{2/3}\,S^{1/3}\,\frac{w_{\mathrm{sat}}(T_b) - w}{w_{\mathrm{sat}}(T_b)}, \tag{41}$$

Where it comes from. This is (26) with the grain area proportional to the salt put in. The rate constant $k = k_d\,\rho_b\,w_{\mathrm{sat}}\,A_{s0}/S$, in s^{-1}, gathers the constants. Whatever k is, **this law makes the one-shot bath faster than the staged one** (Fig. 4): at 10 min, -12.3 against $-9.0\,^{\circ}$C at $k = 2\times10^{-3}$ s^{-1}, because the one-shot bath has more crystal area exposed from the start. The bulk brine at $-7.5\,^{\circ}$C is only 11 wt % against a saturation of about 25 wt %, so in a mixed bath the drive never vanishes and nothing can stall.

So the stall is evidence that the bath was not well mixed. Dense saturated brine around the crust is gravitationally stable and stays on the floor; the crust then dissolves into a layer that is already saturated, and the bulk only receives salt as fast as that layer is exchanged. Stirring sets the exchange rate, and its absence sets it near zero. If this is right, the result is not a dissolution constant but a mixing one, and it predicts something cheap to check: **a probe on the floor of a stalled one-shot bath should read colder than the bulk**, near the liquidus of saturated brine, and stirring should release the stall in seconds. A probe reading the same as the bulk kills the stratification picture and puts the explanation back in surface kinetics.

15 Experimental programme

E1 · Charge check, one afternoon. Mix the rig-note and buffered charges side by side in the tank, logging T_b until it settles. Predicted -17.8 and $-21.2\,^{\circ}$C, with crystals on the floor of the second only. IV: charge recipe. DV: settled T_b, presence of solid salt. Controls: starting temperatures of all three components, recorded.

E2 · The box's leak. Unbuffered charge, lid on, no drink, one hour. Fit $\dot{T}_b$ over the first ten minutes; invert with L_{eff} (13). Predicted 0.62 K h^{-1} for 0.46 W K^{-1} with the stirrer off; the stirrer's power adds to the leak, so measure it once with the stirrer off as well. DV: $(UA)_{\text{box}}$. A simpler measurement of the same number, with no phase change to invert, is the box test of §10.3: cold water only, three hours, (39).

E3 · Drink runs. Can 2 from room temperature in the buffered and unbuffered charges, lying on its side as in the rig. DV: $T_d(t)$, $T_b(t)$. Fit both Newton and the $\Delta T^{1/4}$ law to each run, compare residuals, and report U rather than τ. Log the brine with two probes, one within 5 mm of the can and one at least 5 cm away: fitted against the first, U is the can's alone; the difference between them measures the convection factor f (§8.3), predicted to peak near 6.5 K. Feed the measured $T_b(t)$ of the unbuffered run into (22) as a check of the bath model independent of the drink model.

E4 · Stall test. 150 g ice + 60 g water + 38 g salt at once, unstirred, probes at the floor and mid-depth. Then repeat staged. DV: floor minus bulk temperature; time for the stall to clear after stirring starts.

E5 · The original charge. 1 kg ice, 2 kg water at 25 °C, 0.5 kg salt. Predicted $-10.5\,^{\circ}$C with about 50 g of ice left, so the ice should be visibly almost gone. A direct, memorable test of why `temp()` could not work.

Suggested order: E1, E2, E3, E4, E5. E1 decides whether E3 has a thermostat.

16 Estimates justifying the approximations

16.1 One bath temperature

`b8_sim.py` gives the bath a single T_b; the simulator splits it into rings round the can (§8.4), with the brine by the can 6.5 K above the bulk at the peak. Neither resolves height: §14 argues that layering is what stalls an unstirred one-shot bath. With the stirrer on and probes at least a few centimetres from the can, a single T_b is the right first model for the **FLOOR**.

16.2 The drink as a lump

The drink's Biot number is about 9 (§6.6), so it is not a lump by conduction; internal convection rescues it by a factor of about 9. The lumped model is the volume average of the PDE solved in COMSOL, which is the justification, and the 4 °C density maximum of water is outside it.

16.3 The eutectic solid

At T_E the equilibrium solid is ice + hydrohalite NaCl·2H$_2$O, not ice + halite. The formation enthalpy is $-0.73\,\mathrm{kJ\,mol^{-1}}$ (Drebushchak 2017), 12.5 kJ per kg of salt, against about 1040 kJ per kg of salt consumed on the plateau: 1.2 %. The water locked into hydrohalite matters more for the buffer accounting, but hydrohalite nucleates reluctantly, so on the time scale of one run the halite model is the better description.

16.4 Assumed constants

Changing each placeholder over a plausible range (`run_checks()` section m) moves the unbuffered T_b by at most 0.11 K, the rig-note T_b by at most 0.29 K, and the buffer by at most 16 g of its 193 g. No **FLOOR** conclusion depends on them.

Change	δT_b unbuffered	δ solid salt, buffered	δT_b rig note
$\Delta h_s \times 0.7$ / $\times 1.3$	$+0.11$ / -0.11 K	-15.7 / $+15.1$ g	$+0.28$ / -0.29 K
$c_i = 2.10$ / 1.94	$+0.02$ / -0.01 K	-5.9 / $+3.6$ g	$+0.05$ / -0.03 K
$c_b \mp 0.2$	-0.08 / $+0.08$ K	$+13.9$ / -14.4 g	-0.25 / $+0.24$ K

16.5 Numerics

Against a constant bath the equilibrium model's integrator reproduces the time to 5 °C of (23) to 0.01 s (169.07 s both), and that of the power-law closed form to 0.01 s (154.61 s both). Quartering the step changes $t_{5\,^\circ\mathrm{C}}$ by 10^{-3} s. The heat released by the drink and the time integral of the flux agree to 10^{-3} kJ in 65.44 kJ. The simulator closes its energy bookkeeping to 10^{-9} J (§11).

16.6 What the can's U does

U is the least certain number in the model. Time to target scales as $1/U$ (23); the **FLOOR** results do not depend on it at all. The SINTEF benchmark it came from cannot also be used to validate it. A U fitted against a probe out in the bath includes the spreading of heat through the bath; against a probe at the can wall it does not.

17 Literature

Marked ◐ where only a secondary or abstract-level source has been read.

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Mixing and dissolution. Koefod (2017) — mixing raises the dissolved share of salt, supporting §14 ◐. Xu, Bootsma, Verzicco, Lohse & Huisman, *J. Fluid Mech.* **1019**, A11 (2025), and Yang, Howland, Liu, Verzicco & Lohse, *J. Fluid Mech.* **969**, R2 (2023) — mechanism only; both lie an order of magnitude below the bath's salinity. Pyodide (`https://pyodide.org`) — Python in the browser, for the web simulator.

18 Notes for whoever picks this up

Do first. E1. It costs one afternoon and decides between -17.8 and $-21.2\,^{\circ}$C for every later run, and whether the bath in E3 is a thermostat or a drifting reservoir.

Do not quote the rig-note charge as buffered; do not invert a blank run with L_f; do not report a single latent-heat share without the starting temperature of the water; and do not fit U against a bulk probe and then use it with heat spreading switched on.

The sharpest predictions, cheapest first: the rig-note charge settles near $-17.8\,^{\circ}$C with no crystals left; the original `sypt code` charge leaves only about 50 g of ice; the buffered bath holds $-21.2\,^{\circ}$C for about 2.5 h on the leak alone; the brine at the can runs 6.5 K above the bulk; a stalled one-shot bath is colder at the floor than in the bulk.

Software. `b8_sim.py`: Python 3 with NumPy and SciPy, Matplotlib for the figures; everything runs in about 12 s. The simulator (`b8_model.py`) needs only Python 3.11 or later, plus Matplotlib for its pictures; one run takes about 0.5 s.

A Symbols

Symbol	Unit	Meaning
T, T_b, T_d, T_s	°C	temperature; of the bath (brine), the drink, the ice surface
H, h	J, J kg^{-1}	enthalpy, and per kilogram
W, m_i, m_l	kg	all water; ice; liquid water in the brine
S, s	kg	all salt; dissolved salt
w, w_L, w_{sat}, r	–	salt mass fraction of brine; on the liquidus; at saturation; $w/(1-w)$
T_E, w_E; T_P, w_P	°C, –	eutectic; peritectic
c_w, c_i, c_s, c_b	J kg^{-1}K^{-1}	heat capacity of water, ice, salt, brine
L_0, $L(T)$, L_{eff}	J kg^{-1}	latent heat at 0 °C, at T, as the bath charges it
Δh_s	J kg^{-1}	heat of solution of NaCl
μ, a_w, K_f, b, i	–	chemical potential, water activity, cryoscopic constant, molality, ions per formula
k, ρ, α	W m^{-1}K^{-1}, kg m^{-3}, m^2 s^{-1}	conductivity, density, diffusivity
ν, β, g	m^2 s^{-1}, K^{-1}, m s^{-2}	kinematic viscosity, expansion coefficient, gravity
h, U, G	W m^{-2}K^{-1}, W m^{-2}K^{-1}, W K^{-1}	film coefficient, overall coefficient, conductance
Ra, Nu, Pr, Bi	–	Rayleigh, Nusselt, Prandtl, Biot numbers
C_d, A, τ	J K^{-1}, m^2, s	drink heat capacity, can area, cooling time constant
r_c, H_c, h_s, h_g	m	can radius, length, headspace, gas strip depth
A_f, Z, ε, t_h	m^2, m, –, m	tank floor area, liquid level, heap porosity, heap thickness
f, k_{eff}, ℓ	–, W m^{-1}K^{-1}, m	convection factor, effective conductivity, penetration length
N, r_j, ψ_j, λ	–, m, –, m	rings, ring edges, ring shares, shell length
k_d, A_s, D	m s^{-1}, m^2, m^2 s^{-1}	dissolving coefficient, grain area, salt diffusivity
φ, T_{f0}	–, °C	frozen fraction of the drink's water, its first freezing point

B Using the code

```
python3 b8_sim.py           # checks, then figs/*.pdf and numbers.json
python3 b8_sim.py --quick   # checks only
python3 b8_simple.py        # the simulator, one run: table, summary, picture
python3 b8_app.py           # the simulator behind sliders, each with a box to type into
python3 b8_simple.py --report-figure   # redraws figs/f9_spreading.pdf
python3 test_b8.py          # the simulator's checks
```

```
import b8_sim as B
B.mix(1.0, 2.0, 0.5, T_water=25)        # adiabatic mix -> dict(T, m_ice, w, s_solid, regime)
B.mix_charge(B.BUFFERED)                # a named rig charge
r = B.run_charge(B.UNBUFFERED, t_end=1500, drink=dict(law='power'))
B.time_to(r, 5.0)                       # seconds to 5 C
B.simulate(0.15, 0.06, 0.038, with_drink=False, k_diss=2e-3)   # kinetic dissolution
```

Constants marked `ASSUMED` in the source are the ones to replace with measurements: c_i, $c_b(w)$, Δh_s, the hydrohalite solubility line, the drink's c and can mass, and U_0. To use a measured bath trace instead of the bath model, set `constant_bath=True` and replace the line that sets `T_b` in `simulate()` with an interpolation of the log.

The heart of the equilibrium model, (7) and the solve of §5:

```python
def enthalpy(T, m_ice, m_liq_water, s_dissolved, s_solid):
    """Total enthalpy (kJ) of a bath at a single temperature T."""
    m_brine = m_liq_water + s_dissolved
    w = s_dissolved / m_brine if m_brine > 0 else 0.0
    return (m_brine * c_brine(w) * T          # sensible heat of the brine
            + s_dissolved * DH_SOL             # heat absorbed dissolving the salt
            + m_ice * (C_ICE * T - L_F)        # ice: sensible heat, minus latent heat
            + s_solid * C_SALT * T)            # undissolved salt


def _phases_at(T, W, s_cap, S):
    """Equilibrium phase amounts at a temperature strictly above T_E.
    W = all water (ice + liquid), s_cap = salt allowed to dissolve,
    S = all salt.  Returns (m_ice, m_liq_water, s_dissolved, s_solid)."""
    wl = w_liquidus(T)
    s_noice = min(s_cap, w_saturation(T) / (1.0 - w_saturation(T)) * W)
    w_full = s_noice / (s_noice + W)
    if T >= 0.0 or w_full >= wl:
        # Above the liquidus of the fully melted bath: no ice.
        return 0.0, W, s_noice, S - s_noice
    # On the liquidus: the brine must have composition wl,
    # and all available salt is in it (liquidus lies below saturation).
    m_lw = s_cap * (1.0 - wl) / wl
    return W - m_lw, m_lw, s_cap, S - s_cap


def bath_state(H, W, S, s_cap=None):
    """THE solve.  Given the bath's enthalpy H (kJ), total water W (kg),
    total salt S (kg), and how much salt may dissolve (s_cap, default all),
    return the equilibrium state as a dict.

    Four regimes, in order of decreasing H:
      'liquid'     no ice; T free                    (ice exhausted)
      'liquidus'   ice + brine on the liquidus; T = T_liq(w)
      'eutectic'   ice + brine + solid salt; T = T_E fixed   (F = C - P + 1 = 0)
      'solid'      below the eutectic, no liquid     (not reached in practice)
    """
    if s_cap is None:
        s_cap = S
    s_cap = max(0.0, min(s_cap, S))

    def pack(T, mi, mlw, s, ss, regime):
        mb = mlw + s
        return dict(T=T, m_ice=mi, m_liq_water=mlw, s_dissolved=s, s_solid=ss,
                    w=(s / mb if mb > 0 else 0.0), regime=regime)

    # --- pure water: the only invariant point is 0 C -----------------------
    if s_cap < 1e-9:
        if H > 0.0:
            T = H / (W * C_W + S * C_SALT)
            return pack(T, 0.0, W, 0.0, S, 'liquid')
        if H >= -W * L_F:
            mi = -H / L_F
            return pack(0.0, mi, W - mi, 0.0, S, 'liquidus')
        T = (H + W * L_F) / (W * C_ICE + S * C_SALT)
        return pack(T, W, 0.0, 0.0, S, 'solid')

    # --- the eutectic plateau: T fixed, enthalpy linear in brine mass -----
    r_E = W_E / (1.0 - W_E)                       # kg salt per kg water in eutectic brine
```

```
    m_max = min(W, s_cap / r_E)                  # most water that can be eutectic brine

    def H_eut(m_lw):
        s = r_E * m_lw
        return enthalpy(T_E, W - m_lw, m_lw, s, S - s)

    H_top, H_bot = H_eut(m_max), H_eut(0.0)

    if H >= H_top:                               # above the plateau: monotonic in T
        f = lambda T: enthalpy(T, *_phases_at(T, W, s_cap, S)) - H
        T = brentq(f, T_E + 1e-9, 200.0, xtol=1e-10)
        mi, mlw, s, ss = _phases_at(T, W, s_cap, S)
        return pack(T, mi, mlw, s, ss, 'liquid' if mi == 0.0 else 'liquidus')
    if H >= H_bot:                               # on the plateau
        m_lw = m_max * (H - H_bot) / (H_top - H_bot)
        s = r_E * m_lw
        return pack(T_E, W - m_lw, m_lw, s, S - s, 'eutectic')
    T = T_E + (H - H_bot) / (W * C_ICE + S * C_SALT)   # below the eutectic
    return pack(T, W, 0.0, 0.0, S, 'solid')
```

B8 · Cold drink

Model derivation, step by step

darkpyxlz · SYPT 2027

25 October 2026

The model has two parts. The drink is described by one temperature T_d. The bath — ice, brine and undissolved salt — is described by its total enthalpy $\mathcal{H}$, the heat it holds counted from a reference state. One heat flow, $UA(T_d - T_b)$, joins them, and a second, through the styrofoam box, brings heat in from the room. Nothing stirs the bath or the drink.
This document derives the two equations from the conservation of energy and mass and closes them with the phase diagram of salt water and the laws of heat transfer, explaining every step. Physical constants and handbook properties of water, ice, salt, brine and the container materials are put in as numbers, each with a note on what it is (Table 1). Anything that belongs to this rig — masses, sizes, starting temperatures, film coefficients, the box's conductance — stays a symbol, to be measured. Step 13 collects the results into the closed set of equations; Step 14 solves the cases that have a formula.

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How to read this document

What "the model" is. The 24 Oct salted-ice-bath model, `b8_sim.py` in the repository: a lumped drink; a bath described by its enthalpy, with a map of the five states it can be in (liquid, ice on the freezing curve, the eutectic plateau, saturated brine with salt, solid); Bodnar's freezing curve; and the leak of heat through the box. The $\Delta T^{1/4}$ convection law is an option for the heat path. The buffered eutectic bath (COMSOL v1) is a special case.

No stirrer. Nothing stirs the drink or the bath. Fluid moves only by free convection: liquid warmer than its surroundings is lighter and rises, colder liquid sinks. That fixes which convection laws apply (Step 5) and makes the well-mixed bath (assumption A3) the strongest assumption here.

How each step is laid out. It opens with the question it answers and the assumptions it uses. Each line of algebra is followed by what was done and why. It ends in a boxed **Result** and a short note on what the result means.

Which quantities become numbers. A number is put in only for a constant that would be the same in any laboratory: a physical constant, or a handbook property of a pure substance or a standard material. These are listed, and explained, in Table 1. The heat capacities, latent heat, heat of solution and phase diagram are the values `b8_sim.py` uses, so the bath formulas here reproduce its numbers. Everything set by this rig stays a symbol (Table 2), because it has to be measured.

Conventions.

- SI units throughout: J, kg, m, s, W. An enthalpy is in J, a heat flow in W.
- Temperatures T are in °C, because the enthalpy reference and Bodnar's fit are defined at 0 °C. A temperature *difference* is the same in K and °C. Absolute temperature, in K, appears only in the phase-equilibrium derivation (Step 6), written $\mathcal{T}$.
- Salt content w is a mass fraction: kg of NaCl per kg of brine, between 0 and 1. Bodnar's fit uses weight per cent, $W = 100w$.
- A heat flow is positive in the direction its subscript names: $\dot{Q}_{db}$ flows from the drink to the bath, $\dot{Q}_{\text{leak}}$ from the room to the bath.
- Enthalpy reference: liquid water and solid NaCl, both at 0 °C, have zero enthalpy. Any reference would do, because only differences of enthalpy are ever used; this one makes the enthalpy of ice at 0 °C equal to minus its latent heat.

Constants put in as numbers

Each row is a constant that does not depend on this experiment. Where a property changes with temperature, the value is taken where the model uses it, and the note says how much it changes.

Table 1. Constants substituted in this document

Symbol	Value	What it is
Physical constants		
g	$9.81\,\mathrm{m\,s^{-2}}$	Acceleration due to gravity at the Earth's surface. It sets the buoyancy that drives free convection.
R	$8.314\,\mathrm{J\,mol^{-1}K^{-1}}$	Molar gas constant: Boltzmann's constant times Avogadro's number. It converts temperature into energy per mole in the chemical potential.
$\mathcal{T}_0$	$273.15\,\mathrm{K}$	Melting point of pure ice at atmospheric pressure: 0 °C on the absolute scale.
M_w	$0.018015\,\mathrm{kg\,mol^{-1}}$	Molar mass of water, H_2O.
M_s	$0.05844\,\mathrm{kg\,mol^{-1}}$	Molar mass of sodium chloride, NaCl.
ν_i	2	Particles each formula unit of NaCl gives in water: Na^+ and Cl^-.
$j_{0,1}$	2.405	First zero of the Bessel function J_0; it fixes the slowest cooling pattern of a conducting cylinder (Step 3).
Water, ice and salt		
L_f	$333\,550\,\mathrm{J\,kg^{-1}}$	Latent heat of fusion of ice at 0 °C: the heat that melts 1 kg of ice at 0 °C without warming it.
Δh_{fus}	$6009\,\mathrm{J\,mol^{-1}}$	The same per mole, $L_f M_w$.
c_w	$4180\,\mathrm{J\,kg^{-1}K^{-1}}$	Specific heat of liquid water: the heat that warms 1 kg by 1 K (4180–4220 between 30 and 0 °C).
c_i	$2000\,\mathrm{J\,kg^{-1}K^{-1}}$	Specific heat of ice. It falls with temperature, from 2097 at 0 °C to 1950 at −20 °C (IAPWS R10-06); 2000 is its value near −13 °C, where the bath works.
c_h	$864\,\mathrm{J\,kg^{-1}K^{-1}}$	Specific heat of solid NaCl (halite), $50.5\,\mathrm{J\,mol^{-1}K^{-1}}$.
$c_b(w)$	$4180 - 4400\,w$ $\mathrm{J\,kg^{-1}K^{-1}}$	Specific heat of brine. Dissolved salt lowers it; this straight line gives 3300 at $w = 0.20$ and 3160 at w_E, 2–5 % below the handbook values (3370 and 3310, Melinder 2010).
Δh_s	$66\,400\,\mathrm{J\,kg^{-1}}$	Heat absorbed in dissolving 1 kg of NaCl in water ($3.88\,\mathrm{kJ\,mol^{-1}}$, measured at 25 °C in dilute solution). Positive: dissolving salt cools the water.
The NaCl–water phase diagram		
b_1, b_2, b_3	1.78, −0.0442, 5.572×10^{-4}	Coefficients of Bodnar's (1993) fit to the measured freezing curve of salt water, in wt % against degrees below 0 °C.
T_E, w_E	−21.2 °C, 0.2318	The eutectic: the coldest brine that can stay liquid, and its salt content (the end of Bodnar's fit; handbooks give −21.1 °C and 23.3 wt %).
T_p, w_p	+0.1 °C, 0.26285	The peritectic (Swenne 1983): below it the salt that crystallises from saturated brine is hydrohalite, $NaCl{\cdot}2H_2O$, not halite.
K_f	$1.86\,\mathrm{K\,kg\,mol^{-1}}$	Cryoscopic constant of water: freezing-point drop per mole of dissolved particles per kg of water. Derived in Step 6.
Λ_E	$241\,450\,\mathrm{J\,kg^{-1}}$	Heat needed to turn ice and salt at T_E into 1 kg of eutectic brine. Derived in Step 9.
Liquids moving by free convection		
k_d	$0.589\,\mathrm{W\,m^{-1}K^{-1}}$	Thermal conductivity of water at 15 °C, the middle of the drink's 28-to-5 °C range. A soft drink is about 90 % water, so the drink is given water's properties.
ρ_d	$999.1\,\mathrm{kg\,m^{-3}}$	Density of water at 15 °C.
ν	$1.139 \times 10^{-6}\,\mathrm{m^2\,s^{-1}}$	Kinematic viscosity of water at 15 °C: how quickly momentum diffuses through it.
α	$1.408 \times 10^{-7}\,\mathrm{m^2\,s^{-1}}$	Thermal diffusivity of water at 15 °C, $k_d/(\rho_d c_p)$: how quickly heat diffuses through it.
β	$1.51 \times 10^{-4}\,\mathrm{K^{-1}}$	Thermal expansion coefficient of water at 15 °C: the fractional fall in density per kelvin of warming. It is 0 at 4 °C, 0.88×10^{-4} at 10 and 2.07×10^{-4} at 20 °C.
Pr	8.1	Prandtl number of water at 15 °C, ν/α.

Symbol	Value	What it is
k_b	$0.54\,\mathrm{W\,m^{-1}K^{-1}}$	Thermal conductivity of eutectic brine near $-15\,^\circ$C (Melinder 2010).
ρ_b	$1182\,\mathrm{kg\,m^{-3}}$	Density of eutectic brine.
ν_b	$4.6\times10^{-6}\,\mathrm{m^2\,s^{-1}}$	Kinematic viscosity of eutectic brine near $-15\,^\circ$C: four times water's at $15\,^\circ$C, because it is cold and salty.
α_b	$1.38\times10^{-7}\,\mathrm{m^2\,s^{-1}}$	Thermal diffusivity of the same brine. With ν_b it gives a Prandtl number of about 33.
$C_i,\ C_o$	0.62, 0.64	Constants of laminar free convection on a vertical wall, Churchill & Chu (1975), $0.670/[1+(0.492/\mathrm{Pr})^{9/16}]^{4/9}$, at $\mathrm{Pr}=8.1$ (drink) and 33 (brine).
Container and box materials		
c_c	$900\,\mathrm{J\,kg^{-1}K^{-1}}$	Specific heat of aluminium, the can.
k_w	160 (can) or 0.2 (bottle) $\mathrm{W\,m^{-1}K^{-1}}$	Thermal conductivity of the container wall: can-body aluminium alloy (pure aluminium 237), or PET for a plastic bottle.
k_{box}	$0.035\,\mathrm{W\,m^{-1}K^{-1}}$	Thermal conductivity of expanded polystyrene (styrofoam); 0.033–0.038 depending on the foam's density.

Symbols kept as symbols

Source column: **M** measured on this rig, **C** chosen by the experimenter, **D** derived here, **S** a state variable that changes during a run.

Table 2. Symbols that stay symbols

Symbol	Meaning	Unit	Src
T_d	drink temperature (its well-mixed core)	°C	S
T_b	bath temperature	°C	D (from $\mathcal{H}$)
T_{env}	room air outside the box	°C	M
θ	drink–bath difference, $T_d - T_b$	K	D
$m_d,\ c_d$	drink mass; drink specific heat (about 4000 for a sugary drink, 4180 for water or a diet drink)	kg, $\mathrm{J\,kg^{-1}K^{-1}}$	M
m_c	container mass	kg	M
C_d	heat capacity of drink + container	$\mathrm{J\,K^{-1}}$	D
$r_i,\ r_o,\ H$	container inner radius, outer radius, height of the wetted wall	m	M
t_w	wall thickness	m	M
$A_i,\ A_o$	inner and outer wetted areas	$\mathrm{m^2}$	M
$h_i,\ h_o$	drink-side and bath-side film coefficients	$\mathrm{W\,m^{-2}K^{-1}}$	D/M
R_i, R_w, R_o, R_c	inner film, wall, outer film and contact (label) resistances	$\mathrm{K\,W^{-1}}$	D
$U,\ UA$	overall heat-transfer coefficient; conductance	$\mathrm{W\,m^{-2}K^{-1}}$, $\mathrm{W\,K^{-1}}$	M (fit)
Bi, Nu, Ra	Biot, Nusselt and Rayleigh numbers	—	D
$\mathcal{H}$	total enthalpy of the bath	J	S
C_t	heat capacity of the tank glass, if counted with the bath	$\mathrm{J\,K^{-1}}$	M
$m_W,\ m_S$	all the water (ice + liquid), all the NaCl	kg	M
$m_i,\ m_l,\ m_h$	ice, brine, undissolved salt	kg	S
m_m	mass of ice melted so far	kg	S
$w,\ \bar{w}$	salt fraction of the brine; overall salt fraction $m_S/(m_W+m_S)$	—	S, D
$T_{\mathrm{liq}}(w)$	freezing temperature of brine of salt fraction w	°C	D
$w_{\mathrm{sat}}(T)$	salt fraction of saturated brine at T	—	D
ϑ	degrees below zero on the freezing curve, $-T_{\mathrm{liq}}$	K	D

Symbol	Meaning	Unit	Src
$L_{\text{eff}},\ C_{\text{eff}}$	effective latent heat; effective heat capacity of the bath	$\mathrm{J\,kg^{-1}}$, $\mathrm{J\,K^{-1}}$	D
C_B	heat capacity of the ice and brine together	$\mathrm{J\,K^{-1}}$	D
UA_{box}	conductance of the box from room to bath	$\mathrm{W\,K^{-1}}$	M
$A_{\text{box}},\ \delta_{\text{box}}$	foam area and thickness	$\mathrm{m^2}$, m	M
$\Delta\rho$	density difference driving the brine film	$\mathrm{kg\,m^{-3}}$	D
$a_w,\ \varphi,\ b$	water activity, osmotic coefficient, molality	—, —, $\mathrm{mol\,kg^{-1}}$	D
$\tau,\ \tau_0,\ t^*,\ \tau_c$	time constants	s	D
$n,\ a$	exponent and prefactor in $\dot{Q}_{db} = a\theta^n$	—, $\mathrm{W\,K^{-n}}$	D/M
$T^*,\ T_{fd}$	target drink temperature; freezing point of the drink	°C	C, M
m_{buf}	salt to keep undissolved as a buffer	kg	C

Step 0 — System, control volumes and assumptions

The question. What are the parts of the rig, which of them store heat, and along which paths does heat move between them?

A *control volume* is a region with a boundary drawn round it, kept account of: what crosses the boundary changes what is inside. The rig splits into two volumes that store heat — the drink and the bath — and two that only resist its flow — the container wall and the box. Everything outside the box is the room, so large that the heat it gives up does not change its temperature T_{env}.

Control volume	Contents	State	Exchanges heat with
Drink (D)	liquid + container shell	T_d	the bath, through the wall
Bath (B)	ice, brine, undissolved NaCl	$\mathcal{H}$	the drink; the room, through the box
Container wall	aluminium or PET shell, label	—	resistance only
Box	EPS walls + lid	—	resistance only

Thermal circuit. Heat flows like current in an electrical circuit: temperatures play the part of voltages, heat flows of currents, thermal resistances ($\mathrm{K\,W^{-1}}$) of electrical ones, and heat capacities of capacitors, which store heat as their temperature changes.

$$\underbrace{T_d}_{C_d} \xrightarrow{R_i} T_{wi} \xrightarrow{R_c+R_w} T_{wo} \xrightarrow{R_o} \underbrace{T_b}_{\mathcal{H}} \xleftarrow{1/UA_{\text{box}}} T_{\text{env}}$$

Heat leaves the drink through the film of drink on the inside of the wall (R_i), crosses the label and the wall ($R_c + R_w$), and enters the bath through the film of brine on the outside (R_o). The wall's two faces are at T_{wi} and T_{wo}. From the other side, heat from the room leaks through the box into the bath. The drink stores heat in C_d; the bath stores it in its enthalpy $\mathcal{H}$, much of it as the latent heat of its ice.

Assumptions, each with the reason it is made and where it is tested.

A1 Drink well-mixed. One temperature T_d describes the liquid core; all the change of temperature sits in thin boundary layers at the wall. Free convection keeps the core mixed; Step 3 tests when that is enough.

A2 Wall and tank store no heat, or their heat capacity is added to the system they touch: the can's to C_d, the tank glass's (C_t) to the bath. A can wall 0.1 mm thick holds about 1 % as much heat per kelvin as its drink.

A3 Bath well-mixed and at phase equilibrium at every instant. One temperature T_b and one salt fraction w describe the whole bath, and ice, brine and salt are always in equilibrium. With no stirrer this is the strongest assumption: brine next to the can warms first. The report and the browser simulator relax it, splitting the bath into rings round the

can and melting and dissolving at finite rates.

A4 Quasi-steady resistances. The thin films and the wall reach their steady temperature profile much faster (seconds) than T_d or T_b change (minutes), so each is described by a fixed resistance.

A5 Constant pressure, lid on. Nothing evaporates or condenses, and no frost forms. Radiation inside the box is counted in UA_{box}.

A6 Constant properties. Each specific heat, latent heat and heat of solution is one number; the brine's specific heat depends on its salt content through one fixed straight line. The exceptions are h_i and β in the convection laws (Step 5).

A7 The stable phase diagram, with salt booked as halite. The freezing curve, the solubility and the eutectic are the measured equilibrium ones. Undissolved salt is counted as anhydrous NaCl (halite) in the mass and energy balances. Below +0.1 °C the stable salt solid is really hydrohalite, $\text{NaCl}{\cdot}2\text{H}_2\text{O}$, which locks up $2M_w/M_s = 0.62\,\text{kg}$ of water per kg of salt; that bound water and its heat are left out. Nothing supercools and no metastable phase forms.

A8 Drink fully immersed to a marked waterline, so its wetted area is fixed.

A9 Ideal sensors. Probe lag and offsets are corrections to the measurements, not terms in the model.

Step 1 — First law for the drink

The question. How fast does the drink's temperature fall for a given heat flow out of it?

1.1 The first law at constant pressure. The drink and its can form a closed system: no mass crosses its boundary. For a closed system the first law of thermodynamics says that its internal energy E changes by the heat that enters it minus the work it does, $\mathrm{d}E = \delta Q - p\,\mathrm{d}V$. At constant pressure (A5) the work of expansion is folded in by using the enthalpy $H = E + pV$, since $\mathrm{d}H = \mathrm{d}E + p\,\mathrm{d}V$ when p is fixed. So $\mathrm{d}H = \delta Q$: at constant pressure, *heat in equals the rise of enthalpy.* The only heat crossing the boundary is $\dot{Q}_{db}$, flowing out, so

$$\frac{\mathrm{d}H_D}{\mathrm{d}t} = -\dot{Q}_{db}.$$

1.2 The enthalpy of the drink. The specific heat c is defined by $\mathrm{d}h = c\,\mathrm{d}T$ for 1 kg. Each kilogram of drink holds $c_d(T_d - T_{\text{ref}})$ above a reference temperature, and each kilogram of can $c_c(T_d - T_{\text{ref}})$, since the can is at the drink's temperature (A1, A2). Adding the two masses, with constant specific heats (A6),

$$H_D = (m_d c_d + m_c c_c)(T_d - T_{\text{ref}}) \equiv C_d\,(T_d - T_{\text{ref}}), \qquad \frac{\mathrm{d}H_D}{\mathrm{d}t} = C_d \frac{\mathrm{d}T_d}{\mathrm{d}t}.$$

The reference drops out on differentiating. With aluminium's specific heat, $c_c = 900\,\text{J}\,\text{kg}^{-1}\text{K}^{-1}$:

$$C_d = m_d c_d + 900\,m_c \qquad (\text{J}\,\text{K}^{-1}).$$

The drink's own specific heat c_d stays a symbol because it depends on the drink: sugar lowers it, from water's 4180 to about $4000\,\text{J}\,\text{kg}^{-1}\text{K}^{-1}$ for a sugary soft drink.

Result 1 — drink energy balance

$$C_d \frac{\mathrm{d}T_d}{\mathrm{d}t} = -\dot{Q}_{db}, \qquad C_d = m_d c_d + 900\,m_c$$

What it means. C_d is the heat the drink and can give up for each kelvin they cool, so the drink cools at $\dot{Q}_{db}/C_d$ kelvin per second. A can's 13 g of aluminium adds $12\,\mathrm{J\,K^{-1}}$ to the drink's 1300 or so: about 1 %. Weigh the drink full and empty to get m_d; the outside volume $\pi r_o^2 H$ would include the gas in the headspace.

Step 2 — Heat path drink → bath as series resistances

The question. How does the heat flow out of the drink depend on the two temperatures, and on the films and the wall it crosses?

2.1 Each layer carries the same heat flow. By A4 the films and the wall store no heat, so whatever heat enters one face of a layer leaves the other. The same $\dot{Q}_{db}$ therefore crosses all three, each driven by the temperature drop across it and opposed by its resistance:

$$\dot{Q}_{db} = \frac{T_d - T_{wi}}{R_i} = \frac{T_{wi} - T_{wo}}{R_w} = \frac{T_{wo} - T_b}{R_o}.$$

2.2 Film resistances. Next to a wall, a liquid moving by convection is held still in a thin film, which heat crosses only by conduction. Newton's law of cooling, $\dot{Q} = hA\,\Delta T$, sums that up in a film coefficient h, the heat flow per unit area per kelvin of drop. Comparing with $\dot{Q} = \Delta T/R$,

$$R_i = \frac{1}{h_i A_i}, \qquad R_o = \frac{1}{h_o A_o}.$$

The film coefficients depend on how the liquid moves; Step 5 derives them.

2.3 Wall resistance of a cylinder. Fourier's law says heat flows down a temperature gradient in proportion to it: the flux is $-k_w\,\mathrm{d}T/\mathrm{d}r$ through a solid of conductivity k_w. Through the side wall of a cylinder of height H, it crosses the area $2\pi r H$ at radius r, and the total $\dot{Q}$ is the same at every radius (no storage, A4):

$$\dot{Q} = -k_w(2\pi r H)\frac{\mathrm{d}T}{\mathrm{d}r} \;\Rightarrow\; \int_{r_i}^{r_o} \frac{\mathrm{d}r}{r} = -\frac{2\pi k_w H}{\dot{Q}} \int_{T_{wi}}^{T_{wo}} \mathrm{d}T$$

$$\Rightarrow\; \ln\frac{r_o}{r_i} = \frac{2\pi k_w H}{\dot{Q}}\,(T_{wi} - T_{wo}) \;\Rightarrow\; R_w = \frac{T_{wi} - T_{wo}}{\dot{Q}} = \frac{\ln(r_o/r_i)}{2\pi k_w H}.$$

The variables were separated (r on one side, T on the other) and both sides integrated across the wall. For a thin wall, $t_w = r_o - r_i \ll r_i$, the logarithm is $\ln(1 + t_w/r_i) \approx t_w/r_i$, so $R_w \approx t_w/(k_w A)$ with $A = 2\pi r_i H$: the wall acts as a flat slab. A label sleeve adds a contact resistance $R_c = R_c''/A$, with R_c'' its resistance per unit area, in series.

2.4 Add the series. Rearrange each equality in 2.1 as a temperature drop, $T_d - T_{wi} = \dot{Q}_{db} R_i$ and so on, including the label's, and add them. The wall temperatures cancel in pairs:

$$T_d - T_b = \dot{Q}_{db}\,(R_i + R_c + R_w + R_o).$$

Resistances in series add, because their temperature drops add while the heat flow is common.

2.5 Faces in parallel. The side, the base and (if submerged) the top of the container are separate paths j side by side, each with its own films and wall. Each carries its own heat flow driven by the

same $T_d - T_b$, so the flows add, and with them the conductances (inverse resistances):

$$UA = \sum_j \left(\frac{1}{h_{i,j} A_{i,j}} + R_{c,j} + R_{w,j} + \frac{1}{h_{o,j} A_{o,j}} \right)^{-1},$$
$$\frac{1}{U} = \frac{1}{h_i} + R_c'' + \frac{t_w}{k_w} + \frac{1}{h_o} \quad \text{(thin wall, } A_i \approx A_o \approx A\text{)}.$$

The second line, for one face with a thin wall, multiplies every resistance by the common area: U is the overall coefficient per unit area, and UA the conductance in $\mathrm{W\,K^{-1}}$.

2.6 What a probe on the wall measures. Dividing two of the drops in 2.4 gives the share of the temperature difference each part takes, as in a potential divider:

$$\frac{T_d - T_{wo}}{T_{wo} - T_b} = \frac{R_i + R_c + R_w}{R_o}.$$

A probe taped to the outside of the can therefore reads neither the drink nor the bath.

Can versus bottle. With can-body aluminium, $k_w = 160\,\mathrm{W\,m^{-1}K^{-1}}$, a wall 0.1 mm thick has $t_w/k_w = 6 \times 10^{-7}\,\mathrm{m^2K\,W^{-1}}$; a film with $h = 300$ has $1/h = 3 \times 10^{-3}$, five thousand times more. So R_w drops out for a can. With PET, $k_w = 0.2$, a wall 0.3 mm thick has $t_w/k_w = 1.5 \times 10^{-3}$, half a film's resistance: a bottle's wall matters. Compare containers by U, not by their cooling time, which also depends on how much drink they hold (Step 4).

Result 2 — heat flow from drink to bath

$$\dot{Q}_{db} = UA\,(T_d - T_b), \qquad \frac{1}{UA} = R_i + R_c + R_w + R_o$$

What it means. The whole path from drink to bath is one conductance UA, and the heat flow is proportional to the temperature difference — as long as the film coefficients do not themselves depend on it, which Step 5 revisits.

Step 3 — Biot number and when lumping is allowed

The question. Is it fair to describe the whole drink by one temperature (A1)?

3.1 The usual test. Heat leaving a body meets two resistances in series: inside it, across a length L_c of material of conductivity k, about L_c/k per unit area; outside it, $1/U_{\mathrm{ext}}$. Their ratio is the Biot number,

$$\mathrm{Bi} = \frac{U_{\mathrm{ext}} L_c}{k_d} = 1.70\,U_{\mathrm{ext}} L_c,$$
$$\frac{1}{U_{\mathrm{ext}}} = R_c'' + \frac{t_w}{k_w} + \frac{1}{h_o}, \qquad L_c = \frac{V}{A} \text{ or } r_i,$$

with water's conductivity $k_d = 0.589\,\mathrm{W\,m^{-1}K^{-1}}$ (U_{ext} in $\mathrm{W\,m^{-2}K^{-1}}$, L_c in m). If $\mathrm{Bi} < 0.1$, the inside resistance is small, the temperature inside hardly varies, and a body that conducts can be lumped. A 330 ml can has $V/A \approx 1.3$ cm; with U_{ext} of a few hundred $\mathrm{W\,m^{-2}K^{-1}}$, Bi is about 5 to 10. A still drink fails the test. The tiny Biot number often quoted for a can uses the aluminium's k_w and t_w: it belongs to the wall, not the drink.

3.2 How slowly a still drink would cool. If the drink only conducted, the heat equation $\partial T/\partial t = \alpha \nabla^2 T$ would hold inside it. In a long cylinder whose wall is held at a fixed temperature, separating variables gives patterns $J_0(\lambda r)\, e^{-\alpha \lambda^2 t}$, where J_0 is a Bessel function; the wall condition

$J_0(\lambda r_i) = 0$ allows $\lambda = j_{0,1}/r_i$ for the slowest pattern, $j_{0,1} = 2.405$. That pattern decays with time constant

$$\tau_{\text{cond}} = \frac{r_i^2}{\alpha\, j_{0,1}^2} = \frac{r_i^2}{1.408 \times 10^{-7} \times 5.783} = 1.23 \times 10^6\, r_i^2 \ \text{ s} \qquad (r_i \text{ in m}),$$

22 minutes for a can of radius 3.3 cm: conduction alone would take more than an hour to cool it.

3.3 The rescue: free convection. The drink is not still. Liquid cooled at the wall is denser and sinks in a thin layer down the wall, while the core is stirred by the circulation this sets up. Step 5 shows that the layer is $\delta \sim H\,\mathrm{Ra}_H^{-1/4}$ thick, where Ra_H is the Rayleigh number; with water's properties,

$$\delta \sim \frac{H^{1/4}}{310\,\Delta T_i^{1/4}} = 3.2 \times 10^{-3}\, H^{1/4}\, \Delta T_i^{-1/4} \ \text{ m},$$

with H in m and ΔT_i in K: about 1 mm for a can 11.7 cm tall and a 10 K drop: far thinner than the radius. All the temperature change sits in that layer, whose resistance is R_i, already in the circuit. The measured speed-up over conduction is $E = \tau_{\text{cond}}/\tau_{\text{measured}}$.

Result 3 — lumping criterion

A1 holds if $\delta \sim H\,\mathrm{Ra}_H^{-1/4} \ll r_i$ and $|T_{\text{top}} - T_{\text{bottom}}| \ll T_d - T_b$ (not $\mathrm{Bi} < 0.1$)

What it means. The drink is a mixed core inside a thin cold layer, not a conducting solid. The test is whether the layer is thin and the core uniform, which a probe at the top and one at the bottom of the drink can check: cold liquid collects at the bottom, so the core does stratify slowly.

Step 4 — Fixed-bath solution

The question. If the bath stays at one temperature, how does the drink cool?

4.1 Combine Results 1 and 2. Put $\dot{Q}_{db} = UA(T_d - T_b)$ into $C_d\,\mathrm{d}T_d/\mathrm{d}t = -\dot{Q}_{db}$. With T_b constant, $\mathrm{d}\theta/\mathrm{d}t = \mathrm{d}T_d/\mathrm{d}t$ for the difference $\theta = T_d - T_b$, so

$$\frac{\mathrm{d}\theta}{\mathrm{d}t} = -\frac{\theta}{\tau}, \qquad \tau \equiv \frac{C_d}{UA}.$$

The difference falls at a rate proportional to itself; τ (seconds) is the drink's heat capacity over its conductance.

4.2 Separate and integrate. Put everything in θ on the left and in t on the right, and integrate from the start, where $\theta = \theta_0$:

$$\int_{\theta_0}^{\theta} \frac{\mathrm{d}\theta'}{\theta'} = -\int_0^t \frac{\mathrm{d}t'}{\tau} \ \Rightarrow\ \ln\frac{\theta}{\theta_0} = -\frac{t}{\tau}.$$

4.3 Solve for the temperature. Exponentiate both sides and add back T_b: this gives Result 4. In each time τ the remaining difference shrinks by the factor $e = 2.718$.

4.4 Derived quantities. Setting $T_d = T^*$ and solving for t gives the time to reach a target; differentiating at $t = 0$ gives the initial cooling rate:

$$t_{T^*} = \tau \ln\frac{T_{d0} - T_b}{T^* - T_b}, \qquad \left.\frac{\mathrm{d}T_d}{\mathrm{d}t}\right|_0 = -\frac{T_{d0} - T_b}{\tau}.$$

Two baths at T_{b1} and T_{b2} can be compared in two ways:

$$\frac{t_{T^*}(T_{b1})}{t_{T^*}(T_{b2})} = \frac{\ln[(T_{d0} - T_{b1})/(T^* - T_{b1})]}{\ln[(T_{d0} - T_{b2})/(T^* - T_{b2})]}, \qquad \frac{\dot{T}_d(0)|_{T_{b1}}}{\dot{T}_d(0)|_{T_{b2}}} = \frac{T_{d0} - T_{b1}}{T_{d0} - T_{b2}}.$$

The two ratios differ: "how much faster does the colder bath cool the drink" has two honest answers, one for the time to a target and one for the initial rate.

4.5 Geometry inside τ. The drink's heat capacity is about $\rho_d c_d V$ for a volume V, so $\tau = (\rho_d c_d / U)\,(V/A)$: at the same U, a container with more drink per unit of wetted area cools more slowly.

Result 4 — fixed-bath cooling curve

$$T_d(t) = T_b + (T_{d0} - T_b)\, e^{-t/\tau}, \qquad \tau = \frac{C_d}{UA}$$

What it means. A plot of $\ln(T_d - T_b)$ against time is a straight line of slope $-1/\tau$; from its slope and the weighed C_d, $UA = C_d/\tau$.

Step 5 — Convection closures and the power-law cooling curve

The question. What sets the film coefficients when nothing is stirred, and how does that change the cooling curve?

5.1 Drink side: laminar free convection. In the layer next to the wall the drink is colder by about $\Delta T_i = T_d - T_{wi}$, so denser by $\rho\beta\,\Delta T_i$: a buoyancy force $\rho g\beta\,\Delta T_i$ per unit volume drives it down the wall. In a layer of thickness δ moving at speed u, viscosity resists with a force $\rho\nu u/\delta^2$; balancing the two gives $u \sim g\beta\Delta T_i\delta^2/\nu$. Heat carried along the wall over its height H, $u\Delta T_i/H$, must balance heat conducted across the layer, $\alpha\Delta T_i/\delta^2$. Eliminating u gives $\delta \sim H\,\mathrm{Ra}_H^{-1/4}$, and since the film coefficient is about k_d/δ,

$$\mathrm{Ra}_H = \frac{g\beta\,\Delta T_i H^3}{\nu\alpha}, \qquad \mathrm{Nu}_H = \frac{h_i H}{k_d} = C_i\,\mathrm{Ra}_H^{1/4}.$$

The Rayleigh number compares buoyancy with the two diffusions, of momentum and of heat, that resist it; the Nusselt number compares the film coefficient with conduction over the height. The scaling fixes the power $\frac{1}{4}$; the constant comes from experiments on vertical walls, $C_i = 0.62$ for water (Table 1). With water at 15 °C,

$$\mathrm{Ra}_H = \frac{9.81 \times 1.51 \times 10^{-4}}{1.139 \times 10^{-6} \times 1.408 \times 10^{-7}}\,\Delta T_i H^3 = 9.22 \times 10^9\,\Delta T_i H^3.$$

Solving the Nusselt relation for h_i and putting in the numbers:

$$h_i = C_i k_d \left(\frac{g\beta}{\nu\alpha H}\right)^{1/4} \Delta T_i^{1/4} = 0.62 \times 0.589 \times 310\; H^{-1/4}\,\Delta T_i^{1/4} \equiv K_i\,\Delta T_i^{1/4},$$

$$K_i = 112\, H^{-1/4}$$

in $\mathrm{W\,m^{-2}K^{-5/4}}$ with H in m. For an upright can 11.66 cm tall, $K_i = 192$, and at a 25 K drop $h_i = 430\,\mathrm{W\,m^{-2}K^{-1}}$, close to the 400 measured on a bottle. Three remarks. The law holds while the layer is laminar, $\mathrm{Ra}_H < 10^9$, that is $\Delta T_i H^3 < 0.11\,\mathrm{m^3K}$: always, here. Sugar makes a drink about a third more viscous than water, which lowers K_i by under 10 %, since it enters as a quarter power. And for a can lying on its side, as in the rig, the height of the wall is its diameter, and the constant changes a little (Churchill & Chu give a separate law for horizontal cylinders).

5.2 When the drink's film dominates. If R_i is much larger than $R_c + R_w + R_o$, almost all the temperature difference falls across the drink's film, $\Delta T_i \approx \theta$. Then $\dot{Q}_{db} = h_i A_i \theta = K_i A_i\,\theta^{1/4}\theta$:

$$\dot{Q}_{db} = K_i A_i\,\theta^{5/4} \equiv a\,\theta^{5/4}.$$

The heat flow grows faster than the difference, because a larger difference also thins the film.

5.3 Integrate with T_b fixed. Result 1 becomes $C_d\,\mathrm{d}\theta/\mathrm{d}t = -a\theta^{5/4}$. Separate and integrate as in Step 4, using $\int \theta^{-5/4}\mathrm{d}\theta = -4\theta^{-1/4}$:

$$\int_{\theta_0}^{\theta} \theta'^{-5/4}\mathrm{d}\theta' = -\frac{a}{C_d}\int_0^t \mathrm{d}t' \;\Rightarrow\; \theta^{-1/4} = \theta_0^{-1/4} + \frac{at}{4C_d} \;\Rightarrow\; \theta(t) = \theta_0\left(1 + \frac{t}{t^*}\right)^{-4}.$$

The last step factored out $\theta_0^{-1/4}$ and raised both sides to the power -4, which defines

$$t^* = \frac{4C_d}{a\,\theta_0^{1/4}} = 4\tau_0, \qquad \tau_0 = \frac{C_d}{h_i(\theta_0)A_i}.$$

τ_0 is the time constant the drink would have if its film kept its starting coefficient. The curve starts like an exponential with that time constant and then slows, as the film thickens.

5.4 Any exponent. For $\dot{Q} = a\theta^n$ with $n \neq 1$, the same steps with $\int \theta^{-n}\mathrm{d}\theta = \theta^{1-n}/(1-n)$ give

$$\theta^{1-n} = \theta_0^{1-n} + (n-1)\frac{a}{C_d}t \;\Rightarrow\; \theta(t) = \theta_0\big[1 + (n-1)t/\tau_0\big]^{-1/(n-1)}, \qquad \tau_0 = \frac{C_d}{a\theta_0^{n-1}}.$$

As $n \to 1$, $[1 + (n-1)x]^{-1/(n-1)} \to e^{-x}$ and Result 4 comes back.

5.5 Testing the exponent. Taking logarithms of $C_d\,\mathrm{d}T_d/\mathrm{d}t = -a(T_d - T_b)^n$,

$$\ln\left(-\frac{\mathrm{d}T_d}{\mathrm{d}t}\right) = \ln\frac{a}{C_d} + n\ln(T_d - T_b),$$

so the slope of the log of the cooling rate against the log of the difference is n: 1 for a constant U, $\frac{5}{4}$ for a convection-limited drink.

5.6 Mixed resistances. When the other resistances are not small, the drink's film sees only part of θ: what is left after the drop $\dot{Q}_{db}(R_c + R_w + R_o)$ across the rest. That makes $\dot{Q}_{db}$ implicit,

$$\dot{Q}_{db} = K_i A_i\big[\theta - \dot{Q}_{db}\,(R_c + R_w + R_o)\big]^{5/4}, \qquad 1 \le n_{\text{eff}} \le \tfrac{5}{4},$$

and the effective exponent lies between the two limits.

5.7 Bath side, unstirred. The same reasoning applies to the brine on the outside of the can, except that its density difference $\Delta\rho$ comes from two causes: the brine at the wall is warmer, and the ice it melts dilutes it. Both make it lighter, so it rises along the wall. Writing the Rayleigh number with $\Delta\rho$ directly,

$$\mathrm{Nu}_o = \frac{h_o H}{k_b} = C_o\left(\frac{g\,\Delta\rho\,H^3}{\rho_b\nu_b\alpha_b}\right)^{1/4}$$

$$\Rightarrow\; h_o = C_o k_b\left(\frac{g}{\rho_b\nu_b\alpha_b}\right)^{1/4}\Delta\rho^{1/4}H^{-1/4} = 118\,\Delta\rho^{1/4}H^{-1/4}$$

in $\mathrm{W\,m^{-2}K^{-1}}$, with $\Delta\rho$ in $\mathrm{kg\,m^{-3}}$ and H in m, using eutectic brine's properties ($C_o = 0.64$, $k_b = 0.54$, $\rho_b = 1182$, $\nu_b = 4.6 \times 10^{-6}$, $\alpha_b = 1.38 \times 10^{-7}$). $\Delta\rho$ stays a symbol: it depends on how warm the wall is and how much meltwater it makes. On a can 11.7 cm tall, a few $\mathrm{kg\,m^{-3}}$ gives h_o of 250–300 $\mathrm{W\,m^{-2}K^{-1}}$.

5.8 Caveat: β changes with temperature. Water is densest at 4 °C, where $\beta = 0$; below that it expands on cooling. Dissolved sugar and CO_2 move the maximum a little lower. As the drink approaches it the buoyancy in its film vanishes, and $\beta\,\Delta T_i$ must be replaced by the density difference itself:

$$\mathrm{Ra}_H = \frac{g\,[\rho_d(T_{wi}) - \rho_d(T_d)]\,H^3}{\rho_d\,\nu\,\alpha}.$$

Near the maximum the film coefficient falls sharply, so fit n only over the early part of a run.

Result 5 — convection closure and fixed-bath power law

$$h_i = K_i (T_d - T_{wi})^{1/4}, \qquad K_i = 112\, H^{-1/4}$$

$$\dot{Q}_{db} = a\theta^n,\ 1 \le n \le \tfrac{5}{4}, \qquad \theta(t) = \theta_0 \left(1 + \frac{t}{t^*}\right)^{-4} \quad (n = \tfrac{5}{4})$$

What it means. With nothing stirred, the films set the rate, and each grows as the quarter power of its own temperature drop. A constant U (Result 2) is the simplest fit to a run; the power law is the physically expected shape, and the exponent test of 5.5 tells them apart.

Step 6 — Phase equilibrium of the bath

The question. At what temperature can ice and brine of a given salt content coexist, how much salt can brine hold, and what is the coldest the bath can get?

6.1 Equilibrium condition. Water molecules can pass between the ice and the brine. They stop moving on balance when water has the same *chemical potential* — Gibbs energy per mole — in both. Ice is pure, so its chemical potential is that of pure ice, μ_{ice}. In brine the dissolved ions dilute the water and lower its chemical potential below that of pure liquid water, μ_w^*, by $R\mathcal{T}\ln a_w$, where a_w is the water's activity ($a_w = 1$ for pure water, less in brine) and $R = 8.314\,\text{J mol}^{-1}\text{K}^{-1}$. Equilibrium at absolute temperature $\mathcal{T}$ needs

$$\mu_{\text{ice}}(\mathcal{T}) = \mu_w^*(\mathcal{T}) + R\mathcal{T}\ln a_w \ \Rightarrow\ R\mathcal{T}\ln a_w = -\Delta g_{\text{fus}}(\mathcal{T}),$$

where $\Delta g_{\text{fus}} = \mu_w^* - \mu_{\text{ice}}$ is the Gibbs energy of melting one mole of ice.

6.2 How Δg_{fus} changes with temperature. The Gibbs–Helmholtz relation, $\mathrm{d}(\Delta g/\mathcal{T})/\mathrm{d}\mathcal{T} = -\Delta h/\mathcal{T}^2$, links it to the molar heat of melting, $\Delta h_{\text{fus}} = L_f M_w = 6009\,\text{J mol}^{-1}$. Pure ice and water coexist at $\mathcal{T}_0 = 273.15\,\text{K}$, so $\Delta g_{\text{fus}}(\mathcal{T}_0) = 0$. Integrating from $\mathcal{T}_0$ with Δh_{fus} constant, and using 6.1,

$$\frac{\Delta g_{\text{fus}}}{\mathcal{T}} = \Delta h_{\text{fus}}\left(\frac{1}{\mathcal{T}} - \frac{1}{\mathcal{T}_0}\right) \ \Rightarrow\ \ln a_w = -\frac{\Delta h_{\text{fus}}}{R}\left(\frac{1}{\mathcal{T}} - \frac{1}{\mathcal{T}_0}\right),$$

$$\ln a_w = -722.7\left(\frac{1}{\mathcal{T}} - \frac{1}{273.15}\right)$$

with $\mathcal{T}$ in K. Brine of lower water activity is in equilibrium with ice at a lower temperature. Strictly Δh_{fus} is not constant: it falls by $(c_w - c_i)M_w = 39\,\text{J mol}^{-1}$ per kelvin below 0 °C, about 14 % by the eutectic — one reason the measured curve of 6.4 is used in the model.

6.3 The dilute limit: Blagden's law. In dilute brine, a_w is close to the mole fraction of water, $1 - x$, where x is the mole fraction of dissolved particles, and $\ln(1-x) \approx -x$. Per kg of water there are $\nu_i b$ moles of particles, b being the molality (moles of NaCl per kg of water), against $1/M_w$ moles of water, so $x \approx \nu_i b M_w$. Also $1/\mathcal{T} - 1/\mathcal{T}_0 = (\mathcal{T}_0 - \mathcal{T})/(\mathcal{T}\mathcal{T}_0) \approx \Delta T_f/\mathcal{T}_0^2$ for a small drop ΔT_f. Equating,

$$\Delta T_f = \underbrace{\frac{R\mathcal{T}_0^2 M_w}{\Delta h_{\text{fus}}}}_{K_f} \nu_i b,$$

$$K_f = \frac{8.314 \times 273.15^2 \times 0.018015}{6009} = 1.86\ \text{K kg mol}^{-1}.$$

The molality of brine of salt fraction w is $b = (w/M_s)/(1-w) = 17.11\,w/(1-w)\,\text{mol kg}^{-1}$: per kg of brine, w kg of salt is w/M_s moles, dissolved in $1 - w$ kg of water. With $\nu_i = 2$,

$$\Delta T_f = 2 \times 1.86 \times 17.11\,\frac{w}{1-w} = 63.6\,\frac{w}{1-w}\ \text{K}.$$

At 1 wt % this gives 0.64 K; the measured drop is 0.59 K, because the ions are not independent particles. The osmotic coefficient φ (≈ 0.93 here) corrects for that, $\ln a_w = -\varphi \nu_i b M_w$ exactly, and then 6.2 gives, for real brine,

$$\frac{1}{\mathcal{T}_{\mathrm{liq}}} = \frac{1}{\mathcal{T}_0} + \frac{R}{\Delta h_{\mathrm{fus}}} \varphi\, \nu_i\, b\, M_w,$$

$$\frac{1}{\mathcal{T}_{\mathrm{liq}}} = \frac{1}{273.15} + 8.53 \times 10^{-4}\, \varphi(w)\, \frac{w}{1-w} \quad \mathrm{K}^{-1}.$$

6.4 The measured freezing curve (Bodnar 1993). Rather than tabulate φ, the model uses a fit to measured freezing points. With $\vartheta = -T_{\mathrm{liq}}$ the number of degrees below zero and $W = 100w$,

$$W = 1.78\,\vartheta - 0.0442\,\vartheta^2 + 5.572 \times 10^{-4}\,\vartheta^3.$$

Given w, the cubic is solved for ϑ between 0 and 21.2. Its slope follows from $\mathrm{d}W/\mathrm{d}\vartheta = 1.78 - 0.0884\,\vartheta + 1.672 \times 10^{-3}\,\vartheta^2$, with $\mathrm{d}W = 100\,\mathrm{d}w$ and $\mathrm{d}T_{\mathrm{liq}} = -\mathrm{d}\vartheta$:

$$\frac{\mathrm{d}T_{\mathrm{liq}}}{\mathrm{d}w} = -\frac{100}{1.78 - 0.0884\,\vartheta + 1.672 \times 10^{-3}\,\vartheta^2} \quad \mathrm{K}.$$

It is negative — more salt, colder freezing point — and steepens from 0.56 K per wt % near 0 °C to 1.52 K per wt % near the eutectic.

6.5 The phase rule. Gibbs's phase rule counts how many quantities can still be chosen freely, F, for C components in P phases: $F = C - P + 2$. The components are water and NaCl, $C = 2$, and the pressure is fixed, which uses up one: $F = 3 - P$.

Phases present	P	F	What is fixed
brine	1	2	nothing: T and w are both free
ice + brine	2	1	the temperature by the salt: $T = T_{\mathrm{liq}}(w)$
brine + solid salt	2	1	the salt by the temperature: $w = w_{\mathrm{sat}}(T)$
ice + brine + solid salt	3	0	everything: $T = T_E$, $w = w_E$
ice + solid salt	2	1	T free, below T_E

With ice, brine and salt all present, nothing is left to choose: the temperature is pinned at the eutectic, whatever heat comes in, until one of the three phases runs out.

6.6 Eutectic and solubility. The eutectic is where the freezing curve meets the solubility curve:

$$T_{\mathrm{liq}}(w_E) = T_E, \qquad w_{\mathrm{sat}}(T_E) = w_E, \qquad T_E = -21.2\,^{\circ}\mathrm{C}, \quad w_E = 0.2318.$$

Between the eutectic and the peritectic (+0.1 °C, $w_p = 0.26285$) the solubility is taken as the straight line between them, for $-21.2 \le T \le 0.1$ °C:

$$w_{\mathrm{sat}}(T) \approx w_E + (w_p - w_E)\frac{T - T_E}{T_p - T_E} = 0.2318 + 1.458 \times 10^{-3}\,(T + 21.2).$$

Saturated brine is a little saltier when warmer: 0.0015 per kelvin. At the peritectic it matches the handbook solubility of halite at 0 °C, 35.65 g per 100 g of water, $w = 0.2628$.

Result 6 — phase constraints

ice + brine: $T_b = T_{\mathrm{liq}}(w)$; brine + salt: $w = 0.2318 + 1.458 \times 10^{-3}(T_b + 21.2)$;
all three: $T_b = -21.2\,^{\circ}\mathrm{C}$, $w = 0.2318$

What it means. The phase diagram ties the temperature to the salt content whenever two phases coexist, and pins both at the eutectic when three do. That is what makes a salted ice bath a thermostat.

Step 7 — Bath mass balances

The question. How are the amounts of ice, brine and salt related, given that water and salt are neither created nor destroyed?

Water is in the ice and in the brine; salt is in the brine and in the undissolved crystals. A kilogram of brine holds $1 - w$ kg of water and w kg of salt. Counting each substance,

$$m_W = m_i + (1-w)m_l, \qquad m_S = w\,m_l + m_h, \qquad m_W + m_S = m_i + m_l + m_h,$$

the third being the sum of the first two. While no salt is undissolved ($m_h = 0$), these give the brine mass and its salt fraction from the ice alone:

$$m_l = m_W + m_S - m_i, \qquad w = \frac{m_S}{m_W + m_S - m_i}.$$

Melting a mass $\mathrm{d}m_m$ of ice ($\mathrm{d}m_i = -\mathrm{d}m_m$) adds as much water to the brine, and the same salt is spread through more brine. Differentiating,

$$\frac{\mathrm{d}m_l}{\mathrm{d}m_m} = 1, \qquad \frac{\mathrm{d}w}{\mathrm{d}m_m} = -\frac{m_S}{m_l^2} = -\frac{w}{m_l}.$$

Melting always dilutes the brine. Whether the overall salt fraction $\bar{w} = m_S/(m_W + m_S)$ lies below or above w_E decides which phase runs out first on the eutectic plateau (Step 9).

Result 7 — mass constraints

$$m_W = m_i + (1-w)m_l, \quad m_S = w\,m_l + m_h, \quad 0 \le m_i \le m_W, \quad 0 \le m_h \le m_S$$

What it means. Two numbers, the water and the salt put in, fix the bath's make-up; the state only decides how they are shared out between ice, brine and crystals.

Step 8 — Bath enthalpy and the bath energy equation

The question. How much heat does the bath hold, and how does it change?

8.1 Specific enthalpies. Each is counted from the reference (liquid water and solid salt at 0 °C), in J kg^{-1}, with T in °C:

$$h_{\mathrm{ice}} = 2000\,T - 333\,550, \qquad h_{\mathrm{salt}} = 864\,T, \qquad h_{\mathrm{brine}} = (4180 - 4400\,w)\,T + 66\,400\,w.$$

Ice at 0 °C holds less heat than water at 0 °C by the latent heat L_f, and cooling it removes $c_i = 2000$ per kelvin. Salt starts at the reference and has only its sensible heat, $c_h = 864$ per kelvin. Brine is made from water and salt at the reference: dissolving its w kg of salt absorbs $\Delta h_s = 66\,400$ per kg of salt, and it then warms with its own specific heat $c_b(w)$. The heat of diluting brine further is left out.

The straight line $c_b(w) = (1-w)\,c_w + w\,(c_w - 4400)$ can be read as water at $c_w = 4180$ plus dissolved salt at $c_w - 4400 = -220$ J kg^{-1}K^{-1}. The negative number is not a heat capacity of salt: it is how much dissolved ions lower the heat capacity of the water round them. It has one useful consequence: at fixed salt, $m_l c_b(w) = 4180\,m_l - 4400\,m_S$, so each kilogram of water added to the brine adds exactly c_w to its heat capacity.

8.2 Total enthalpy. Enthalpy is extensive: the bath's is the sum of mass times specific enthalpy over its parts, all at the one temperature T_b (A3):

$$\begin{aligned}\mathcal{H} = {} & m_i(2000\,T_b - 333\,550) + m_l\left[(4180 - 4400\,w)\,T_b + 66\,400\,w\right] \\ & + 864\,m_h T_b \quad [+\,C_t T_b].\end{aligned}$$

The last term is there if the tank glass is counted with the bath (A2).

8.3 First law for the bath. The bath is at constant pressure and does no work: as in Step 1, its enthalpy rises by the heat that enters it, from the drink and from the room. There is no stirrer, so no other energy enters.

$$\frac{\mathrm{d}\mathcal{H}}{\mathrm{d}t} = \dot{Q}_{db} + \dot{Q}_{\mathrm{leak}}.$$

8.4 The leak through the box. The box is a resistance between the room and the bath (A4), with conductance UA_{box} (Step 12):

$$\dot{Q}_{\mathrm{leak}} = UA_{\mathrm{box}}(T_{\mathrm{env}} - T_b).$$

8.5 Energy check. Adding Result 1, $C_d\,\mathrm{d}T_d/\mathrm{d}t = -\dot{Q}_{db}$, the drink's term cancels: $\mathrm{d}(C_dT_d + \mathcal{H})/\mathrm{d}t = \dot{Q}_{\mathrm{leak}}$. If the box were perfect ($UA_{\mathrm{box}} = 0$), $C_dT_d + \mathcal{H}$ would stay constant — what the drink loses, the bath gains. Any computation must pass this check.

Result 8 — bath energy equation

$$\frac{\mathrm{d}\mathcal{H}}{\mathrm{d}t} = UA(T_d - T_b) + UA_{\mathrm{box}}(T_{\mathrm{env}} - T_b)$$

$$\mathcal{H} = m_i(2000\,T_b - 333\,550) + m_l\big[(4180 - 4400\,w)\,T_b + 66\,400\,w\big] + 864\,m_hT_b$$

What it means. The bath's state variable is the heat it holds, not its temperature. Heat in raises $\mathcal{H}$; what that does to T_b depends on whether it warms the bath or melts ice, which Step 9 decides.

Step 9 — Inverting $\mathcal{H}$: the regime map

The question. Given the bath's enthalpy and what was put in, what is its temperature, and how much ice, brine and salt does it hold?

There are five unknowns, T_b, m_i, m_l, m_h and w. The two mass balances (Step 7) and the enthalpy (Step 8) give three equations; the phase rule (Step 6) supplies the other two from whichever phases are present (a missing phase supplies its own zero mass). So the state is fixed:

$$(T_b,\, m_i,\, m_l,\, m_h,\, w) = \mathcal{S}(\mathcal{H};\, m_W,\, m_S).$$

Which phases are present depends on $\mathcal{H}$; the five possibilities are the regimes below, from warm to cold.

R-I — liquid only. No ice, no crystals: $m_i = m_h = 0$, $m_l = m_W + m_S$, $w = \bar{w}$. The enthalpy is that of the brine alone, which is linear in T_b:

$$T_b = \frac{\mathcal{H} - 66\,400\,m_S}{(m_W + m_S)(4180 - 4400\,\bar{w})}.$$

R-II — ice and brine on the freezing curve. No crystals ($m_h = 0$); this regime exists only if $\bar{w} < w_E$. The brine sits on the freezing curve, $T_b = T_{\mathrm{liq}}(w)$, and Step 7 gives $m_l = m_S/w$ and $m_i = m_W + m_S - m_S/w$. Putting these into $\mathcal{H}$ makes it a function of w alone:

$$\mathcal{H}_{\mathrm{II}}(w) = \Big(m_W + m_S - \frac{m_S}{w}\Big)(2000\,T_{\mathrm{liq}}(w) - 333\,550) + \frac{m_S}{w}(4180 - 4400\,w)\,T_{\mathrm{liq}}(w) + 66\,400\,m_S.$$

It is solved for w between $\bar{w}$ and w_E. There is exactly one root: adding heat melts ice ($\mathrm{d}\mathcal{H}/\mathrm{d}m_m = L_{\mathrm{eff}} > 0$, Step 10), and melting dilutes the brine ($\mathrm{d}w/\mathrm{d}m_m < 0$), so $\mathcal{H}_{\mathrm{II}}$ falls steadily as w rises. Then $T_b = T_{\mathrm{liq}}(w)$.

R-III — the eutectic plateau. Ice, brine and crystals together: $T_b = -21.2\,^{\circ}\mathrm{C}$ and $w = 0.2318$ (Step 6). Only the brine mass is free, and $\mathcal{H}$ is linear in it. With no brine at all, everything is ice and salt at T_E:

$$\mathcal{H}_{E0} = m_W(c_i T_E - L_f) + m_S c_h T_E = -375\,950\,m_W - 18\,317\,m_S.$$

Each kilogram of eutectic brine is made from $1 - w_E$ kg of ice and w_E kg of salt at T_E; the enthalpy this takes is the brine's minus theirs, $\Lambda_E = h_{\mathrm{brine}} - (1 - w_E)h_{\mathrm{ice}} - w_E h_{\mathrm{salt}}$:

$$\Lambda_E = (1 - w_E)L_f + w_E\Delta h_s + T_E\big[c_b(w_E) - (1 - w_E)c_i - w_E c_h\big].$$

The first term melts 0.768 kg of ice, 256 230 J; the second dissolves 0.232 kg of salt, 15 390 J; the third, $-21.2 \times (3160 - 1536 - 200) = -30\,180$ J, is negative because the brine has a larger heat capacity than the ice and salt it came from, and at T_E, below the reference, a larger heat capacity means less enthalpy. In all, $\Lambda_E = 241\,450\,\mathrm{J\,kg^{-1}}$, and

$$m_l = \frac{\mathcal{H} - \mathcal{H}_{E0}}{241\,450}, \qquad m_i = m_W - 0.7682\,m_l, \qquad m_h = m_S - 0.2318\,m_l.$$

Heat moves the bath along the plateau by making or freezing eutectic brine. If $\bar{w} < w_E$ the salt runs out first, and the bath leaves into R-II; if $\bar{w} > w_E$ the ice runs out first, and it leaves into R-IV.

R-IV — saturated brine and crystals, no ice. Only if $\bar{w} > w_E$. All the water is liquid, $m_l = m_W/(1 - w)$, and the brine is saturated, $w = w_{\mathrm{sat}}(T_b) = 0.2318 + 1.458 \times 10^{-3}(T_b + 21.2)$; the rest of the salt is crystals, $m_h = m_S - w\,m_l$. Then $\mathcal{H}$ is a function of T_b alone:

$$\begin{aligned}\mathcal{H}_{\mathrm{IV}}(T_b) = {} & \frac{m_W}{1 - w}\big[(4180 - 4400\,w)\,T_b + 66\,400\,w\big] \\ & + \Big(m_S - \frac{w\,m_W}{1 - w}\Big)864\,T_b,\end{aligned}$$

solved for T_b between T_E and $T_\star$, the temperature at which all the salt would just dissolve, $w_{\mathrm{sat}}(T_\star) = \bar{w}$, that is $T_\star = -21.2 + (\bar{w} - 0.2318)/(1.458 \times 10^{-3})$.

R-V — fully solid. No brine ($m_l = 0$): all the water is ice and all the salt crystals, below T_E. The enthalpy is linear in T_b:

$$T_b = \frac{\mathcal{H} + 333\,550\,m_W}{2000\,m_W + 864\,m_S}.$$

Regime boundaries. Each threshold is the enthalpy at which a phase appears or disappears, found by putting that state into the formula of either neighbouring regime; the state is continuous across every boundary.

Bath type	Regime	Range	Threshold
$\bar{w} < w_E$	R-I	$\mathcal{H} \geq \mathcal{H}_A$	$\mathcal{H}_A = (m_W + m_S)(4180 - 4400\,\bar{w})T_{\mathrm{liq}}(\bar{w}) + 66\,400\,m_S$: first ice
	R-II	$\mathcal{H}_B \leq \mathcal{H} < \mathcal{H}_A$	$\mathcal{H}_B = \mathcal{H}_{E0} + 1\,041\,600\,m_S$: all salt in eutectic brine
	R-III	$\mathcal{H}_{E0} \leq \mathcal{H} < \mathcal{H}_B$	
	R-V	$\mathcal{H} < \mathcal{H}_{E0}$	
$\bar{w} > w_E$	R-I	$\mathcal{H} \geq \mathcal{H}_C$	$\mathcal{H}_C = (m_W + m_S)(4180 - 4400\,\bar{w})T_\star + 66\,400\,m_S$: first crystal
	R-IV	$\mathcal{H}_D \leq \mathcal{H} < \mathcal{H}_C$	$\mathcal{H}_D = \mathcal{H}_{E0} + 314\,300\,m_W$: all water in eutectic brine
	R-III	$\mathcal{H}_{E0} \leq \mathcal{H} < \mathcal{H}_D$	
	R-V	$\mathcal{H} < \mathcal{H}_{E0}$	

The numbers are $\Lambda_E/w_E = 1\,041\,600$ and $\Lambda_E/(1 - w_E) = 314\,300\,\mathrm{J\,kg^{-1}}$: the enthalpy of the brine that holds all the salt, or all the water, at the eutectic.

Result 9 — bath closure

$$T_b = T_b(\mathcal{H}), \quad m_i = m_i(\mathcal{H}), \quad m_h = m_h(\mathcal{H}), \quad w = w(\mathcal{H}) \quad \text{via R-I to R-V}$$

What it means. The enthalpy alone, with the two masses, tells everything about the bath. This is why the model steps $\mathcal{H}$ forward in time rather than T_b: $\mathcal{H}$ always changes smoothly, while T_b stops dead on the plateau.

Step 10 — Effective latent heat, bath drift and the blank run

The question. On the freezing curve, how much heat melts a kilogram of ice, and how fast does the bath warm?

10.1 Differentiate along the freezing curve. In R-II all the salt is dissolved, so $m_l w = m_S$ is fixed, and by 8.1 $m_l(4180 - 4400w) = 4180\,m_l - 4400\,m_S$. The enthalpy is $\mathcal{H} = m_i(c_i T_b - L_f) + (4180\,m_l - 4400\,m_S)T_b + 66\,400\,m_S$, with $m_l = m_W + m_S - m_i$. A small change of ice and temperature changes it by

$$\mathrm{d}\mathcal{H} = [h_{\mathrm{ice}}(T_b) - c_w T_b]\,\mathrm{d}m_i + \underbrace{(m_i c_i + m_l c_b(w))}_{C_B}\,\mathrm{d}T_b.$$

The first term: a kilogram of ice that melts becomes a kilogram of water in the brine, at heat content $c_w T_b$ in place of $h_{\mathrm{ice}}(T_b)$. The second warms the ice and brine already there; C_B is their heat capacity.

10.2 Heat per kilogram melted. With $\mathrm{d}m_m = -\mathrm{d}m_i$ and $h_{\mathrm{ice}} = c_i T_b - L_f$:

$$L_{\mathrm{eff}} \equiv \frac{\mathrm{d}\mathcal{H}}{\mathrm{d}m_m} = L_f + (c_w - c_i)T_b + C_B \frac{\mathrm{d}T_b}{\mathrm{d}m_m},$$
$$L_{\mathrm{eff}} = 333\,550 + 2180\,T_b + C_B \frac{\mathrm{d}T_b}{\mathrm{d}m_m}.$$

Three parts: the latent heat at 0 °C; a correction because the ice melts at T_b, not 0 °C (below zero the latent heat is smaller by $c_w - c_i = 2180\,\mathrm{J\,kg^{-1}}$ per kelvin, Kirchhoff's law, so this term is negative); and the heat that warms the whole bath, because melting dilutes the brine and so raises its freezing point. The last part makes L_{eff} larger than L_f: for the 20 Oct rig-note charge at −17.8 °C, $L_{\mathrm{eff}} = 411\,500\,\mathrm{J\,kg^{-1}}$, 23 % more.

10.3 Warming per kilogram melted. By the chain rule, with $\mathrm{d}w/\mathrm{d}m_m = -w/m_l$ (Step 7) and the slope of the freezing curve (6.4),

$$\frac{\mathrm{d}T_b}{\mathrm{d}m_m} = \frac{\mathrm{d}T_{\mathrm{liq}}}{\mathrm{d}w}\,\frac{\mathrm{d}w}{\mathrm{d}m_m} = \frac{100\,w}{m_l\,(1.78 - 0.0884\,\vartheta + 1.672 \times 10^{-3}\,\vartheta^2)} > 0.$$

Melting always warms a bath on the freezing curve, and more so the less brine there is.

10.4 Effective heat capacity in every regime. The heat that raises the bath's temperature by one kelvin, $C_{\mathrm{eff}} = \mathrm{d}\mathcal{H}/\mathrm{d}T_b$, follows from differentiating each regime's enthalpy:

$$C_{\mathrm{eff}} = \begin{cases} (m_W + m_S)(4180 - 4400\,\bar{w}) & \text{R-I} \\ C_B + \dfrac{333\,550 + 2180\,T_b}{\mathrm{d}T_b/\mathrm{d}m_m} & \text{R-II} \\ \infty & \text{R-III} \\ m_l c_b(w) + 864\,m_h + \dfrac{1.458 \times 10^{-3}\,m_W}{(1-w)^2}\,(66\,400 - 1084\,T_b) & \text{R-IV} \\ 2000\,m_W + 864\,m_S & \text{R-V} \end{cases}$$

R-I and R-V are plain heat capacities. In R-II, warming by one kelvin needs $1/(\mathrm{d}T_b/\mathrm{d}m_m)$ kg of ice melted, each costing $L_f + (c_w - c_i)T_b$, on top of C_B: this is usually much larger than C_B. On the plateau no amount of heat changes the temperature. In R-IV, warming dissolves more salt, $\mathrm{d}(w\,m_l)/\mathrm{d}T_b = m_W w'_{\mathrm{sat}}/(1-w)^2$ with $w'_{\mathrm{sat}} = 1.458 \times 10^{-3}\,\mathrm{K}^{-1}$, each kilogram absorbing Δh_s and changing the sensible heat by $(-220 - 864)T_b = -1084\,T_b$ (dissolved instead of solid salt, 8.1).

10.5 The plateau as a thermostat. On the plateau the heat coming in, $\dot{Q}_{\mathrm{in}} = \dot{Q}_{db} + \dot{Q}_{\mathrm{leak}}$, does not warm the bath; it makes eutectic brine at Λ_E per kilogram, using ice and salt in the ratio $1 - w_E$ to w_E:

$$\frac{\mathrm{d}m_l}{\mathrm{d}t} = \frac{\dot{Q}_{\mathrm{in}}}{241\,450}, \qquad \frac{\mathrm{d}m_i}{\mathrm{d}t} = -\frac{\dot{Q}_{\mathrm{in}}}{314\,300}, \qquad \frac{\mathrm{d}m_h}{\mathrm{d}t} = -\frac{\dot{Q}_{\mathrm{in}}}{1\,041\,600} \quad (\mathrm{kg\,s^{-1}}).$$

The undissolved salt $m_h(0)$ is the buffer: it lasts

$$t_{\mathrm{buffer}} = \frac{m_h(0)\,\Lambda_E}{w_E\,\bar{\dot{Q}}_{\mathrm{in}}} = \frac{1.042 \times 10^6\, m_h(0)}{\bar{\dot{Q}}_{\mathrm{in}}} \ \mathrm{s},$$

with $m_h(0)$ in kg and the mean heat flow in W. Each 100 g of spare salt holds the bath at $-21.2\,^\circ\mathrm{C}$ against 1 W for 29 hours.

10.6 The blank run. With no drink in the bath and no stirrer, only the leak brings heat in. On the freezing curve, $\dot{Q}_{\mathrm{leak}} = L_{\mathrm{eff}}\,\mathrm{d}m_m/\mathrm{d}t$, and since $\mathrm{d}m_m/\mathrm{d}t = \dot{T}_b/(\mathrm{d}T_b/\mathrm{d}m_m)$,

$$\dot{Q}_{\mathrm{leak}} = L_{\mathrm{eff}}\frac{\mathrm{d}T_b/\mathrm{d}t}{\mathrm{d}T_b/\mathrm{d}m_m} = C_{\mathrm{eff}}\frac{\mathrm{d}T_b}{\mathrm{d}t} \ \Rightarrow\ UA_{\mathrm{box}} = \frac{C_{\mathrm{eff}}\,\dot{T}_b}{T_{\mathrm{env}} - T_b}.$$

A logged warming rate $\dot{T}_b$ therefore measures the box. Using L_f in place of L_{eff} would miss the heat that warms the whole bath and underestimate the leak.

Result 10 — explicit bath temperature equation

$$C_{\mathrm{eff}}(T_b)\frac{\mathrm{d}T_b}{\mathrm{d}t} = UA(T_d - T_b) + UA_{\mathrm{box}}(T_{\mathrm{env}} - T_b) \qquad (\dot{T}_b = 0 \text{ in R-III})$$

What it means. The bath behaves like a body of heat capacity C_{eff}, but one that changes with the regime: huge on the freezing curve, infinite on the plateau. That is why a bath with ice in it warms so slowly.

Step 11 — Initial state from premixing

The question. What state does the bath start in, once the ice, water and salt are mixed?

11.1 Enthalpy of the ingredients. Before mixing, each ingredient is at its own temperature: ice at T_{i0}, water at T_{w0}, salt at T_{s0}. Using 8.1 for each,

$$\mathcal{H}_0 = m_{i0}(2000\,T_{i0} - 333\,550) + 4180\,m_{w0}T_{w0} + 864\,m_S T_{s0},$$

and all the water is $m_W = m_{i0} + m_{w0}$.

11.2 Adiabatic mixing. In the insulated box, mixing exchanges heat only among the ingredients, so the enthalpy does not change (Step 8 with no heat in). The starting state is the one Step 9 gives for $\mathcal{H}_0$: $(T_{b0}, m_i, m_l, m_h, w)(0) = \mathcal{S}(\mathcal{H}_0; m_W, m_S)$. Enthalpy is a state function, so this does not depend on the order of mixing: adding the salt at once, in stages, or pre-dissolved reaches the same equilibrium, only at different rates.

11.3 Start on the eutectic. The bath starts on the plateau if $\mathcal{H}_0$ falls in its range (Step 9):

$$\mathcal{H}_{E0} \le \mathcal{H}_0 < \mathcal{H}_{E0} + 1\,041\,600\,m_S \quad \text{if } \bar{w} < w_E,$$
$$\mathcal{H}_{E0} \le \mathcal{H}_0 < \mathcal{H}_{E0} + 314\,300\,m_W \quad \text{if } \bar{w} > w_E.$$

11.4 Keep a salt buffer. To start with at least m_{buf} of salt undissolved, the plateau formula $m_h = m_S - 0.2318\, m_l$ with $m_l = (\mathcal{H}_0 - \mathcal{H}_{E0})/241\,450$ must give $m_h \geq m_{\text{buf}}$:

$$\mathcal{H}_0 \leq \mathcal{H}_{E0} + 1\,041\,600\,(m_S - m_{\text{buf}}).$$

11.5 Minimum ice fraction. Write the ice as a fraction $f = m_{i0}/m_W$ of the water. Then $\mathcal{H}_0 = m_W[f(2000\,T_{i0} - 333\,550) + (1-f)\,4180\,T_{w0}] + 864\,m_S T_{s0}$, which falls as f rises. For $\mathcal{H}_0$ to stay below any bound $\mathcal{H}_{\max}$ (from 11.3 or 11.4),

$$f \geq f_{\text{crit}} = \frac{4180\,m_W T_{w0} + 864\,m_S T_{s0} - \mathcal{H}_{\max}}{m_W\,(4180\,T_{w0} - 2000\,T_{i0} + 333\,550)}.$$

11.6 Melting against dissolving. When the mixture cools itself, the heat comes from melting Δm of ice and dissolving $m_{S,\text{dis}}$ of salt. The share due to melting is

$$\chi_{\text{lat}} = \frac{333\,550\,\Delta m}{333\,550\,\Delta m + 66\,400\,m_{S,\text{dis}}}.$$

Per kilogram, melting ice absorbs five times as much heat as dissolving salt.

Result 11 — initial conditions

$$T_d(0) = T_{d0},$$
$$\mathcal{H}(0) = m_{i0}(2000\,T_{i0} - 333\,550) + 4180\,m_{w0}T_{w0} + 864\,m_S T_{s0}$$

What it means. How cold the bath starts, and whether it starts on the plateau with salt to spare, is set by a single number, the enthalpy of what goes in; a drier, colder charge has less of it.

Step 12 — The box leak

The question. How fast does heat from the room get into the bath?

Heat from the room crosses three layers in series: the film of room air on the outside of the box, the foam, and the air between the foam and the tank. As in Step 2 their resistances add:

$$\frac{1}{U A_{\text{box}}} = \frac{1}{h_{\text{air}} A_{\text{out}}} + \frac{\delta_{\text{box}}}{0.035\,A_{\text{box}}} + \frac{1}{h_{\text{in}} A_{\text{in}}} \approx \frac{\delta_{\text{box}}}{0.035\,A_{\text{box}}},$$

with expanded polystyrene's conductivity, $0.035\,\text{W m}^{-1}\text{K}^{-1}$. The foam dominates: 30 mm of it has $\delta/k = 0.86\,\text{m}^2\text{K W}^{-1}$, against about $1/9 = 0.11$ for room air (free convection plus radiation) and $1/6 = 0.17$ for the air gap inside. Dropping the films overestimates the leak by about a third; gaps round the lid, where heat bypasses the foam, push it the other way.

Two variants for when an assumption is broken. With the lid off (A5), the bath's open surface also takes heat from the air and from water condensing on it: $\dot{Q}_{\text{surf}} = h_s A_s (T_{\text{air}} - T_b) + \dot{m}_v h_{sg}$, with $\dot{m}_v$ the condensation rate and h_{sg} its latent heat. With the drink not fully immersed (A8), its exposed part takes heat from the air: $C_d \dot{T}_d = -UA(T_d - T_b) + UA_{da}(T_{\text{air}} - T_d)$.

Result 12 — the leak

$$\dot{Q}_{\text{leak}} = U A_{\text{box}} (T_{\text{env}} - T_b), \qquad U A_{\text{box}} \approx \frac{0.035\,A_{\text{box}}}{\delta_{\text{box}}}$$

What it means. The leak is proportional to how much colder the bath is than the room, so it is largest exactly when the bath is most useful. It is best measured, with the blank run of 10.6.

Step 13 — The closed model

The question. What is the complete set of equations to integrate?

Two first-order differential equations for the state $(T_d, \mathcal{H})$, closed by one law for the heat flow and one map from $\mathcal{H}$ to the bath's state.

13.1 Governing equations (Results 1 and 8):

$$C_d \frac{\mathrm{d}T_d}{\mathrm{d}t} = -\dot{Q}_{db}, \qquad \frac{\mathrm{d}\mathcal{H}}{\mathrm{d}t} = \dot{Q}_{db} + UA_{\mathrm{box}}(T_{\mathrm{env}} - T_b).$$

13.2 Heat-flow closure — choose one (Results 2 and 5):

$$\text{A (constant conductance):} \quad \dot{Q}_{db} = UA(T_d - T_b), \quad \frac{1}{UA} = R_i + R_c + R_w + R_o$$

$$\text{B (drink-film convection):} \quad \dot{Q}_{db} = 112\, H^{-1/4} A_i \left[(T_d - T_b) - \dot{Q}_{db}(R_c + R_w + R_o)\right]^{5/4}$$

13.3 Bath closure (Results 6, 7 and 9), $(T_b, m_i, m_l, m_h, w) = \mathcal{S}(\mathcal{H}; m_W, m_S)$, in J, kg and °C:

$$\mathcal{S}: \begin{cases} \text{R-I:} & m_i = m_h = 0,\ w = \bar{w},\ T_b = \dfrac{\mathcal{H} - 66\,400\, m_S}{(m_W + m_S)(4180 - 4400\,\bar{w})} \\ \text{R-II:} & m_h = 0,\ m_l = m_S/w,\ T_b = T_{\mathrm{liq}}(w),\ w \text{ from } \mathcal{H} = \mathcal{H}_{\mathrm{II}}(w) \text{ (Step 9)} \\ \text{R-III:} & T_b = -21.2,\ w = 0.2318,\ m_l = \dfrac{\mathcal{H} - \mathcal{H}_{E0}}{241\,450},\ m_i = m_W - 0.7682\, m_l,\ m_h = m_S - 0.2318\, m_l \\ \text{R-IV:} & m_i = 0,\ w = w_{\mathrm{sat}}(T_b),\ T_b \text{ from } \mathcal{H} = \mathcal{H}_{\mathrm{IV}}(T_b) \text{ (Step 9)} \\ \text{R-V:} & m_l = 0,\ T_b = \dfrac{\mathcal{H} + 333\,550\, m_W}{2000\, m_W + 864\, m_S} \end{cases}$$

with the regime chosen by the thresholds of Step 9, and

$$W = 1.78\,\vartheta - 0.0442\,\vartheta^2 + 5.572 \times 10^{-4}\,\vartheta^3 \quad (W = 100w,\ \vartheta = -T_{\mathrm{liq}}),$$

$$w_{\mathrm{sat}}(T) = 0.2318 + 1.458 \times 10^{-3}(T + 21.2),$$

$$\mathcal{H}_{E0} = -375\,950\, m_W - 18\,317\, m_S, \qquad \bar{w} = \frac{m_S}{m_W + m_S}.$$

13.4 Equivalent explicit form. For a hand calculation (Desmos, Mathematica), Result 10 replaces the $\mathcal{H}$ equation:

$$C_d \dot{T}_d = -\dot{Q}_{db}, \qquad C_{\mathrm{eff}}(T_b)\, \dot{T}_b = \dot{Q}_{db} + UA_{\mathrm{box}}(T_{\mathrm{env}} - T_b) \quad (\dot{T}_b = 0 \text{ in R-III}).$$

A program should integrate the $\mathcal{H}$ form, which crosses the plateau without dividing by $C_{\mathrm{eff}} = \infty$.

13.5 Initial conditions (Result 11):

$$T_d(0) = T_{d0}, \qquad m_W = m_{i0} + m_{w0},$$

$$\mathcal{H}(0) = m_{i0}(2000\, T_{i0} - 333\,550) + 4180\, m_{w0} T_{w0} + 864\, m_S T_{s0}.$$

13.6 When to stop, and when the model stops being valid.

- T_d reaches the target T^*: the measured output.
- T_d reaches the drink's freezing point T_{fd} (about $-0.9\,°\mathrm{C}$ for a sugary fizzy drink): beyond it the drink needs its own enthalpy model, like the bath's.
- In a buffered run, $m_h \to 0$: the thermostat is gone and the bath leaves the plateau.

- T_d approaches the density maximum near 4 °C: closure B and assumption A1 weaken (5.8).

13.7 What has to be measured.

Group	Parameters	How they are fixed
Drink	m_d, c_d, m_c	weighing; the drink's specific heat from its label (sugar)
Heat path	UA (A), or A_i, H, R_c, R_w, R_o (B)	fitted on a buffered run, where $T_b = -21.2\,°\mathrm{C}$ is known
Bath	m_W, m_S, m_{i0}, m_{w0}	weighing
Environment	UA_{box}, $T_{\mathrm{env}}(t)$	a blank run (10.6); room temperature logged on channel 4
Start	T_{d0}, T_{i0}, T_{w0}, T_{s0}	logged

Every other constant is in Table 1. Once UA is fixed on a buffered run, an unbuffered run has no free parameter left: it is a prediction.

Result 13 — the model, in one block

$$
\begin{aligned}
C_d\,\dot{T}_d &= -\,UA\,(T_d - T_b), \qquad C_d = m_d c_d + 900\,m_c \\
\dot{\mathcal{H}} &= UA\,(T_d - T_b) + UA_{\mathrm{box}}(T_{\mathrm{env}} - T_b) \\
T_b &= T_b(\mathcal{H};\, m_W,\, m_S) \quad \text{via } \mathcal{S} \\
T_d(0) &= T_{d0}, \qquad \mathcal{H}(0) = \mathcal{H}_0
\end{aligned}
$$

What it means. Five numbers describe the rig (C_d, UA, m_W, m_S, UA_{box}), plus the starting temperatures; the rest is physics that is the same in any laboratory.

Step 14 — Closed-form special cases

The question. When can the equations be solved with a formula, rather than step by step?

14.1 Buffered bath. If the bath stays on the plateau (R-III throughout), $T_b = -21.2\,°\mathrm{C}$ is constant and Result 4 applies directly:

$$T_d(t) = -21.2 + (T_{d0} + 21.2)\,e^{-t/\tau}, \qquad \tau = \frac{C_d}{UA},$$

valid while $t < t_{\mathrm{buffer}}$ (10.5).

14.2 Bath on the freezing curve, sealed box. In R-II over a small range of temperature, C_{eff} is roughly constant; take $UA_{\mathrm{box}} = 0$. Then drink and bath are two heat capacities joined by one conductance. Dividing the two equations of 13.4 by their heat capacities and subtracting,

$$\frac{\mathrm{d}\theta}{\mathrm{d}t} = -UA\Big(\frac{1}{C_d} + \frac{1}{C_{\mathrm{eff}}}\Big)\theta \;\Rightarrow\; \theta = \theta_0 e^{-t/\tau_c}, \qquad \tau_c = \frac{C_d C_{\mathrm{eff}}}{UA\,(C_d + C_{\mathrm{eff}})} = \frac{\tau}{1+\Gamma}, \quad \Gamma = \frac{C_d}{C_{\mathrm{eff}}}.$$

The difference closes faster than with a fixed bath, because the bath warms to meet the drink. Energy conservation (8.5), $C_d T_d + C_{\mathrm{eff}} T_b = \mathrm{const}$, fixes the common final temperature, and each temperature approaches it with the same time constant:

$$T_\infty = \frac{C_d T_{d0} + C_{\mathrm{eff}} T_{b0}}{C_d + C_{\mathrm{eff}}}, \qquad T_d(t) = T_\infty + (T_{d0} - T_\infty)e^{-t/\tau_c}, \qquad T_b(t) = T_\infty - (T_\infty - T_{b0})e^{-t/\tau_c}.$$

The drink reaches T^* only if $T_\infty < T^*$, after $t_{T^*} = \tau_c \ln\left[(T_{d0} - T_\infty)/(T^* - T_\infty)\right]$.

14.3 With the box leak. With T_{env} constant, the two equations of 13.4 are linear in $\mathbf{x} = (T_d, T_b)^{\mathsf{T}}$: $\dot{\mathbf{x}} = \mathsf{A}\mathbf{x} + \mathbf{f}$, with $\mathbf{f} = (0,\ UA_{\text{box}}T_{\text{env}}/C_{\text{eff}})^{\mathsf{T}}$ and

$$\mathsf{A} = \begin{pmatrix} -\dfrac{UA}{C_d} & \dfrac{UA}{C_d} \\[2ex] \dfrac{UA}{C_{\text{eff}}} & -\dfrac{UA + UA_{\text{box}}}{C_{\text{eff}}} \end{pmatrix}.$$

The solution is a steady state plus two decaying exponentials, $e^{\lambda_\pm t}$, where $\lambda_\pm$ are the eigenvalues of A, the roots of $\lambda^2 - (\operatorname{tr}\mathsf{A})\lambda + \det\mathsf{A} = 0$:

$$\lambda_\pm = \tfrac{1}{2}\Big[\operatorname{tr}\mathsf{A} \pm \sqrt{(\operatorname{tr}\mathsf{A})^2 - 4\det\mathsf{A}}\Big],$$
$$\operatorname{tr}\mathsf{A} = -\Big(\frac{UA}{C_d} + \frac{UA + UA_{\text{box}}}{C_{\text{eff}}}\Big), \qquad \det\mathsf{A} = \frac{UA \cdot UA_{\text{box}}}{C_d\, C_{\text{eff}}}.$$

The trace is negative and the determinant positive, so both eigenvalues are real and negative. When the leak is small, the fast one is nearly $\operatorname{tr}\mathsf{A}$: the drink and bath coming together as in 14.2. The slow one is nearly $\det\mathsf{A}/\operatorname{tr}\mathsf{A}$: the room warming both. The steady state is $T_d = T_b = T_{\text{env}}$, reached only after the ice is gone — by which time C_{eff} has changed, so this solution holds only within one regime.

14.4 Fixed bath, convection-limited drink. Result 5: $\theta = \theta_0(1 + t/t^*)^{-4}$, $t^* = 4\tau_0$.

What it means. Each special case isolates one physical effect: the thermostat (14.1), the bath warming to meet the drink (14.2), and the room warming both (14.3). A full run strings them together as the bath moves from regime to regime.

Unified Salt–Ice–Water Model, with the Can

Full equation compendium and step-by-step derivations: second edition

Styrofoam heat transfer, ice melting, NaCl dissolution, and a drink can that may be part-filled and may contain ethanol

SYPT modelling notes

29 October 2026

Abstract

This second edition expands every step of the first (26 October 2026) and adds the can of drink that the bath is there to cool. Each equation is derived from the ones before it, line by line, with the assumption it needs, a check, and a worked number. Steps 1–7 are the bath, as before: the first law, the heat that leaks in through the Styrofoam, the ice, the water, the heat of dissolving salt, the freezing point and the rate at which salt dissolves. Steps 8–10 are new: the drink and its can as a second body that gives heat to the bath; a can that is not completely full, standing or lying, held under or floating; and an ethanol model of the drink, which fixes its heat capacity and how cold it can get before it starts to turn to slush. Steps 11 and 12 rebuild the time-dependent and equilibrium models with the can in them. As in the first edition, equations that need qualification or correction are labelled; they should not be used as exact identities.

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What is new in this edition, and how to read it

The first edition modelled the bath alone: ice, brine and salt in a Styrofoam box. A bath is only there to cool something, so this edition adds the can and its drink, and expands every step.

- **The can** (Steps 8–10). The drink and its can are a second body, with its own temperature T_d, that gives heat to the bath through the can wall. Step 8 derives that heat flow. Step 9 works out which part of the wall passes it when the can is not full, stands or lies, is held under or floats. Step 10 models the drink as a solution of ethanol, sugar and carbon dioxide in water, which fixes its heat capacity and the temperature at which it starts to freeze.
- **The coupled models** (Steps 11–12). The time-dependent model and the equilibrium of the first edition are rebuilt with the drink's heat in them, and solved for the rig.
- **Every step expanded.** Each step says what it does and what it assumes, derives each equation from the one before, checks it, and works a number.
- **New corrections**, labelled like those of the first edition: the Styrofoam's air films (Step 2); the salt's own heat capacity (Step 5); the error of the dilute freezing law in strong brine, and the temperature at which the model's bath stops cooling (Step 6); and starting temperatures other than 0 °C in the equilibrium (Step 12).

How each step is laid out. The idea says in words what the step does. **Assumptions** lists what it takes for granted. **Derivation** goes line by line: every equation follows from the ones before it, and the text says how. Key results are boxed. **Check** tests a result against its units or a limiting case. A shaded box works a number. **Where it fails** says when not to use the result.

Two sets of numbers. The first edition illustrated its equations with a small demonstration bath. A can needs a bath big enough to cool it, so the new steps use the rig of the B8 experiment with Can 2. Both sets are in Table 1; each worked example says which it uses.

Table 1: The two sets of numbers used in the worked examples.

	Demonstration (first edition)	Rig, with Can 2
Box	$k = 0.033\,\mathrm{W\,m^{-1}K^{-1}}$, $A = 0.05\,\mathrm{m^2}$, $d = 0.02\,\mathrm{m}$, so $H = 0.0825\,\mathrm{W\,K^{-1}}$	30 mm of expanded polystyrene: $H \approx 0.46\,\mathrm{W\,K^{-1}}$ for the foam alone
Room	25 °C	28 °C
Ice	0.200 kg at 0 °C	4.0 kg at 0 °C
Water	0.100 kg at 0 °C	1.6 kg at 4 °C
Salt	0.050 kg	1.0 kg at 28 °C, grains 0.5 mm across
Drink	—	330 g at 28 °C: a cola, or a 5 % beer (Step 10)
Can	—	aluminium, 13 g; diameter 6.6 cm, height 11.66 cm; headspace 1.2 cm when upright
Films	—	$h_i = 400$, $h_o = 270\,\mathrm{W\,m^{-2}K^{-1}}$ upright; both $\times 1.2$ lying (Step 8)

1 System, notation and assumptions

A closed Styrofoam box holds a container of ice, liquid water (brine, once salt dissolves), solid NaCl and dissolved NaCl: the *bath*. A sealed can of drink sits in the bath. The room, at T_{room}, leaks heat into the bath through the box. The drink, at T_d, gives heat to the bath through the can wall. The bath, at T, absorbs both by warming, by melting ice and by dissolving salt.

Let $m_i(t)$, $m_w(t)$ and $m_s(t)$ be the masses of ice, liquid water and *dissolved* NaCl (kg). Let $m_{s,0}$ be the solid salt added at the start and $m_{s,d,0}$ any salt dissolved at the start. Then $M_w = m_{i,0} + m_{w,0}$

is all the water and $S_0 = m_{s,0} + m_{s,d,0}$ all the salt. Temperatures in sensible-heat terms are in °C (a temperature difference is the same in K); absolute temperature appears only in Step 6, written $\mathcal{T}$.

Assumptions about the bath.

B1 One bath temperature. One temperature T describes the whole bath. With no stirrer this is an idealisation: brine next to the can warms first (the B8 report models that spreading).

B2 Ice and brine in equilibrium. Ice melts or freezes fast enough that, while both are present, the bath sits at the freezing point of its brine (Step 6).

B3 Salt dissolves at a finite rate (Step 7), so the dissolved salt is a variable of its own.

B4 Constant properties. Specific heats, the latent heat and the heat of solution are constants. The brine's heat capacity is that of its water.

B5 The box is a steady conductor (Step 2): it stores no heat and passes heat in proportion to the temperature difference.

B6 Closed and still. The lid is on, nothing evaporates, nothing is added during a run, and nothing does work on the bath.

Assumptions about the can and drink.

C1 One drink temperature T_d: free convection keeps the drink mixed (Step 8).

C2 Steady wall and films: the can's wall and the liquid films on it store no heat.

C3 A sealed can: the drink's mass is fixed, and the gas in the headspace holds negligible heat.

C4 Heat passes only where the wall has drink inside and brine outside (Step 9).

C5 The drink's properties follow from its composition (Step 10).

Table 2: Notation. Numbers are the constants used throughout; the rest is set by the experiment.

Symbol	Meaning	Value or unit
The bath (first edition)		
k	Styrofoam conductivity	$\mathrm{W\,m^{-1}K^{-1}}$
$A,\ d$	area and thickness of the insulating layer	$\mathrm{m^2}$, m
$H = kA/d$	conductance of the box	$\mathrm{W\,K^{-1}}$
$T,\ T_{\mathrm{room}}$	bath and room temperatures	°C
$m_i,\ m_w,\ m_s$	ice, liquid water, dissolved NaCl	kg
$M_w,\ S_0$	all the water, all the salt	kg
$W = M_w - m_i$	liquid water, equal to m_w	kg
$c_i,\ c_w$	specific heats of ice and liquid water	2100, $4180\,\mathrm{J\,kg^{-1}K^{-1}}$
c_s	specific heat of solid NaCl	$864\,\mathrm{J\,kg^{-1}K^{-1}}$
L_f	latent heat of melting ice at 0 °C	$333\,500\,\mathrm{J\,kg^{-1}}$
M_{NaCl}	molar mass of NaCl	$0.05844\,\mathrm{kg\,mol^{-1}}$
ΔH_{sol}	molar heat of solution of NaCl	$3900\,\mathrm{J\,mol^{-1}}$
$\beta = \Delta H_{\mathrm{sol}}/M_{\mathrm{NaCl}}$	heat of solution per kg	$66\,735\,\mathrm{J\,kg^{-1}}$
K_f	cryoscopic constant of water	$1.86\,\mathrm{K\,kg\,mol^{-1}}$
i_v	van 't Hoff factor: ions per formula unit (named i_v to avoid the ice index)	2
$\alpha = i_v K_f/M_{\mathrm{NaCl}}$	freezing-point coefficient	63.65 K
b	molality: moles of solute per kg of liquid water	$\mathrm{mol\,kg^{-1}}$
k_m	dissolution mass-transfer coefficient	$\mathrm{m\,s^{-1}}$
A_s	exposed surface area of the salt crystals	$\mathrm{m^2}$
$C,\ C_{\mathrm{sat}}$	salt per volume of water; the same at saturation	$\mathrm{kg\,m^{-3}}$; $C_{\mathrm{sat}} \approx 357$
ρ_w	density of water	$1000\,\mathrm{kg\,m^{-3}}$

Symbol	Meaning	Value or unit
$C_{\text{eff}} = m_i c_i + W c_w$	heat capacity of the ice and water	$\mathrm{J\,K^{-1}}$
The can and drink (new)		
T_d	drink temperature	°C
$m_d,\ c_d$	drink mass; drink specific heat (Step 10)	kg; $\mathrm{J\,kg^{-1}K^{-1}}$
$m_c,\ c_c$	can mass; specific heat of aluminium	kg; $900\,\mathrm{J\,kg^{-1}K^{-1}}$
$C_d = m_d c_d + m_c c_c$	heat capacity of the drink and can	$\mathrm{J\,K^{-1}}$
$r,\ \ell$	radius and height of the can	m
$t_w,\ k_w$	wall thickness; conductivity of can aluminium	m; $160\,\mathrm{W\,m^{-1}K^{-1}}$
$h_i,\ h_o$	drink-side and brine-side film coefficients	$\mathrm{W\,m^{-2}K^{-1}}$
h_{ov}	overall coefficient through both films and the wall	$\mathrm{W\,m^{-2}K^{-1}}$
A_c	area of wall with drink inside and brine outside	$\mathrm{m^2}$
$G = h_{\text{ov}} A_c$	conductance from the drink to the bath	$\mathrm{W\,K^{-1}}$
$\dot{Q}_d = G(T_d - T)$	heat flow from the drink to the bath	W
h_g	headspace height, can upright	m
f	fill fraction: share of the can's volume that is drink	—
$z_d,\ z_b$	upright can: drink height; depth of brine on the can	m
$y_d,\ y_g,\ y_b$	lying can: drink depth; gas-strip depth; depth of brine on the can	m
$A_{\text{seg}}(y),\ P(y)$	area and arc length of a circular segment of depth y	$\mathrm{m^2}$, m
$\rho_d,\ \rho_b$	densities of the drink and of strong brine	$\mathrm{kg\,m^{-3}}$; $\rho_b \approx 1182$
The drink as a solution (new)		
ABV	alcohol by volume	—
$w_e,\ w_{su},\ w_c,\ w_x$	mass fractions of ethanol, sugar, CO_2 and other extract in the drink	—
$w_{wd},\ m_{wd}$	mass fraction and mass of water in the drink	—, kg
$M_e,\ M_{su},\ M_c$	molar masses: ethanol, sucrose, CO_2	0.04607, 0.3423, $0.04401\,\mathrm{kg\,mol^{-1}}$
ρ_e	density of ethanol at 20 °C	$789.2\,\mathrm{kg\,m^{-3}}$
T_{fd}	temperature at which the drink starts to freeze	°C
ψ	share of the drink's water that is frozen	—
$C_{d,\text{eff}}$	effective heat capacity of a freezing drink	$\mathrm{J\,K^{-1}}$

2 Step 1: energy in = energy out

The idea. Energy is not created or destroyed. Heat that enters the bath either stays in it, as a change of its energy, or leaves again. The can adds a second body to keep account of.

The first edition's framing.

$$E_{\text{in}} = E_{\text{out}}, \tag{1}$$

$$E_{\text{in}} - E_{\text{out}} = \Delta U, \qquad U_{\text{initial}} = U_{\text{final}} \ \text{ if insulated and no work.} \tag{2}$$

Derivation. The first law of thermodynamics for a *closed* system, one that no mass crosses, says that its internal energy U changes by the heat Q that enters it minus the work W_{by} it does on its surroundings: $\Delta U = Q - W_{\text{by}}$. Writing the net heat as what comes in minus what goes out, $Q = E_{\text{in}} - E_{\text{out}}$, and taking $W_{\text{by}} = 0$ gives (2).

Why the work is zero: the bath does no work except expanding against the atmosphere, and for liquids and solids that is tiny. Melting 1 kg of ice shrinks it by $1/917 - 1/1000 = 9.1 \times 10^{-5}\,\mathrm{m^3}$; at atmospheric pressure that is $p\,\Delta V \approx 9\,\mathrm{J}$, against the 333 500 J the melting takes. So here the

internal energy U and the enthalpy $H = U + pV$ change by the same amount, and either may be used.

(1) is the special case $\Delta U = 0$. It holds only when nothing accumulates, for example in a steady state. While the bath cools or melts, its energy changes, and (2) is the correct statement. Dividing by a short time Δt and letting it shrink turns energies into rates:

$$\frac{\mathrm{d}U}{\mathrm{d}t} = \dot{Q}_{\mathrm{in}}, \tag{3}$$

with $\dot{Q}_{\mathrm{in}}$ the net heat flow into the system, in watts.

Two bodies (new). With the can there are two closed systems. The bath B receives heat through the box, $\dot{Q}_{\mathrm{box}}$, and from the drink, $\dot{Q}_d$; the drink and can D lose $\dot{Q}_d$. Applying (3) to each, and adding:

$$\frac{\mathrm{d}U_B}{\mathrm{d}t} = \dot{Q}_{\mathrm{box}} + \dot{Q}_d, \tag{4}$$

$$\frac{\mathrm{d}U_D}{\mathrm{d}t} = -\dot{Q}_d, \tag{5}$$

$$\frac{\mathrm{d}(U_B + U_D)}{\mathrm{d}t} = \dot{Q}_{\mathrm{box}}. \tag{6}$$

$\dot{Q}_d$ is counted positive from the drink to the bath, so it leaves one body and enters the other, and it cancels in the sum: only the box lets energy into the pair.

Check. Every term is in watts. With a perfect box, $\dot{Q}_{\mathrm{box}} = 0$, (6) says $U_B + U_D$ stays constant: what the drink loses, the bath gains.

Where it fails. If mass crosses the boundary — the lid is lifted to add ice, or water evaporates — the systems are open, and the energy that mass carries must be added. A stirrer would add its work; there is none here.

3 Step 2: heat entering through Styrofoam

The idea. Heat flows from the warm room to the cold bath through the foam. Foam conducts poorly, so the flow is small, and it is proportional to the temperature difference.

Assumptions. The foam wall is thin compared with its width, so heat flows straight through it (one dimension). The foam stores no heat, so the flow is steady (B5). Its conductivity k is uniform. And, as in the first edition, its faces are at T_{room} and T.

Derivation. Fourier's law says that heat flows down a temperature gradient in proportion to it. Take a coordinate x through the wall from the bath side ($x = 0$) to the room side ($x = d$); the heat flux toward the bath is $q = k\,\mathrm{d}T/\mathrm{d}x$ ($\mathrm{W\,m^{-2}}$). If the foam stores no heat, whatever enters a slice of it leaves the other side, so q is the same at every x. Then $\mathrm{d}T/\mathrm{d}x$ is constant, and the temperature rises in a straight line from T to T_{room}:

$$\frac{\mathrm{d}T}{\mathrm{d}x} = \frac{T_{\mathrm{room}} - T}{d}, \qquad q = \frac{k}{d}\,(T_{\mathrm{room}} - T).$$

Over an area A the heat flow is qA. Adding it up over time, naming the conductance, and using

(3) for the bath alone:

$$\dot{Q}_{\text{Styrofoam}}(t) = \frac{kA}{d}\,(T_{\text{room}} - T(t)), \tag{7}$$

$$Q_{\text{in}}(t) = \int_0^t \frac{kA}{d}\,[T_{\text{room}} - T(\tau)]\,\mathrm{d}\tau, \tag{8}$$

$$H = \frac{kA}{d}, \tag{9}$$

$$\frac{\mathrm{d}U}{\mathrm{d}t} = H\,(T_{\text{room}} - T(t)). \tag{10}$$

H, in $\mathrm{W\,K^{-1}}$, is the heat flow for each kelvin of difference. Walls, base and lid of different thickness d_j and area A_j are separate paths side by side, so their flows add:

$$\dot{Q}_{\text{in}} = \sum_j \frac{k_j A_j}{d_j}\,(T_{\text{room}} - T). \tag{11}$$

Air films (correction). The foam's faces are not at T_{room} and T: still air on each side adds a film resistance $1/(hA)$ in series with the foam's $d/(kA)$. Resistances in series add, because the temperature drops add while the heat flow is the same through each:

$$\frac{1}{H_j} = \frac{1}{h_{\text{out}} A_j} + \frac{d_j}{k_j A_j} + \frac{1}{h_{\text{in}} A_j}. \tag{12}$$

Still room air gives $h_{\text{out}} \approx 9\,\mathrm{W\,m^{-2}K^{-1}}$ (about 5 by convection and 4 by radiation), and the air gap inside the box about 6. For 2 cm of foam with $k = 0.033$, $d/k = 0.61\,\mathrm{m^2K\,W^{-1}}$ while the films add $1/9 + 1/6 = 0.28$: the films cut the leak by about 30 %, and by 25 % for the rig's 3 cm.

Is the foam steady? A temperature change crosses a layer of thickness d in a time of order d^2/κ_f, where $\kappa_f = k/(\rho_f c_f)$ is the foam's thermal diffusivity. For expanded polystyrene, $\kappa_f = 0.035/(20 \times 1300) = 1.35 \times 10^{-6}\,\mathrm{m^2\,s^{-1}}$: 300 s for 2 cm, 670 s for 3 cm. After the first ten minutes or so the foam is steady. Before that it is still giving up heat it held at room temperature, which (7) leaves out.

Worked example: the leak

Demonstration: $H = 0.033 \times 0.05/0.02 = 0.0825\,\mathrm{W\,K^{-1}}$; with the bath at −10 °C and the room at 25 °C, $\dot{Q} = 0.0825 \times 35 = 2.9\,\mathrm{W}$. Rig: 30 mm of foam gives about $0.46\,\mathrm{W\,K^{-1}}$ alone and about 0.34 with its films and the gaps round the lid; with the bath at −21 °C and the room at 28 °C, 17 to 23 W.

Where it fails. Corners conduct more than flat walls, gaps round the lid let heat bypass the foam, and the foam's own heat matters early on. Measuring H directly, with the box test of the B8 README, is more reliable than computing it.

4 Step 3: the ice and the latent heat

The idea. Heat reaching the bath either warms what is there (sensible heat) or melts ice at constant temperature (latent heat).

Derivation. The specific heat c is the heat that warms 1 kg by 1 K, so warming a mass m_i of ice by $\mathrm{d}T$ takes $m_i c_i\,\mathrm{d}T$; dividing by $\mathrm{d}t$ gives the rate. The latent heat L_f is the heat that melts 1 kg of ice without changing its temperature, so melting $\mathrm{d}m_{\text{melt}}$ takes $L_f\,\mathrm{d}m_{\text{melt}}$. Ice that melts is ice

lost, $\dot{m}_{\text{melt}} = -\dot{m}_i$:

$$\mathrm{d}Q_i = m_i c_i \,\mathrm{d}T, \tag{13}$$
$$\dot{Q}_{i,\text{sensible}} = m_i c_i \,\dot{T}, \tag{14}$$
$$\mathrm{d}Q_{\text{melt}} = L_f \,\mathrm{d}m_{\text{melt}}, \tag{15}$$
$$\dot{Q}_{\text{melt}} = L_f \,\dot{m}_{\text{melt}}, \tag{16}$$
$$\dot{m}_{\text{melt}} = -\dot{m}_i, \tag{17}$$
$$\dot{Q}_{\text{melt}} = -L_f \,\dot{m}_i. \tag{18}$$

The heat entering through the box warms the ice, warms the liquid water ($m_w c_w \dot{T}$, by the same reasoning) and melts ice. Adding the three gives the first edition's combined balance:

$$H(T_{\text{room}} - T) = (m_i c_i + m_w c_w)\,\dot{T} - L_f \,\dot{m}_i. \tag{19}$$

When ice melts, $\dot{m}_i < 0$, so $-L_f \dot{m}_i$ is positive: melting absorbs heat.

Check. Pure ice and water coexist only at 0 °C, so there $\dot{T} = 0$ and (19) gives $\dot{m}_i = -H(T_{\text{room}} - T)/L_f$. Demonstration: $0.0825 \times 25/333\,500 = 6.2 \times 10^{-6}\,\mathrm{kg\,s^{-1}}$, 22 g of ice an hour.

Where it fails. L_f is the latent heat at 0 °C. Ice melting at −15 °C takes less, by $c_w - c_i$ for each kelvin below zero (Section 14). Ice straight from a freezer at −18 °C must first be warmed: $2100 \times 18 = 37.8\,\mathrm{kJ\,kg^{-1}}$, 11 % of L_f.

5 Step 4: liquid water and mass conservation

The idea. Water only changes form, ice to liquid or back, so ice and liquid always add up to the same total.

Derivation. The liquid's sensible heat follows exactly as the ice's:

$$Q_w = m_w c_w \,\Delta T, \tag{20}$$
$$\dot{Q}_{w,\text{sensible}} = m_w c_w \,\dot{T}. \tag{21}$$

If melting and freezing are the only changes of form, and no water enters or leaves (B6), every kilogram of ice lost is a kilogram of liquid gained. Integrating that from the start:

$$\mathrm{d}m_w = -\mathrm{d}m_i, \tag{22}$$
$$\dot{m}_w = -\dot{m}_i, \tag{23}$$
$$m_w(t) + m_i(t) = m_{w,0} + m_{i,0} = M_w, \tag{24}$$
$$m_w(t) = M_w - m_i(t). \tag{25}$$

Substituting (25) into (19) leaves the ice as the only mass:

$$H(T_{\text{room}} - T) = \left[m_i c_i + (M_w - m_i)\,c_w\right]\dot{T} - L_f \,\dot{m}_i. \tag{26}$$

Where it fails. Dissolved salt makes the liquid heavier than its water but is given no heat capacity here (B4): the brine's heat capacity is taken as that of its water.

6 Step 5: enthalpy of dissolution of NaCl

The idea. Dissolving salt takes heat: pulling the crystal apart costs more energy than the water molecules give back by gathering round the ions. A bath cools itself a little by dissolving its salt.

Derivation. A mass m_s of dissolved salt is $n_s = m_s/M_{\mathrm{NaCl}}$ moles. The heat of solution ΔH_{sol} is measured per mole, so dissolving at a rate $\dot{n}_s$ absorbs $\Delta H_{\mathrm{sol}}\,\dot{n}_s$:

$$n_s = \frac{m_s}{M_{\mathrm{NaCl}}}, \tag{27}$$

$$\dot{Q}_{\mathrm{sol}} = \Delta H_{\mathrm{sol}}\,\dot{n}_s = \frac{\Delta H_{\mathrm{sol}}}{M_{\mathrm{NaCl}}}\,\dot{m}_s = \beta\,\dot{m}_s, \tag{28}$$

$$\beta = \frac{3900}{0.05844} = 66\,735\ \mathrm{J\,kg^{-1}}.$$

Dissolving is endothermic, $\beta > 0$: a positive $\dot{m}_s$ takes heat, so it joins the other heat sinks on the right of (19):

$$\boxed{H(T_{\mathrm{room}} - T) = (m_i c_i + m_w c_w)\,\dot{T} - L_f\,\dot{m}_i + \frac{\Delta H_{\mathrm{sol}}}{M_{\mathrm{NaCl}}}\,\dot{m}_s} \tag{29}$$

or, eliminating m_w with (25),

$$H(T_{\mathrm{room}} - T) = \left[m_i c_i + (M_w - m_i)\,c_w\right]\dot{T} - L_f\,\dot{m}_i + \beta\,\dot{m}_s. \tag{30}$$

Worked example: dissolving

Dissolving 1 kg of salt absorbs 66.7 kJ, as much as melting 0.20 kg of ice. The demonstration's 50 g absorbs 3.3 kJ, the heat of melting 10 g of ice.

The salt's own heat capacity (correction). Undissolved salt also warms and cools with the bath, at $c_s = 864\,\mathrm{J\,kg^{-1}K^{-1}}$, and (29) leaves it out. To include it, add $(S_0 - m_s)\,c_s$ to the heat capacity in front of $\dot{T}$. For the rig, 1 kg of salt adds $864\,\mathrm{J\,K^{-1}}$ to the $15\,\mathrm{kJ\,K^{-1}}$ of its ice and water: 6 %. Steps 6–11 follow the first edition and leave it out; the equilibrium of Step 12 includes it, with the heat the salt brings in from the room.

Where it fails. ΔH_{sol} is measured in dilute solution at 25 °C, and it changes with concentration and temperature: one number is an approximation.

7 Step 6: freezing-point depression $i_v K_f b$

The idea. Ice melts in brine below 0 °C because the dissolved ions dilute the water: ice and brine can only coexist at a lower temperature than ice and pure water. While both are present the bath sits at that temperature (B2), so the salt in the brine fixes the bath temperature.

Where the law comes from. Ice and liquid water exchange molecules until water has the same *chemical potential* — Gibbs energy per mole — in both. Ice is pure water. In the brine the ions lower water's chemical potential below that of pure water, μ_w^*, by $R\mathcal{T}\ln a_w$, where $a_w < 1$ is the water's activity, $R = 8.314\,\mathrm{J\,mol^{-1}K^{-1}}$ and $\mathcal{T}$ the absolute temperature:

$$\mu_{\mathrm{ice}}(\mathcal{T}) = \mu_w^*(\mathcal{T}) + R\mathcal{T}\ln a_w.$$

At $\mathcal{T}_0 = 273.15\,\mathrm{K}$ pure ice and water coexist. The Gibbs–Helmholtz relation, with the molar heat of melting $\Delta h_{\mathrm{fus}} = L_f M_{\mathrm{H_2O}} = 333\,500 \times 0.018015 = 6008\,\mathrm{J\,mol^{-1}}$, turns the condition into

$$\ln a_w = -\frac{\Delta h_{\mathrm{fus}}}{R}\left(\frac{1}{\mathcal{T}} - \frac{1}{\mathcal{T}_0}\right).$$

In dilute brine, $\ln a_w$ is minus the moles of dissolved particles per mole of water. Each formula unit of NaCl gives $i_v = 2$ ions, and b moles of NaCl are dissolved in each kilogram of water,

which is $1/M_{\mathrm{H_2O}}$ moles of water, so $\ln a_w \approx -i_v\, b\, M_{\mathrm{H_2O}}$. For a small drop $\Delta T_f = \mathcal{T}_0 - \mathcal{T}$, $1/\mathcal{T} - 1/\mathcal{T}_0 = \Delta T_f/(\mathcal{T}\mathcal{T}_0) \approx \Delta T_f/\mathcal{T}_0^2$. Equating the two expressions for $\ln a_w$:

$$\Delta T_f = \underbrace{\frac{R\mathcal{T}_0^2 M_{\mathrm{H_2O}}}{\Delta h_{\mathrm{fus}}}}_{K_f}\, i_v\, b, \tag{31}$$

$$K_f = \frac{8.314 \times 273.15^2 \times 0.018015}{6008} = 1.86\ \mathrm{K\,kg\,mol^{-1}}.$$

The ions are not quite independent: the osmotic coefficient, about 0.93 between 0.1 and 1 $\mathrm{mol\,kg^{-1}}$, corrects for that, and in strong brine the logarithm can no longer be expanded.

Derivation. With b in moles of NaCl per kilogram of *liquid water*:

$$\Delta T_f = i_v K_f b, \tag{32}$$

$$b(t) = \frac{m_s(t)/M_{\mathrm{NaCl}}}{m_w(t)}, \tag{33}$$

$$T_f(t) = -i_v K_f b(t) = -\frac{i_v K_f\, m_s(t)}{M_{\mathrm{NaCl}}\, m_w(t)}, \tag{34}$$

$$T(t) = T_f(t) \quad \text{only when ice and brine coexist in equilibrium.} \tag{35}$$

(33) is moles of salt over kilograms of water. (34) is the freezing point in °C, minus the drop. (35): the bath sits at the freezing point only while there is ice to melt or freeze. Substituting the mass balance (25) and the shortcut $\alpha = i_v K_f/M_{\mathrm{NaCl}} = 63.65\,\mathrm{K}$:

$$\boxed{T = -\frac{i_v K_f\, m_s}{M_{\mathrm{NaCl}}\,(M_w - m_i)} = -\alpha\,\frac{m_s}{M_w - m_i}} \tag{36}$$

The bath gets colder as salt dissolves, and warmer as ice melts and dilutes the brine.

Worked example: the first edition's example, and how accurate it is

0.005 kg of salt dissolved in 0.100 kg of water:

$$T_f = -2 \times 1.86 \times \frac{0.005/0.05844}{0.100} = -3.18\,^{\circ}\mathrm{C}. \tag{37}$$

The measured freezing curve (Bodnar 1993) gives −2.87 °C for this brine, 4.76 wt % salt: the dilute law is 11 % too cold. In strong brine the error changes sign:

Salt in the brine (wt %)	5	10	15	16.5	20	23.2
Dilute law, ΔT_f (K)	3.35	7.07	11.23	12.58	15.91	19.21
Measured (Bodnar), ΔT_f (K)	3.03	6.61	11.03	12.57	16.71	21.20

How much salt the brine can hold. Brine holds at most C_{sat} of salt; once it is saturated, extra salt stays solid. With ice and saturated brine together, (36) with $m_s/W = C_{\mathrm{sat}}/\rho_w$ gives the coldest the model's bath can get:

$$T_{E,\mathrm{model}} = -\alpha\,\frac{C_{\mathrm{sat}}}{\rho_w}. \tag{38}$$

With halite's solubility, 357 kg per $\mathrm{m^3}$ of water (35.7 g per 100 g), this is −22.7 °C. The real coldest point, the eutectic, is −21.2 °C at 23.2 wt % (0.302 kg of salt per kg of water), where the dilute law gives −19.2 °C.

Correction. The dilute law (32) is good to about 10 % up to 15 wt %, but at the eutectic composition it is 2 K too warm, and with halite's solubility it puts the bath's floor at −22.7 °C instead of −21.2 °C. For strong brine, replace it with the measured freezing curve, $W = 1.78\,\vartheta - 0.0442\,\vartheta^2 + 5.572\times10^{-4}\,\vartheta^3$ (wt % of salt against ϑ degrees below zero; Bodnar 1993), and the measured eutectic.

8 Step 7: salt dissolution rate and shrinking crystals

The idea. Salt dissolves from the surface of each grain. Right at the surface the brine is saturated; further out it is weaker, and dissolved salt diffuses across the thin layer between. The rate is the area of the grains times how fast salt crosses that layer.

Derivation. Fick's law: salt diffuses down its concentration gradient at a flux D times the gradient, with D the diffusion coefficient. Across a layer of thickness δ, from saturated brine at the surface (C_{sat}) to the bulk (C), the flux is $D(C_{\mathrm{sat}} - C)/\delta$ in $\mathrm{kg\,m^{-2}s^{-1}}$. Calling $k_m = D/\delta$ the mass-transfer coefficient and multiplying by the exposed area A_s:

$$\dot m_s = k_m A_s\,(C_{\mathrm{sat}} - C), \tag{39}$$

$$C(t) \approx \frac{\rho_w\, m_s(t)}{m_w(t)}, \tag{40}$$

$$\dot m_s \approx k_m A_s\Big[C_{\mathrm{sat}}(T) - \frac{\rho_w m_s}{m_w}\Big]. \tag{41}$$

(40) measures the concentration as salt per volume of water, $m_s/(m_w/\rho_w)$, close to salt per volume of brine while the brine is dilute. For NaCl, $D \approx 1.5\times10^{-9}\,\mathrm{m^2s^{-1}}$; a layer 0.1 mm thick, as when brine flows past the grains, gives $k_m \approx 10^{-5}\,\mathrm{m\,s^{-1}}$, and a still layer ten times thicker 10^{-6}.

Shrinking crystals. Take N grains of the same shape, each a cube of side a. Their mass is $m_{\mathrm{solid}} = N\rho_{\mathrm{salt}}a^3$ and their area $6Na^2$. Eliminating a, the area is $6N^{1/3}(m_{\mathrm{solid}}/\rho_{\mathrm{salt}})^{2/3}$: for a fixed number of similar grains it goes as the solid mass to the power $\frac{2}{3}$, whatever their shape. With no salt dissolved at the start:

$$m_{\mathrm{solid}}(t) = m_{s,0} - m_s(t), \tag{42}$$

$$A_s(t) = A_{s,0}\Big(\frac{m_{s,0} - m_s(t)}{m_{s,0}}\Big)^{2/3}, \tag{43}$$

so

$$\boxed{\dot m_s = k_m A_{s,0}\Big(\frac{m_{s,0} - m_s}{m_{s,0}}\Big)^{2/3}\Big[C_{\mathrm{sat}}(T) - \frac{\rho_w m_s}{m_w}\Big]} \tag{44}$$

Worked example: how fast the rig's salt dissolves

1 kg of grains 0.5 mm across is 3.7×10^6 cubes, with $A_{s,0} = 5.5\,\mathrm{m^2}$. In fresh water, with $C_{\mathrm{sat}} = 357\,\mathrm{kg\,m^{-3}}$, the starting rate $k_m A_{s,0} C_{\mathrm{sat}}$ is $20\,\mathrm{g\,s^{-1}}$ for $k_m = 10^{-5}\,\mathrm{m\,s^{-1}}$ (the lot in about a minute, were the rate to stay up) and $2\,\mathrm{g\,s^{-1}}$ for 10^{-6} (about eight minutes). Both slow as the grains shrink and the brine strengthens.

Where it fails (from the first edition). It needs $0 \le m_s < m_{s,0}$: undissolved salt must remain, and the brine must be undersaturated. An extension that forbids precipitation uses $\max(0, C_{\mathrm{sat}} - C)$ and stops at complete dissolution. With salt dissolved at the start, use $S_0 - m_s$ in place of $m_{s,0} - m_s$. For concentrated brine use $C = m_s/V_{\mathrm{brine}}$ with the measured brine density. k_m and $A_{s,0}$ depend on grain size, on how brine moves through the ice and on whether a crust forms, and are best fitted to a run.

9 Step 8: the can and its drink (new)

The idea. The drink and its can are a second body in the bath. It has one temperature T_d and a heat capacity C_d, and it gives heat to the bath through the can wall at a rate proportional to $T_d - T$.

9.1 The drink as one body

Derivation. Apply (5) to the drink and can. With one temperature (C1) and constant specific heats, their energy measured from a reference temperature is $U_D = (m_d c_d + m_c c_c)(T_d - T_{\mathrm{ref}})$; the reference drops out on differentiating:

$$C_d \frac{\mathrm{d}T_d}{\mathrm{d}t} = -\dot{Q}_d, \qquad C_d = m_d c_d + m_c c_c. \tag{45}$$

The gas in the headspace is left out (C3). A 41 mL headspace of CO_2 at two atmospheres is $pV/(R\mathcal{T}) \approx 0.0035\,\mathrm{mol}$, 0.15 g, holding $0.13\,\mathrm{J\,K^{-1}}$: a ten-thousandth of C_d.

Worked example: Can 2

For a sugary soft drink, $C_d = 0.330 \times 4000 + 0.013 \times 900 = 1320 + 12 = 1332\,\mathrm{J\,K^{-1}}$. The can adds 1 %.

9.2 The heat path through the wall

Derivation. Heat leaves the drink through three layers in series: the film of drink on the inside of the wall, the aluminium, and the film of brine on the outside. By C2 none of them stores heat, so the same heat flow crosses each, driven by the temperature drop across it. Per unit area a film passes $h\,\Delta T$ — Newton's law of cooling, which defines the film coefficient h — and the wall $k_w \Delta T / t_w$, by Fourier's law as in Step 2:

$$\frac{\dot{Q}_d}{A_c} = h_i\,(T_d - T_{wi}) = \frac{k_w}{t_w}\,(T_{wi} - T_{wo}) = h_o\,(T_{wo} - T),$$

with T_{wi} and T_{wo} the wall's inner and outer faces and A_c the area that passes heat (Step 9). Rearranging each to a temperature drop and adding them, the wall temperatures cancel:

$$\dot{Q}_d = G\,(T_d - T), \qquad G = h_{\mathrm{ov}} A_c, \qquad \frac{1}{h_{\mathrm{ov}}} = \frac{1}{h_i} + \frac{t_w}{k_w} + \frac{1}{h_o}. \tag{46}$$

The aluminium hardly matters: $t_w/k_w = 10^{-4}/160 = 6 \times 10^{-7}\,\mathrm{m^2K\,W^{-1}}$, against $1/h_i = 2.5 \times 10^{-3}$ and $1/h_o = 3.7 \times 10^{-3}$. A plastic sleeve label would add its own resistance in series.

9.3 The film coefficients

Derivation. Nothing stirs the drink or the bath, so the films are set by free convection. Next to the wall the liquid is ΔT warmer or colder than the rest, so lighter or denser by $\rho \beta_T \Delta T$, with β_T its thermal expansion coefficient, and buoyancy drives it along the wall. Balancing buoyancy against viscosity, and the heat carried along the wall against the heat conducted across the layer, gives a layer thickness $\delta_T \sim \mathcal{L}\,\mathrm{Ra}^{-1/4}$, where $\mathcal{L}$ is the height of the wall and $\mathrm{Ra} = g\beta_T \Delta T \mathcal{L}^3/(\nu\kappa)$ the Rayleigh number (ν and κ are the liquid's viscosity and thermal diffusivity). The film coefficient is about k/δ_T, so

$$h \propto \left(\frac{\Delta T}{\mathcal{L}}\right)^{1/4}. \tag{47}$$

The constant comes from experiments. This document takes $h_i = 400$ and $h_o = 270\,\mathrm{W\,m^{-2}K^{-1}}$ for an upright can at a 25 K difference (a study of bottles, and a fit to a measured cooling time), and multiplies both by 1.2 for a can lying on its side, whose relevant length is its 6.6 cm diameter instead of its 11.7 cm height: $(11.66/6.6)^{1/4} = 1.15$ from (47) alone. Then $h_{\mathrm{ov}} = 161$ upright and 193 lying. In a part-filled upright can the drink's wall is only as tall as the drink, so h_i is a few per cent larger.

9.4 Is one drink temperature enough?

A still drink would not be uniform: its Biot number, $h_i(V/A)/k = 400 \times 0.0129/0.58 \approx 9$, is far above the 0.1 below which conduction alone keeps a body uniform. It is uniform enough because free convection stirs it: the cold layer on the wall is about 1 mm thick and sinks, and the core is mixed. Cold drink does collect at the bottom, so the core stratifies slowly; a probe at the top of the drink and one at the bottom test C1.

9.5 The fixed-bath solution

Derivation. If the bath stays at T, (45) with (46) is $C_d\dot{T}_d = -G(T_d - T)$: the difference $T_d - T$ falls at a rate proportional to itself. Separating the variables and integrating from $T_{d,0}$, then solving for the time to reach a target T^*:

$$T_d(t) = T + (T_{d,0} - T)\,e^{-t/\tau}, \qquad \tau = \frac{C_d}{G}, \qquad t_{T^*} = \tau \ln \frac{T_{d,0} - T}{T^* - T}. \tag{48}$$

Worked example: if the whole can passed heat

The whole surface of Can 2 is $A_c = 2\pi r\ell + 2\pi r^2 = 242 + 68 = 310\,\mathrm{cm^2}$. Upright, $G = 161 \times 0.0310 = 5.0\,\mathrm{W\,K^{-1}}$ and $\tau = 1332/5.0 = 266\,\mathrm{s}$; from 28 °C to 5 °C in a bath at −18 °C, $t = 266 \ln(46/23) = 185\,\mathrm{s}$. Step 9 shows why less of the wall passes heat.

What it does to the bath. The drink's heat enters the bath: $G(T_d - T)$ joins $H(T_{\mathrm{room}} - T)$ on the left of the bath balance (29) (Step 11).

10 Step 9: a can that is not completely full (new)

The idea. A sealed can holds gas above its drink. Heat crosses the wall well only where there is liquid on both sides: drink inside, brine outside. The gas also decides whether the can floats, and how much room the drink has to expand if it freezes.

10.1 What the headspace changes

- **The drink's mass** is whatever is in the can: weigh it full and empty. The headspace does not change C_d (Step 8).
- **The wall with gas behind it passes almost no heat.** Heat would have to reach it through the gas, which conducts twenty to thirty times worse than water and barely convects in so thin a layer: a film coefficient of a few $\mathrm{W\,m^{-2}K^{-1}}$, a hundred times less than h_i. So the area that passes heat, A_c, is the wall wetted by drink inside *and* by brine outside (C4).
- **Conduction along the aluminium** carries a little heat past the drink's surface. The wall acts as a fin whose reach is $\lambda = \sqrt{k_w t_w / h_o} = \sqrt{160 \times 10^{-4}/270} = 7.7\,\mathrm{mm}$: a strip about that wide along the drink's surface passes some heat, adding up to about 6 % to G for the upright can. It is left out, so the model slightly underestimates G.

10.2 Measuring the fill

Stand the can upright and measure the headspace height h_g, from the top of the cylinder to the drink's surface; or fill an empty can with water to find its volume and subtract the drink's volume m_d/ρ_d. Treating the can as a cylinder of radius r and height ℓ, the gas volume is $V_g = \pi r^2 h_g$ and the fill fraction $f = 1 - h_g/\ell$. A real can's neck and domed base hold less than $\pi r^2 \ell$: Can 2's cylinder would hold 399 mL, against 330 mL of drink and about 41 mL of headspace. Measure volumes rather than compute them.

10.3 Upright can

Derivation. The drink stands $z_d = \ell - h_g$ high. Inside, the side wall is wet up to z_d and the base all over; the lid has gas behind it. Outside, the brine reaches a depth z_b up the can ($z_b \geq \ell$ when the can is held under). The wall is wet on both sides up to the lower of the two heights, so the side wall passes heat up to that height, and the base passes heat everywhere:

$$A_c = 2\pi r \, \min(z_d, z_b) + \pi r^2. \tag{49}$$

10.4 Lying can: the circular segment

Derivation. Lying on its side, the can's cross-section is a circle of radius r, and the drink fills the part below a horizontal chord: a circular *segment*. A chord at height y above the lowest point is $r - y$ below the centre, so it subtends a half-angle θ at the centre with $\cos\theta = (r-y)/r$. The segment is the sector of angle 2θ, of area $r^2\theta$, minus the triangle between the chord and the centre, of area $(r-y)\,r\sin\theta = (r-y)\sqrt{2ry - y^2}$; and the arc below the chord is $2r\theta$ long:

$$A_{\text{seg}}(y) = r^2 \arccos\frac{r-y}{r} - (r-y)\sqrt{2ry - y^2}, \tag{50}$$
$$P(y) = 2r \arccos\frac{r-y}{r}.$$

The formula holds for $y > r$ too, where $r - y$ is negative and the "triangle" adds to the sector. Laying the can down does not change the volume of its gas, so the gas strip along the top has the same volume as the upright headspace. That fixes the strip's depth y_g, and the drink's depth is what is left:

$$\ell\, A_{\text{seg}}(y_g) = \pi r^2 h_g, \qquad y_d = 2r - y_g. \tag{51}$$

With brine y_b up the can, the wall is wet on both sides below $y = \min(y_d, y_b)$: along the length of the can (the arc times ℓ) and on the two flat ends (a segment each):

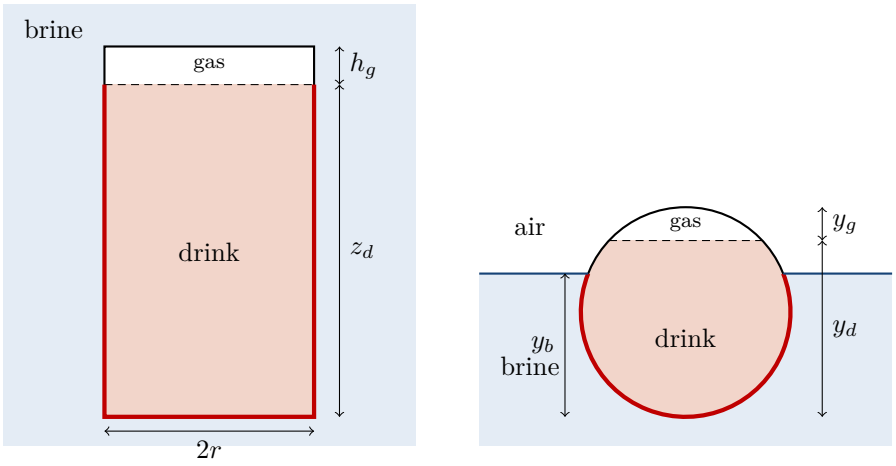
$$A_c = \ell\, P(y) + 2\, A_{\text{seg}}(y), \qquad y = \min(y_d, y_b). \tag{52}$$

Worked example: Can 2 held under the brine

$r = 3.3\,\text{cm}$, $\ell = 11.66\,\text{cm}$, $h_g = 1.2\,\text{cm}$. Upright: $z_d = 10.46\,\text{cm}$ ($f = 0.897$) and (49) gives $A_c = 217 + 34 = 251\,\text{cm}^2$, 81 % of the whole surface. Lying: (51) needs a segment of $\pi r^2 h_g/\ell = 3.52\,\text{cm}^2$, 10 % of the circle, so the gas strip is $y_g = 1.05\,\text{cm}$ deep and the drink $y_d = 5.55\,\text{cm}$; (52) gives $A_c = 178.5 + 61.4 = 240\,\text{cm}^2$, 77 %.

10.5 Does the can float?

Derivation. By Archimedes' principle, a floating body sinks until the liquid it displaces weighs as much as it does. The can and drink weigh $(m_d + m_c)g$; a volume V_{imm} of brine of density ρ_b



(a) upright, held under (b) lying, floating

Figure 1: Can 2 to scale, part-filled. (a) Upright and held under the brine: 1.2 cm of headspace above 10.46 cm of drink. (b) End-on, lying and floating in strong brine: a gas strip 1.05 cm deep, drink 5.55 cm deep, and the brine 4.51 cm up the can. Heat passes only where the wall has drink inside and brine outside (red).

weighs $\rho_b g V_{\text{imm}}$. So $V_{\text{imm}} = (m_d + m_c)/\rho_b$, and if that is less than the can's volume, the can floats. Upright, the immersed volume is $\pi r^2 z_b$; lying, it is $\ell\, A_{\text{seg}}(y_b)$:

$$z_b = \frac{m_d + m_c}{\rho_b\, \pi r^2}, \qquad A_{\text{seg}}(y_b) = \frac{m_d + m_c}{\rho_b\, \ell}. \tag{53}$$

For Can 2, 343 g in all, in strong brine ($\rho_b = 1182\,\text{kg m}^{-3}$): $z_b = 8.48$ cm upright, $y_b = 4.51$ cm lying. A part-filled can is lighter for its size, so it floats higher than a full one.

Which way up? Tilt a floating upright can by a small angle. The displaced brine shifts to the low side and moves the centre of buoyancy B sideways, as if the can hung from a point M, the *metacentre*, a height $BM = I/V_{\text{imm}}$ above B, where $I = \pi r^4/4$ is the second moment of the can's cross-section at the brine surface. The can rights itself if M lies above its centre of mass. Measuring from the bottom of the can, B is at $z_b/2$ and the centre of mass at about $z_d/2$, the middle of the drink, so the height of M above the centre of mass is

$$GM = \frac{z_b}{2} + \frac{r^2}{4 z_b} - \frac{z_d}{2}. \tag{54}$$

For Can 2, $GM = 4.24 + 0.32 - 5.25 = -0.69$ cm (the centre of mass at 5.25 cm includes the can). The drink sloshing inside, with a free surface of its own, lowers it further, by $(\rho_d/\rho_b)\, BM$, to -0.97 cm. $GM < 0$: an upright can floating in brine is unstable and **tips onto its side**. In fresh water it is marginal (+0.03 cm, and -0.25 cm with the drink's free surface) and falls over too. A can that is not held floats lying down, as in the rig.

Worked example: Can 2 floating on its side

The brine reaches $y_b = 4.51$ cm up a can 6.6 cm across, below the drink's 5.55 cm, so the wall is wet on both sides only below 4.51 cm: $A_c = 200\,\text{cm}^2$, 64 % of the surface. The top centimetre of drink faces the air in the box instead, which exchanges heat at only 5–10 $\text{W m}^{-2}\text{K}^{-1}$, and is left out.

10.6 What the fill and the immersion do to the cooling

Table 3 puts the areas into (46) and (48), for a bath held at $-18\,^\circ$C and a drink starting at $28\,^\circ$C.

Table 3: Can 2 with a cola, cooled from 28 °C to 5 °C in a bath held at −18 °C.

Can	A_c (cm^2)	G (W K^{-1})	τ (s)	time to 5 °C (s)
Whole surface passes heat (first estimate), upright	310	5.00	266	185
Part-filled, upright, held under	251	4.05	329	228
Part-filled, lying, held under	240	4.64	287	199
Part-filled, lying, floating	200	3.86	345	239

Ignoring the headspace makes an upright can cool about 20 % too fast. Lying down loses wetted area but gains stronger films, and ends only 8 % slower than the whole-surface estimate. Floating costs another 20 %: hold the can under.

10.7 Room to freeze

If the drink starts to freeze (Step 10), its ice takes more room than the water it came from: $1/917 - 1/1000 = 9.05 \times 10^{-5}\,\mathrm{m^3}$ per kg, about 0.09 mL per gram of ice. The headspace takes up the expansion by compressing its gas, whose pressure at constant temperature goes as $1/V$ (Boyle's law):

$$\frac{p}{p_0} = \frac{V_g}{V_g - \Delta V_{\mathrm{ice}}}, \qquad \Delta V_{\mathrm{ice}} = 9.05 \times 10^{-5}\, m_{\mathrm{ice}}. \tag{55}$$

240 g of ice, Step 10's cola at −5 °C, needs 22 mL. In a 41 mL headspace that more than doubles the pressure; in a real can's smaller headspace the can may bulge or burst.

11 Step 10: the drink as a solution, an ethanol model (new)

The idea. A drink is water with things dissolved in it: sugar, carbon dioxide and, in a beer or a cider, ethanol. They set two things: the drink's heat capacity c_d, which says how much heat must come out, and its freezing point T_{fd}, below which it starts to turn to slush. Dissolved molecules lower the freezing point exactly as salt lowers the bath's (Step 6), but each counts once, since none splits into ions.

11.1 Composition from the label

Derivation. Alcohol by volume, ABV, is the volume of ethanol per volume of drink. Per cubic metre of drink, the ethanol's mass is $\rho_e\,\mathrm{ABV}$, with $\rho_e = 789.2\,\mathrm{kg\,m^{-3}}$, and the drink's mass is ρ_d; their ratio is the ethanol's mass fraction. Sugar and CO_2 are given as mass per volume, s and c in $\mathrm{g\,L^{-1}}$ (equal to $\mathrm{kg\,m^{-3}}$); dividing by ρ_d gives their mass fractions. Beer also holds about 3.5 % of unfermented extract, w_x, mostly large molecules. The rest is water:

$$w_e = \frac{\rho_e\,\mathrm{ABV}}{\rho_d}, \qquad w_{su} = \frac{s}{\rho_d}, \qquad w_c = \frac{c}{\rho_d}, \tag{56}$$
$$w_{wd} = 1 - w_e - w_{su} - w_c - w_x.$$

The molality of each solute is its moles per kilogram of the drink's water: its mass fraction over its molar mass, divided by the water's mass fraction:

$$b_k = \frac{w_k}{M_k\, w_{wd}}, \qquad k = e,\ su,\ c. \tag{57}$$

11.2 Where the drink starts to freeze

Derivation. The derivation of (31) counted dissolved particles per kilogram of water, whatever they were. Ethanol, sugar and CO_2 stay whole molecules in water, $i_v = 1$, so their molalities simply add:

$$\boxed{T_{fd} = -K_f\,(b_e + b_{su} + b_c)} \tag{58}$$

The extract's molecules are large, so few per gram, and it is left out.

Table 4: Three drinks, from (56)–(58) and (62).

	Cola	Beer, 5 % ABV	Hard seltzer, 5 % ABV
Density ρ_d ($\mathrm{kg\,m^{-3}}$)	1040	1008	993
Sugar; CO_2 ($\mathrm{g\,L^{-1}}$)	106 (sucrose); 6.5	0; 5.0	0; 6.0
Ethanol w_e; water w_{wd}	0; 0.892	0.039; 0.921	0.040; 0.954
b_e, b_{su}, b_c ($\mathrm{mol\,kg^{-1}}$)	0, 0.334, 0.159	0.923, 0, 0.122	0.904, 0, 0.144
Freezing point T_{fd}	−0.92 °C	−1.94 °C	−1.95 °C
Specific heat c_d ($\mathrm{kJ\,kg^{-1}K^{-1}}$)	3.95	4.14	4.24

5 % of ethanol lowers the freezing point by 1.7 K, more than the 10 % of sugar in a cola (0.6 K): an ethanol molecule has a seventh of the mass of a sucrose molecule, so per gram it counts seven times. Measured freezing points of beers are close to −2 °C. Glucose–fructose syrup, whose molecules are half the mass of sucrose's, counts twice per gram. The law is the dilute one; for spirits, such as 40 % ABV, use measured freezing points.

11.3 Freezing gradually: slush

Derivation. Ice that forms in the drink is pure water; the ethanol, sugar and CO_2 stay in the liquid, which gets stronger as it loses water. If a share ψ of the drink's water has frozen, every molality is $1/(1-\psi)$ times its starting value, and by (58) so is the freezing-point drop. The liquid that is left is in equilibrium with the ice when

$$T_d = \frac{T_{fd}}{1-\psi} \qquad \Longleftrightarrow \qquad \psi(T_d) = 1 - \frac{T_{fd}}{T_d} \quad (T_d < T_{fd}). \tag{59}$$

Below T_{fd} the drink is a slush whose ice grows as it cools: half its water frozen at $2T_{fd}$, three quarters at $4T_{fd}$.

Freezing releases latent heat, which the bath must take away. The drink's energy, measured as in Step 8, gains a latent term for its ice; neglecting the small difference between the heat capacities of its ice and its water,

$$U_D = C_d\,(T_d - T_{\mathrm{ref}}) - L_f\,m_{wd}\,\psi(T_d), \tag{60}$$

with $m_{wd} = w_{wd} m_d$ the water in the drink. Differentiating with respect to T_d, with $\mathrm{d}\psi/\mathrm{d}T_d = T_{fd}/T_d^2$ from (59) (negative, since $T_{fd} < 0$: more ice as it cools), gives the heat that cools the drink by one kelvin, its effective heat capacity:

$$\boxed{\begin{aligned} &C_{d,\mathrm{eff}}(T_d) = C_d + \frac{L_f\,m_{wd}\,|T_{fd}|}{T_d^2} && (T_d < T_{fd}),\\ &C_{d,\mathrm{eff}}(T_d) = C_d && (T_d \geq T_{fd}) \end{aligned}} \tag{61}$$

and the drink's equation (45) becomes $C_{d,\mathrm{eff}}(T_d)\,\dot{T}_d = -\dot{Q}_d$.

Worked example: a beer and a cola left in

Beer: 304 g of water, $T_{fd} = -1.94\,^\circ\mathrm{C}$. Just below T_{fd}, $C_{d,\mathrm{eff}}$ jumps from $1.38\,\mathrm{kJ\,K^{-1}}$ to $53\,\mathrm{kJ\,K^{-1}}$, forty times larger, so the drink lingers near $-2\,^\circ\mathrm{C}$ while it freezes. At $-5\,^\circ\mathrm{C}$ it is 61 % frozen (186 g of ice, 17 mL of expansion); at $-17\,^\circ\mathrm{C}$, 89 %. A cola, $T_{fd} = -0.92\,^\circ\mathrm{C}$ with 294 g of water, is 82 % frozen at $-5\,^\circ\mathrm{C}$ (240 g of ice, 22 mL). Ethanol keeps a beer liquid a kelvin lower, but a drink left in a $-20\,^\circ\mathrm{C}$ bath ends mostly ice either way.

11.4 The drink's heat capacity

Derivation. A solution's heat capacity is a sum over what it contains, each part weighted by its mass fraction and counted with its *partial* specific heat: what a kilogram of it adds to the solution, which is not its heat capacity as a pure substance:

$$c_d = w_{wd}\,c_w + w_e\,\bar{c}_e + w_{su}\,\bar{c}_{su} + w_x\,\bar{c}_x. \tag{62}$$

In dilute solution ethanol adds about $5.4\,\mathrm{kJ\,kg^{-1}K^{-1}}$ ($0.25\,\mathrm{kJ\,mol^{-1}K^{-1}}$), more than twice pure ethanol's 2.44 and more than water's: water molecules order themselves round the ethanol's ethyl group, and that order melts as the solution warms, which takes heat. Sucrose adds about $1.9\,\mathrm{kJ\,kg^{-1}K^{-1}}$ and beer's extract about 1.5; the CO_2 is too little to matter. So a little ethanol *raises* c_d slightly above water's, while sugar lowers it: the values in Table 4. A mass-weighted average with pure ethanol's 2.44 would wrongly put a beer below water.

These are estimates, good to a few per cent. A drink's own value is easily measured by mixing: pour a mass m_1 of drink at T_1 into a mass m_2 of water at T_2 in an insulated cup and read the final temperature T_3. The heat the drink loses is the heat the water gains, so $c_d = m_2 c_w (T_3 - T_2)/[m_1(T_1 - T_3)]$.

11.5 Convection inside the can

At 5 % ABV the drink is about a fifth more viscous than water and slightly less conductive; through the quarter powers of Step 8 these change h_i by a few per cent. More important near the end of a run: water is densest at 4 °C, and as the drink at the wall approaches that temperature it stops getting denser, so the free convection that mixes it weakens. Dissolved sugar and CO_2 move that point lower. The usual target, 5 °C, is just above it.

11.6 Supercooling

A still drink in a smooth can often cools several kelvin below T_{fd} before any ice appears, then freezes part of the way within seconds, its latent heat warming it back onto the curve (59). The model assumes ice appears at T_{fd}.

11.7 What ethanol changes

In the coupled model of Step 11 a beer takes 3 % longer than a cola to reach 5 °C (the worked example of Step 11), because its c_d is larger. Ethanol matters little for the cooling time; it matters for what happens after, by keeping the drink liquid a kelvin lower.

11.8 Ethanol in the bath instead of salt

The same law describes ice with ethanol in place of salt, with $i_v = 1$ and $M_e = 0.04607\,\mathrm{kg\,mol^{-1}}$: per unit of the mass ratio m_s/W, ethanol lowers the freezing point by $K_f/M_e = 40.4\,\mathrm{K}$ against

salt's 63.7 K, so in dilute solution salt is the stronger by weight. Two things differ more. Ethanol mixes with water in any proportion, so there is no saturation and no eutectic plateau within reach: such a bath has no thermostat, and warms as its ice melts. And mixing ethanol with water releases heat, where dissolving salt absorbs it. This document does not model an ethanol bath further.

12 Step 11: fully unified time-dependent model, with the can

The idea. Put the pieces together: the bath's energy balance (Step 5) with the drink's heat in it, the freezing point (Step 6) tying the bath temperature to its salt and water, the dissolution rate (Step 7), and the drink's equation (Steps 8 and 10). Then solve for the rates.

Shortcuts (the first edition's, and the can's):

$$W = M_w - m_i = m_w, \tag{63}$$

$$C_{\text{eff}} = m_i c_i + W c_w, \tag{64}$$

$$\alpha = \frac{i_v K_f}{M_{\text{NaCl}}}, \qquad \beta = \frac{\Delta H_{\text{sol}}}{M_{\text{NaCl}}}, \qquad H = \frac{kA}{d}, \qquad G = h_{\text{ov}} A_c. \tag{65}$$

α is in kelvin per unit of the mass ratio m_s/W, and β in $\mathrm{J\,kg^{-1}}$.

Derivation. The bath now receives heat from the room and from the drink. Call their sum

$$\dot{Q}_{\text{in}} = H(T_{\text{room}} - T) + G(T_d - T); \tag{66}$$

it replaces the left side of (29), whose right side is unchanged. The freezing point is (36):

$$\dot{Q}_{\text{in}} = C_{\text{eff}}\,\dot{T} - L_f\,\dot{m}_i + \beta\,\dot{m}_s, \tag{67}$$

$$T = -\alpha\,\frac{m_s}{W}. \tag{68}$$

Differentiate (68) with the quotient rule, using $\dot{W} = -\dot{m}_i$ from (63):

$$\dot{T} = -\alpha\,\frac{\mathrm{d}}{\mathrm{d}t}\Big(\frac{m_s}{W}\Big) \tag{69}$$

$$= -\alpha\,\frac{\dot{m}_s W - m_s \dot{W}}{W^2} \tag{70}$$

$$= -\alpha\Big(\frac{\dot{m}_s}{W} + \frac{m_s\,\dot{m}_i}{W^2}\Big). \tag{71}$$

Dissolving salt cools the bath (the first term); melting ice warms it, since $\dot{m}_i < 0$ (the second). Substituting (71) into (67), then collecting the terms in $\dot{m}_s$ and in $\dot{m}_i$:

$$\dot{Q}_{\text{in}} = -\alpha C_{\text{eff}}\Big(\frac{\dot{m}_s}{W} + \frac{m_s\,\dot{m}_i}{W^2}\Big) - L_f\,\dot{m}_i + \beta\,\dot{m}_s \tag{72}$$

$$= \Big(\beta - \frac{\alpha C_{\text{eff}}}{W}\Big)\dot{m}_s - \Big(L_f + \frac{\alpha C_{\text{eff}} m_s}{W^2}\Big)\dot{m}_i. \tag{73}$$

Solving for the ice, and writing $\dot{Q}_{\text{in}}$ out again:

$$\boxed{\dot{m}_i = \frac{\Big(\beta - \dfrac{\alpha C_{\text{eff}}}{W}\Big)\dot{m}_s - H(T_{\text{room}} - T) - G(T_d - T)}{L_f + \dfrac{\alpha C_{\text{eff}} m_s}{W^2}}} \tag{74}$$

What the terms mean. The denominator is the heat it takes to melt a kilogram of ice *in this bath*: L_f, plus the heat that warms the whole bath, because the meltwater dilutes the brine

and raises its freezing point. In the numerator, heat from the room and from the drink melts ice. Dissolving salt melts ice too, through the bracket, which is negative in practice: the salt lowers the freezing point, the bath must cool to follow it, and the heat for that cooling comes from melting ice. This is how a salted ice bath cools itself.

Worked example: the rig bath as the can goes in

After 15 minutes with no stirrer (the run below): $m_s = 0.633\,\mathrm{kg}$ and $T = -16.8\,^\circ\mathrm{C}$, so $W = \alpha m_s/|T| = 2.40\,\mathrm{kg}$, $m_i = 3.20\,\mathrm{kg}$ and $C_{\mathrm{eff}} = 16.8\,\mathrm{kJ\,K^{-1}}$. The denominator is $333\,500 + 116\,800 = 450\,300\,\mathrm{J\,kg^{-1}}$, 35 % more than L_f. The bracket is $66\,700 - 443\,800 = -377\,100\,\mathrm{J\,kg^{-1}}$: each kilogram of salt that dissolves melts $377\,100/450\,300 = 0.84\,\mathrm{kg}$ of ice.

The drink. From Steps 8 and 10:

$$C_{d,\mathrm{eff}}(T_d)\,\dot{T}_d = -G\,(T_d - T). \tag{75}$$

The full reduced system.

$$\boxed{\begin{aligned}
\dot{m}_s &= k_m A_{s,0}\Big(\frac{m_{s,0} - m_s}{m_{s,0}}\Big)^{2/3}\Big[C_{\mathrm{sat}}(T) - \frac{\rho_w m_s}{W}\Big],\\
\dot{m}_i &= \frac{(\beta - \frac{\alpha C_{\mathrm{eff}}}{W})\dot{m}_s - H(T_{\mathrm{room}} - T) - G(T_d - T)}{L_f + \frac{\alpha C_{\mathrm{eff}} m_s}{W^2}},\\
\dot{T}_d &= -\frac{G\,(T_d - T)}{C_{d,\mathrm{eff}}(T_d)},\\
T &= -\alpha\,\frac{m_s}{W}, \qquad W = M_w - m_i, \qquad C_{\mathrm{eff}} = m_i c_i + W c_w.
\end{aligned}} \tag{76}$$

Three state variables, m_s, m_i and T_d, move by the three rate equations; the bath temperature follows from the last line. Start with $m_i(0)$ and $m_s(0)$ consistent with (68), and $T_d(0)$ the drink's starting temperature; before the can goes in, set $G = 0$. This *ice–brine equilibrium branch* stops applying when the ice runs out, when the brine saturates (then $\dot{m}_s = 0$ and the bath sits near (38)), or when the bath is far from phase equilibrium.

Alternative single implicit equation. Eliminating T from the heat balance with (68), without solving for $\dot{m}_i$:

$$\begin{aligned}
&H\Big(T_{\mathrm{room}} + \frac{\alpha m_s}{W}\Big) + G\Big(T_d + \frac{\alpha m_s}{W}\Big)\\
&\qquad = -\alpha C_{\mathrm{eff}}\Big(\frac{\dot{m}_s}{W} + \frac{m_s\,\dot{m}_i}{W^2}\Big) - L_f\,\dot{m}_i + \beta\,\dot{m}_s.
\end{aligned} \tag{77}$$

It is equivalent to (74) paired with (68).

Worked example: the rig with Can 2, solved

(76) integrated numerically with the rig's numbers: a box of $0.46\,\mathrm{W\,K^{-1}}$ and a room at $28\,^\circ\mathrm{C}$; the $4\,^\circ\mathrm{C}$ water first melts 80 g of ice so that the bath starts at $0\,^\circ\mathrm{C}$; grains 0.5 mm across, $A_{s,0} = 5.5\,\mathrm{m^2}$; $C_{\mathrm{sat}} = 357\,\mathrm{kg\,m^{-3}}$ held constant; the can goes in at $28\,^\circ\mathrm{C}$ after 15 minutes.

Can	G (W K^{-1})	time to 5 °C (s)	
		$k_m = 10^{-6}$	$k_m = 10^{-5}$
Whole surface passes heat, upright	5.00	192	165
Part-filled, upright, held under	4.05	235	202
Part-filled, lying, held under	4.64	207	176
Part-filled, lying, floating	3.86	246	213
As above, with a beer	4.64	213	183
Bath when the can goes in		−16.8 °C	−22.4 °C
Salt dissolved by then		0.63 kg	0.93 kg

k_m is in m s^{-1}. With still brine ($k_m = 10^{-6}$) the salt dissolves slowly and the bath is only at −16.8 °C after 15 minutes; with brine moving past the grains (10^{-5}) it has nearly reached the model's floor, −22.7 °C. A drink left in for two hours ends at −19.2 °C: a cola 95 % frozen, a beer 90 %.

13 Step 12: time-independent equilibrium, with the can

The idea. There are two different things "time-independent" can mean: energy accounting between two moments, and a true steady state that nothing changes any more. They are not interchangeable.

13.1 Insulated equilibrium

Derivation. With no heat in, (2) says the energy at the end equals the energy at the start, $U_f = U_0$. Measure energies from ice and liquid water at $T_0 = 0$ °C, with the first edition's reference: liquid water holds L_f per kilogram more than ice. Between start and end, ice and water warm or cool to T_f (sensible heat), $m_{i,0} - m_{i,f}$ of ice melts (latent energy gained) and $m_{s,f} - m_{s,d,0}$ of salt dissolves (heat of solution absorbed); their sum is zero:

$$\boxed{0 = (m_{i,f} c_i + m_{w,f} c_w)\, T_f - L_f\,(m_{i,f} - m_{i,0}) + \beta\,(m_{s,f} - m_{s,d,0})} \tag{78}$$

With ice and brine together at the end, the mass balance and the freezing point close it:

$$m_{w,f} = M_w - m_{i,f}, \tag{79}$$

$$T_f = -\alpha\,\frac{m_{s,f}}{M_w - m_{i,f}}. \tag{80}$$

If solid salt remains, the brine is saturated, which is a third condition:

$$C_f = C_{\mathrm{sat}}(T_f); \tag{81}$$

if all the salt has dissolved, $m_{s,f} = S_0$ instead. These are alternative regimes, not simultaneous conditions: try one, and check that its answer is consistent.

With the can, and other starting temperatures (new). The drink cools from $T_{d,0}$ to T_f; if $T_f < T_{fd}$ part of its water freezes, by (59), giving up latent heat. Water starting at $T_{w,0}$ and salt at $T_{s,0}$ bring their own sensible heat, measured from 0 °C, on the starting side; the salt that is still solid at the end is counted at T_f (Step 5's correction). With the ice starting at 0 °C:

$$\begin{aligned} 0 = {} & \left[m_{i,f} c_i + m_{w,f} c_w + (S_0 - m_{s,f})\, c_s\right] T_f - L_f\,(m_{i,f} - m_{i,0}) \\ & + \beta\,(m_{s,f} - m_{s,d,0}) + C_d\,(T_f - T_{d,0}) - L_f\, m_{wd}\, \psi(T_f) \\ & - m_{w,0} c_w T_{w,0} - S_0 c_s T_{s,0}. \end{aligned} \tag{82}$$

Worked example: the rig and Can 2, left until everything settles

Rig numbers, with the drink at 28 °C; "measured diagram" means the freezing curve, solubility and eutectic of the B8 report's model (`b8_sim.py`) instead of (80)–(81).

	T_f	ice left	solid salt left	drink frozen
Bath alone, dilute law	−22.7 °C	2.85 kg	19 g (saturated)	—
Bath alone, measured diagram	−21.2 °C	2.93 kg	193 g	—
With a cola, dilute law	−20.3 °C	2.46 kg	0	95 %
With a cola, measured diagram	−21.2 °C	2.42 kg	40 g	96 %
With a beer, dilute law	−20.3 °C	2.46 kg	0	90 %
With a beer, measured diagram	−21.2 °C	2.42 kg	40 g	91 %

The drink gives the bath about 160 kJ, cooling from 28 °C and freezing, which melts 0.4–0.5 kg of ice. With the dilute law that dissolves the last of the salt and leaves the bath at −20.3 °C; with the measured diagram the bath stays on its eutectic plateau at −21.2 °C and uses 150 g of its spare salt. (A beer and a cola give almost the same heat: the beer's larger heat capacity is offset by its smaller frozen share.) Either way a can left until everything settles is mostly ice: take it out at 5 °C.

13.2 Integrated heat leakage over a finite interval

Derivation. Integrating the bath balance (67) from 0 to t_f, for constant c_i, c_w, L_f and β, the latent and dissolution terms integrate exactly, but the sensible term does not, because C_{eff} changes as ice melts:

$$\begin{aligned}\int_0^{t_f} \{H[T_{\text{room}} - T(t)] + G[T_d(t) - T(t)]\}\,\mathrm{d}t \\ = \int_0^{t_f} C_{\text{eff}}(t)\,\dot{T}(t)\,\mathrm{d}t - L_f(m_{i,f} - m_{i,0}) + \beta(m_{s,f} - m_{s,d,0}).\end{aligned} \tag{83}$$

The first edition's endpoint formula,

$$\begin{aligned}\int_0^{t_f} H[T_{\text{room}} - T(t)]\,\mathrm{d}t \\ \overset{\text{earlier approximation}}{\approx} C_f T_f - C_0 T_0 - L_f(m_{i,f} - m_{i,0}) + \beta(m_{s,f} - m_{s,d,0}),\end{aligned} \tag{84}$$

is not generally valid when C_{eff} varies.

Correction (first edition). Integrating by parts,

$$\int_0^{t_f} C_{\text{eff}}\,\dot{T}\,\mathrm{d}t = \left[C_{\text{eff}}T\right]_0^{t_f} - \int_0^{t_f} T\,\dot{C}_{\text{eff}}\,\mathrm{d}t, \qquad \dot{C}_{\text{eff}} = (c_i - c_w)\,\dot{m}_i, \tag{85}$$

so (83), not the endpoint replacement (84), follows the rate equation literally.

The drink's part, by contrast, integrates exactly: from (75), $G(T_d - T) = -C_{d,\text{eff}}\dot{T}_d$, so the heat it gives the bath between two moments is $C_d(T_{d,0} - T_d)$, plus $L_f m_{wd}\psi$ if it has started to freeze. Its heat capacity, unlike the bath's, does not change.

13.3 True steady state with heat leakage

Derivation. In a steady state nothing changes: $\dot{T} = \dot{m}_i = \dot{m}_s = \dot{T}_d = 0$. Then (75) gives $G(T_d - T) = 0$, and (67) gives $H(T_{\text{room}} - T) = 0$:

$$H(T_{\text{room}} - T_{\text{eq}}) = 0, \qquad G(T_{d,\text{eq}} - T_{\text{eq}}) = 0, \tag{86}$$

$$T_{\text{eq}} = T_{d,\text{eq}} = T_{\text{room}} \qquad (H > 0,\ G > 0). \tag{87}$$

Above 0 °C no ice can be in equilibrium, so the true steady state comes only after all the ice has melted; it cannot be found by imposing the ice–brine freezing condition on a warm room.

14 A thermodynamically more consistent refinement

The idea. The model so far treats L_f as a constant while giving ice and water different specific heats, $c_i \neq c_w$. That is inconsistent away from 0 °C. A consistent model starts from an energy that is a function of the state, and differentiates it.

Derivation. Choose $T_{\text{ref}} = 0\,°\text{C}$; give ice zero energy there, and liquid water L_f. In a first ideal-mixture approximation, the bath's energy is

$$U_{\text{approx}} = m_i c_i\,(T - T_{\text{ref}}) + m_w\,[L_f + c_w(T - T_{\text{ref}})] + \beta\, m_s. \tag{88}$$

Differentiating with the product rule, and using $\dot{m}_w = -\dot{m}_i$:

$$\dot{U}_{\text{approx}} = (m_i c_i + m_w c_w)\,\dot{T} + [(c_i - c_w)(T - T_{\text{ref}}) - L_f]\,\dot{m}_i + \beta\,\dot{m}_s. \tag{89}$$

Setting this equal to the heat flowing in, now including the drink's:

$$\begin{aligned} &H(T_{\text{room}} - T) + G(T_d - T) \\ &\quad = (m_i c_i + m_w c_w)\,\dot{T} - [L_f + (c_w - c_i)(T - T_{\text{ref}})]\,\dot{m}_i + \beta\,\dot{m}_s. \end{aligned} \tag{90}$$

At $T = T_{\text{ref}}$ this is the simpler balance (67). Elsewhere the heat to melt a kilogram of ice is $L_f + (c_w - c_i)(T - T_{\text{ref}})$, smaller below zero (Kirchhoff's law): at −15 °C, $333\,500 - 2080 \times 15 = 302\,300\,\text{J kg}^{-1}$, 9 % less. A more precise model would add the brine's salt-dependent heat capacity and density, non-ideal mixing, the measured NaCl liquidus, and the hydrated solid phase where it forms.

14.1 Consistent integrated energy at any starting and final temperature

Derivation. With an energy function, the integrated balance is exact: the heat let in equals the change of energy. The drink's heat flows inside the system and cancels (6), so for bath and drink together, with U_D from (60) and the undissolved salt's sensible heat included:

$$\boxed{\int_0^{t_f} H(T_{\text{room}} - T)\,\mathrm{d}t = \left[U_{\text{approx}} + (S_0 - m_s)\,c_s(T - T_{\text{ref}}) + U_D\right]_0^{t_f}} \tag{91}$$

Written out, with the bath at T_0 and the drink at $T_{d,0}$ at the start:

$$\begin{aligned} \int_0^{t_f} H(T_{\text{room}} - T)\,\mathrm{d}t = {} & m_{i,f} c_i (T_f - T_{\text{ref}}) - m_{i,0} c_i (T_0 - T_{\text{ref}}) \\ & + m_{w,f}\,[L_f + c_w(T_f - T_{\text{ref}})] - m_{w,0}\,[L_f + c_w(T_0 - T_{\text{ref}})] \\ & + \beta\,(m_{s,f} - m_{s,d,0}) + (S_0 - m_{s,f})\,c_s(T_f - T_{\text{ref}}) - S_0\,c_s(T_0 - T_{\text{ref}}) \\ & + C_d\,(T_{d,f} - T_{d,0}) - L_f\,m_{wd}\,\psi(T_{d,f}). \end{aligned} \tag{92}$$

This is the preferred endpoint equation for the idealised, constant-coefficient model. For a perfectly insulated box, set the left side to zero. For a real box over a finite time, the left side depends on the unknown history of T and cannot be found from the endpoints alone.

15 Model limits and practical next steps

1. **Initial temperature consistency.** The condition $T = -\alpha m_s/m_w$ cannot hold for pure ice and fresh water above 0 °C, or for brine that has not reached its ice yet. The first moments of a run are out of equilibrium; start the model once ice and brine have met, or add a non-equilibrium start.
2. **Salt concentration.** The $i_v K_f b$ law suits dilute brine only (Step 6). Use the measured NaCl freezing curve and eutectic for the bath's working range.
3. **Dissolution kinetics.** Rates depend on grain size, crust, and whether brine flows through the ice. Measure or fit k_m and $A_{s,0}$.
4. **The box.** $H = kA/d$ assumes flat, steady walls with faces at room and bath temperature. Calibrating H with a box test is more reliable than computing it.
5. **Switching of phases.** Enforce $m_i \geq 0$, $0 \leq m_s \leq S_0$ and saturation. Ice–brine equilibrium, saturated brine with salt, all-liquid and eutectic regimes each need their own closure.
6. **The can is not a cylinder.** Its neck and domed base change the volumes and areas of Step 9 by several per cent: measure the brimful volume and the headspace.
7. **Hold the can under.** A free can floats lying down with a third of its wall out of the brine, and cools 20 % more slowly (Table 3).
8. **The bath next to the can.** With no stirrer the brine near the can warms first, and the ice there melts first; one bath temperature overstates the cooling. The B8 report splits the bath into rings to follow this.
9. **The drink.** It stratifies slowly; its free convection weakens near 4 °C; it may supercool, then partly freeze at once; and a frozen drink may burst its can.
10. **Drink data.** The freezing points and heat capacities of Step 10 come from dilute laws and approximate partial heat capacities. Measure T_{fd} from a slow cooling curve (the plateau where ice appears) and c_d by mixing.
11. **Validation.** Compare predicted temperature–time curves and end states with experiments at several salt-to-water ratios, starting masses and temperatures, and insulation thicknesses; and, for the can, with probes in the drink and in the brine next to it, for full and part-filled cans, upright and lying, held and floating, and for drinks with and without ethanol.

16 Quick index of principal equations

First law with the can: (4)–(6). Conduction through the box: (7), with films (12). Ice and water conservation: (25). Latent energy: (19). NaCl energy: (28) and (29). Freezing depression: (34)–(36); the model's floor (38). Dissolution kinetics: (39)–(44). The drink and its can: (45), (46), (48). A part-filled can: (49)–(52); floating (53), (54); room to freeze (55). The ethanol model: (56)–(62). Dynamic derivatives: (71) and (74); the full system (76). Insulated equilibrium: (78)–(82). Finite-time energy: (83). True steady state: (87). Thermodynamically consistent update: (90)–(92).

SYPT 2027 · PROBLEM B8 "COLD DRINK" · THE STRONG CASE

Salt, water, ethanol and ice in an unstirred cooler: how cold does it get, and how fast does a part-filled can cool with no convection?

A closed-form model. You choose four masses: water, salt and ethanol at room temperature, and ice at 0 °C. Everything else is measured once or follows from an explicit formula.

§1 The system, the variables and the assumptions
§2 Part I · the bath in closed form: temperature, ice melted, composition, regimes, design rules
§3 Part II · the part-filled can, its air space, buoyancy, and where the ice and brine sit in the tank
§4 Part III · heat flow with no convection: the diluted brine, the slush, the cooling law
§5 The whole chain from your four masses to the drink temperature; design; what to measure
§6 Appendix: Laplace inversions, the classical Stefan limit, verification log, sources

The answer on one page

Headline numbers for your rig

Recipe (water / salt / ethanol / ice, kg)	T_b (°C)	t_5 (s): 25 → 5 °C			rate at 5 °C
		ice floats	lid board	ice at floor	(K/min, board)
B0 plain ice water (1.60 / 0 / 0 / 3.0)	0.0	643	640	640	0.22
B1 eutectic salt bath (1.60 / 1.00 / 0 / 4.0)	−21.2	223†	203	151	1.95
B2 salt + ethanol (1.00 / 1.00 / 0.30 / 4.0)	−24.4*	198	185	138	2.23
B3 ethanol only (1.00 / 0 / 0.80 / 4.0)	−13.2	332	324	222	1.06
B4 your original rig charge (2.60 / 0.95 / 0 / 3.0)	−14.7	477	447	287	0.81

Strict lower bound (no convection inside the can either), B1: $t_5 \approx 813$ s, rate at 5 °C ≈ 0.66 K/min.

Can 2 with 330 mL in your 14.2 × 29.0 × 17.5 cm tank, base 1 cm off the floor; drink well mixed; bath purely conductive. †at the brim (pack top 17.6 cm). *cotectic below −23.6 °C: extrapolated liquidus.

Six things the strong case says that a stirred model does not.

1. **Two closed forms carry everything.** The bath temperature is the positive root of a quadratic, Eq. (2.7). The drink temperature is $T_d(t) = T_b + \Theta_0\,[p_2\,\mathrm{erfcx}(p_2\sqrt{t}) - p_1\,\mathrm{erfcx}(p_1\sqrt{t})]/(p_2 - p_1)$, Eq. (4.10). It is not exponential (rate $\propto t^{-1/2}$ early, excess $\propto t^{-3/2}$ late).
2. **Your "coefficient lowered" appears by itself.** With nothing flowing, the meltwater from the ice next to the can cannot be carried away, so the brine there is diluted. The last ice beside the can melts at $T_L = -8.14$ °C, not at the bath's −21.2 °C. The no-convection coefficient is $\eta_c = \theta_L/\theta_b = 0.384$, Eq. (4.2); your mixing coefficient η_s (salt that never reached the ice) multiplies on top.
3. **The slush is still an excellent heat sink.** Its latent heat is spread through a partly melted ("mushy") zone, so for heat conduction it behaves like a material with effusivity $E \approx 2\,575$ W s$^{1/2}$ m^{-2} K^{-1}. That is 1.78 × the effusivity of clear brine (1 447).
4. **Where the ice sits matters more than ethanol.** Ice held down around the can: 151 s; floating ice leaves clear brine at the bottom: 223 s. Ethanol melts ice, so B3 is 1.59 × slower than B1.
5. **The air space matters less than you would think.** The drink's heat capacity and its wetted wall both scale with the fill, so almost doubling the drink (200→375 mL) moves t_5 only from 122 to 166 s. The air space matters for *buoyancy*: a 330 mL can floats even in fresh water and needs 1.11 N to hold it under brine.
6. **If even the drink cannot convect,** the wall jumps at once to the contact temperature $T_c = -7.38$ °C, below the drink's freezing point, and the drink takes about 813 s: the true minimum cooling rate,

4.0 × slower than well mixed.

How to read this

Every symbol is defined where it first appears. In the tables, C marks the four *control* variables, M a quantity you *measure* once (or take from a handbook), and D a *derived* quantity given by an explicit formula. The equation numbers in §5 chain them in order. Every number quoted is computed from those formulas. They were checked against the earlier numerical bath model over 116 recipes (worst error 0.73 K) and against a full numerical solution of the conduction equations (16 cases, rms error 4.4 %, worst 10.7 %).

1 The system, the variables and the assumptions

The experiment. An insulated tank (yours: glass, inner floor 14.2 × 29.0 cm, height 17.5 cm, in an EPS box). You tip in ice at 0 °C, then water, salt (NaCl) and ethanol at room temperature T_r, and wait for the bath to settle. Nothing stirs it. A can of drink at T_0 is lowered in and held upright with its base z_0 above the floor. Two questions:

(a) what temperature T_b does the bath settle at, and what is it made of (ice left, brine composition, solid salt left)?
(b) how does the drink temperature $T_d(t)$ fall when nothing moves except heat, by conduction?

The "strong case" combines everything at once: salt *and* water *and* ethanol with ice; no stirrer and no convection; the real can with its air space; and salt that is not perfectly mixed into the ice.

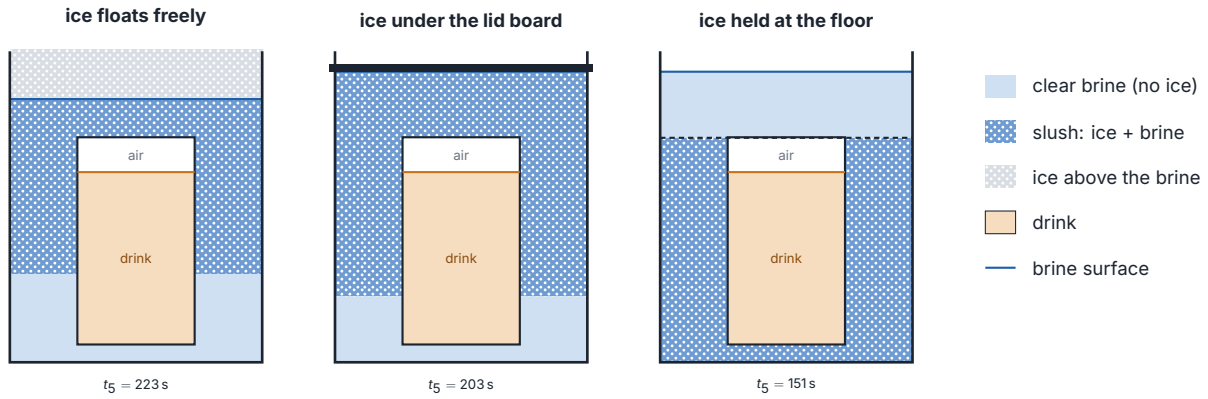


Figure 1. Your tank drawn to scale after recipe B1 has settled (4.0 kg ice, 1.60 kg water, 1.00 kg salt). Ice floats in brine, so where it ends up depends on how you hold it. Left: free-floating ice leaves 5.0 cm of clear (ice-free) brine at the bottom. Middle: ice pushed under a lid board. Right: ice held down on the floor by a mesh, with the clear brine on top. The can's drink (amber) fills 97.0 mm of its 116.6 mm height; the rest is air. Times are for 25 → 5 °C from §4.

1.1 The variables

Control variables (what you change between runs):

symbol	meaning	enters as
C m_w	mass of liquid water added, at T_r	sensible heat; solvent
C m_s	mass of NaCl added, at T_r	solute; dissolution heat
C m_e	mass of ethanol added, at T_r	solute; heat of mixing
C m_i	mass of ice added, at 0 °C	latent-heat store

Measured or tabulated once (M). Values are the ones used for every number in this document.

symbol	meaning	value	how to get it
thermodynamics			
c_w, c_i, c_s, c_e	specific heats: water, ice, NaCl (solid), pure ethanol (0–25 °C)	4 190, 2 060, 860, 2 345 J kg^{-1}K^{-1}	handbook
L_0	latent heat of fusion of ice at 0 °C	333.55 kJ/kg	handbook
$\Delta c = c_w - c_i$	makes the latent heat temperature dependent, $L(\theta) = L_0 - \Delta c\,\theta$	2 130 J kg^{-1}K^{-1}	derived
Δh_s	molar enthalpy of solution of NaCl (endothermic)	3.88 kJ/mol	handbook (placeholder)
$h_E(b_e)$	heat of mixing per mole of ethanol at 0 °C, $h_E = h_\infty + g\,b_e$	$h_\infty = -14.52$ kJ/mol, $g = 0.752$	fitted to H^E table
$c_{ea}(b_e)$	apparent specific heat of ethanol dissolved in water below 0 °C	$\min(6.31,\ 7.10 - 0.349\,b_e)$ kJ kg^{-1}K^{-1}	fitted to Melinder
M_s, M_e	molar masses of NaCl, ethanol	58.44, 46.07 g/mol	
liquidus (freezing-point) constants, fitted in §2.3			
K_s, ω_s	NaCl: depression constant and bound water per mole	3.173 K kg/mol, 0.0440 kg/mol	Bodnar (1993)
K_e, ω_e	ethanol: depression constant, bound water	2.116 K kg/mol, 0	Lange, Melinder
b_E, θ_E	eutectic brine molality and depression	5.163 mol/kg, 21.2 K	Bodnar (1993)
λ	drop in NaCl solubility per mol/kg of ethanol (along the cotectic)	0.0461 kg/mol	NIST ThermoML
mixing			
η_s, η_e	fraction of the salt / ethanol that actually reaches the liquid around the ice	1 (perfect)	§2.1: from T_b
slush, brine, drink, can, tank			
φ_p	volume fraction of ice in the packed slush	0.55	weigh ice in a known volume
ρ_i, k_i	ice density, conductivity	917 kg/m^3, 2.25 W m^{-1}K^{-1}	handbook
k_b, k_l	conductivity of cold brine; of the diluted melt next to the can	0.52, 0.52 W m^{-1}K^{-1}	handbook
ρ_0, β_s, β_e	brine density $\rho_b = \rho_0 + \beta_s w_s - \beta_e w_e$	1000, 765, 120 kg/m^3	weigh 100 mL of brine
ρ_d, c_d, k_d	drink density, specific heat, conductivity	1 040, 4 050, 0.56	10 wt% sugar solution
R, H, t_w, m_{Al}	can: outer radius, height, wall thickness, empty mass	33.00, 116.6, 0.10 mm; 13 g	callipers, balance
V_d	volume of drink in the can	330 mL	weigh full and empty
k_{Al}, c_{Al}	aluminium	205 W m^{-1}K^{-1}, 900 J kg^{-1}K^{-1}	handbook
A_t, H_t, z_0	tank floor area and height; height of the can base above the floor	412 cm^2, 17.5 cm, 1.0 cm	ruler
T_r, T_0, T_1	room temperature; drink start; target	25, 25, 5 °C	thermometer

Everything else is D *derived*. Each derived quantity is defined by one numbered equation, in the order in which you evaluate them (the full list is in §5).

1.2 Assumptions

A1 The bath reaches equilibrium before the can goes in (without a stirrer this takes a while; see A4). The tank exchanges no heat with the room during one run; a heat leak can be added as a known heat Q_x in Eq. (2.2).

A2 Heat capacities are constant within each step of the energy bookkeeping. The ethanol heat of mixing and apparent heat capacity depend on concentration and are updated once (a two-pass evaluation, §2.10).

A3 The ice liquidus of the mixed brine follows the Zdanovskii–Stokes–Robinson (ZSR) water-activity rule, with each binary liquidus written in a form fitted to data (§2.3).

A4 Imperfect mixing: only the fractions η_s and η_e of the salt and ethanol are dissolved in the liquid that touches the ice. The rest stays apart (salt crust on the floor, an ethanol layer on top).

A5 **No convection in the bath.** Heat moves only by conduction. Dissolved salt and ethanol do not move during a run: their diffusion is about 90 times slower than heat diffusion (§4.1).

A6 The drink is well mixed (one temperature T_d) in the main model. The limit in which the drink cannot

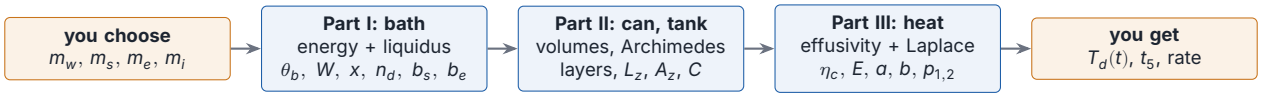
convect either is treated in §4.9.

A7 The can is a cylinder with a 0.1 mm aluminium wall, whose resistance ($t_w/k_{Al} \approx 5 \times 10^{-7}$ m^{2}K/W) is negligible. Its dome, neck and coatings are ignored. The air space conducts no heat.

A8 Ice packs with ice volume fraction φ_p, and brine fills the pores below the brine surface. The part of the can above the brine surface exchanges no heat (ice touching dry aluminium through air).

A9 On the time scale of one cooling the slush and brine around the can act as unbounded. The thermal disturbance reaches about 2 cm, less than the 3.8 cm clearance to the tank wall.

A10 The drink (about 10 wt% sugar) does not freeze in the main model. Its freezing point is −0.77 °C; see §4.9 for when the wall goes colder than that.



2 Part I · The bath in closed form

What decides the bath temperature

Two conditions must hold at the same time.

- **Energy.** The heat that the warm water, salt and ethanol bring in, plus the heat of solution and of mixing, has nowhere to go but into melting ice.
- **Phase equilibrium.** Ice can only sit in a liquid that is on its *liquidus*: the more dissolved solute per kilogram of liquid water, the colder that liquid must be.

Melting ice adds water and dilutes the brine, so the two conditions pull against each other. Solving them together gives one temperature.

Notation. Write the bath temperature as a depression below 0 °C, $\theta \equiv -T_b$ (in K, positive when colder than 0 °C). W is the mass of liquid water in the bath at the end, $x = W - m_w$ the mass of ice that melted, and n_d the moles of salt dissolved.

2.1 The mixing coefficient

With no stirrer, some of the salt you pour in never meets the ice: it sinks as a crust, or sits in a pocket of already saturated brine. Ethanol is lighter than brine and can float as a layer. Let the fractions that do end up in the liquid touching the ice be η_s and η_e (between 0 and 1). The *active* amounts are

$$n_s' = \eta_s \frac{m_s}{M_s}, \qquad n_e' = \eta_e \frac{m_e}{M_e}. \tag{2.1}$$

This is the "lowered coefficient" in its simplest form. In every formula below only n_s' and n_e' lower the freezing point and absorb or release heat of solution, so the effective depression constant per kilogram of salt you added drops from K_s/M_s to $\eta_s K_s/M_s$. The inactive salt still has to be cooled, so it stays in the heat capacities. How to measure η_s is in §2.9.

2.2 Energy balance

Enthalpy is a function of state, so any imaginary path from the start to the end gives the same answer. Take this one:

(i) cool the water, salt and ethanol from T_r to 0 °C, releasing $C_r\,\Delta T_r$;

(ii) at 0 °C, melt x of ice (absorbs xL_0), dissolve n_d moles of salt (absorbs $n_d \Delta h_s$), and mix n_e' moles of ethanol into the water (absorbs $n_e' h_E$; $h_E < 0$, so it actually releases heat);

(iii) cool everything from 0 °C to $-\theta$, releasing $\theta\,[\,(m_w + x)c_w + (m_i - x)c_i + m_s c_s + \eta_e m_e c_{ea} + (1-\eta_e)m_e c_e\,]$.

In an insulated tank, heat released equals heat absorbed. Collecting the terms in x:

$$\boxed{x\,(L_0 - \Delta c\,\theta) = C_r\,\Delta T_r + C_\theta\,\theta - n_d\,\Delta h_s - n'_e\,h_E + Q_x} \tag{2.2}$$

$$C_r = m_w c_w + m_s c_s + m_e c_e, \qquad C_\theta = m_w c_w + m_s c_s + \eta_e m_e c_{ea} + (1-\eta_e) m_e c_e + m_i c_i, \qquad \Delta T_r = T_r - 0\,^\circ\mathrm{C}. \tag{2.3}$$

Reading Eq. (2.2) term by term:

- **left side:** the heat needed to melt x of ice at the bath temperature. $L_0 - \Delta c\,\theta$ is the latent heat at $-\theta$ (Kirchhoff's law): 333.55 kJ/kg at 0 °C, about 13 % less at the eutectic.
- $C_r\Delta T_r$**:** the heat the warm inputs bring.
- $C_\theta\theta$**:** the heat released as everything cools below 0 °C. This includes the ice, which starts at 0 °C and must be cooled too.
- $n_d\Delta h_s$**:** the heat dissolution of salt takes.
- $-n'_e h_E$**:** the heat mixing ethanol gives.
- Q_x**:** any extra heat in (heat leak, or the can's heat; 0 for the bath alone).

2.3 The liquidus with bound water

Why not Blagden's law. $\theta = iK_f b$ with $i = 2$ assumes an ideal, dilute solution. Against Bodnar's (1993) measured NaCl liquidus it is wrong by up to 2.0 K: too cold below about 16 wt% and too warm above, because the ions are not ideal.

A form that fits and keeps the algebra closed. Suppose each mole of solute ties up ω kilograms of water in its hydration shell, so that only $W - \omega n$ of the liquid water acts as free solvent. A dilute-law depression on the free water then reads

$$\theta = \frac{K\,n}{W - \omega\,n} \quad\Longleftrightarrow\quad \theta = \frac{K\,b}{1 - \omega\,b}, \qquad b = \frac{n}{W}. \tag{2.4}$$

Fitted to Bodnar's liquidus and forced through the eutectic ($b_E = 5.163$ mol/kg at $\theta_E = 21.2$ K), NaCl gives $K_s = 3.173$ K kg/mol and $\omega_s = 0.0440$ kg/mol. That is 2.44 water molecules per NaCl, and the worst error from 0 to -21.2 °C is 0.05 K, against 2.0 K for Blagden. For ethanol the fitted ω_e comes out as zero (it is not an ion), leaving $\theta = K_e b_e$ with $K_e = 2.116$ K kg/mol (worst error 0.77 K over 0–28 K; Fig. 2).

Mixing the two (ZSR). The ZSR rule says a mixed brine freezes at the temperature where each solute's share of "water activity" adds up. If $b_{j,0}(\theta)$ is the molality of solute j alone that freezes at θ, then $\sum_j b_j / b_{j,0}(\theta) = 1$. Inverting Eq. (2.4) gives $b_{j,0} = \theta/(K_j + \omega_j\theta)$, so

$$\frac{b_s(K_s + \omega_s\theta) + b_e(K_e + \omega_e\theta)}{\theta} = 1 \quad\Longrightarrow\quad \theta = \frac{K_s b_s + K_e b_e}{1 - \omega_s b_s - \omega_e b_e}.$$

With $b = n/W$:

$$\boxed{\theta\,(W - W_b) = N, \qquad N = K_s\,n'_s + K_e\,n'_e, \qquad W_b = \omega_s\,n'_s + \omega_e\,n'_e} \tag{2.5}$$

N (in K kg) is the "depression strength" of what is dissolved. W_b is the bound water. Against the numerical ZSR model built on the full data tables, Eq. (2.5) is within 1.31 K everywhere (rms 0.36 K), and the worst case is at -30 °C.

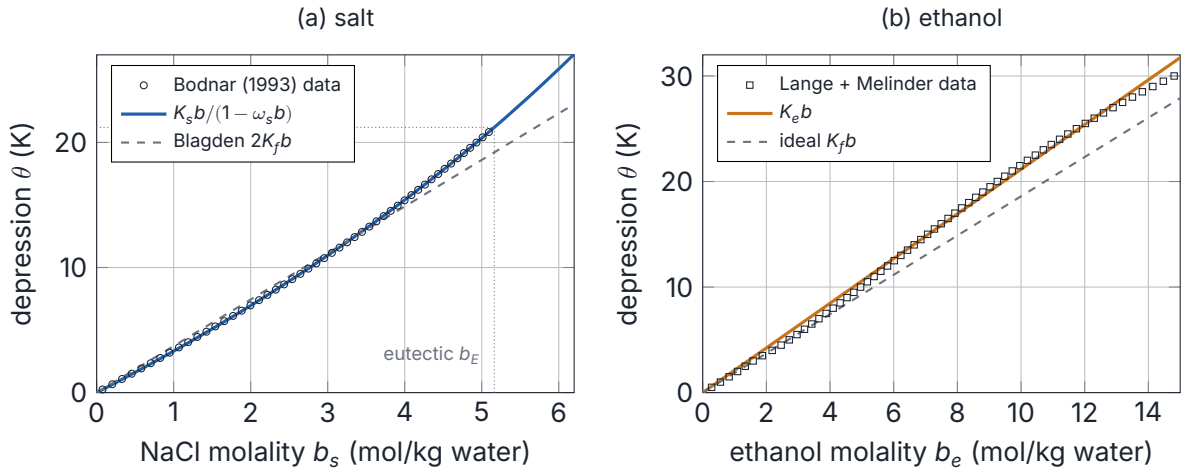


Figure 2. The two liquidus fits. (a) NaCl: the bound-water form follows Bodnar's data to 0.05 K; Blagden's law misses by up to 2.0 K. (b) Ethanol: a straight line $K_e b_e$ with an effective $K_e = 2.116$ (ideal: 1.86).

2.4 Regime U: salt all dissolved, ice left

All the active salt is in solution, $n_d = n'_s$, and $W = m_w + x$. Eq. (2.5) gives $W = W_b + N/\theta$. Substituting this into Eq. (2.2) and multiplying by $\theta\,(L_0 - \Delta c\,\theta)$ leaves a quadratic in θ:

$$A\,\theta^2 + B\,\theta - NL_0 = 0, \qquad \begin{aligned} A &= C_\theta - \Delta c\,(m_w - W_b), \\ B &= (m_w - W_b)\,L_0 + Q_0 + \Delta c\,N, \\ Q_0 &= C_r \Delta T_r - n'_s \Delta h_s - n'_e h_E + Q_x. \end{aligned} \tag{2.6}$$

$A > 0$ (it equals $m_w c_i + m_s c_s + m_e[\eta_e c_{ea} + (1-\eta_e)c_e] + m_i c_i + \Delta c\,W_b$) and $NL_0 > 0$, so exactly one root is positive:

$$\boxed{\theta_U = \frac{2NL_0}{B + \sqrt{B^2 + 4ANL_0}}}, \qquad T_b = -\theta_U, \qquad W = W_b + \frac{N}{\theta_U}, \qquad x = W - m_w. \tag{2.7}$$

(This form avoids the cancellation in $(-B + \sqrt{\cdot})/2A$.) It holds while two checks pass: ice is left ($x \le m_i$), and the brine is not saturated ($n'_s \le b_E\,(W - \lambda n'_e)$, see Eq. (2.9)). If $N = 0$ (plain ice water), $\theta = 0$ and $x = Q_0/L_0$.

2.5 Regime S: saturated, solid salt and ice both left

Salt only: the eutectic. Ice, solid salt and brine coexist: three phases of two components, so the phase rule gives $F = C - P + 1 = 2 - 3 + 1 = 0$ degrees of freedom. The temperature is pinned at $\theta = \theta_E$ and the brine at b_E, so $n_d = b_E W$. Eq. (2.2) becomes linear in W:

$$\boxed{W_S = \frac{m_w L_E + C_r \Delta T_r + C_\theta \theta_E + Q_x}{L_E + b_E \Delta h_s}}, \qquad L_E = L_0 - \Delta c\,\theta_E, \qquad m_{\text{salt,left}} = m_s - b_E W_S M_s. \tag{2.8}$$

With ethanol: the cotectic. Three components and three phases give $F = 1$, so the temperature now drifts with the ethanol content. Two facts fix it. Per kilogram of water, salt is less soluble when ethanol is present, $b_{\text{sat}} = b_E(1 - \lambda b_e)$, so $n_d = b_E(W - \lambda n'_e)$. Put that into Eq. (2.5):

$$\theta = \frac{PW + P'}{QW + Q'}, \qquad P = K_s b_E, \quad P' = n'_e(K_e - \lambda K_s b_E), \quad Q = 1 - \omega_s b_E, \quad Q' = n'_e(\lambda \omega_s b_E - \omega_e). \tag{2.9}$$

Multiplying Eq. (2.2) by $QW + Q'$ gives a quadratic $a_2 W^2 + a_1 W + a_0 = 0$:

$$\begin{aligned} a_2 &= \alpha_1 + b_E \Delta h_s Q, & \alpha_1 &= L_0 Q - \Delta c\,P, \\ a_1 &= \alpha_0 - m_w \alpha_1 - C_\theta P + b_E \Delta h_s Q' - GQ, & \alpha_0 &= L_0 Q' - \Delta c\,P', \\ a_0 &= -m_w \alpha_0 - C_\theta P' - GQ', & G &= C_r \Delta T_r + \lambda b_E n'_e \Delta h_s - n'_e h_E + Q_x. \end{aligned} \tag{2.10}$$

Since $a_2 > 0$ and $a_0 < 0$, there is one positive root, $W_S = (-a_1 + \sqrt{a_1^2 - 4a_2a_0})/2a_2$, and θ_S follows from Eq. (2.9). With $n'_e = 0$ this reduces exactly to Eq. (2.8). The cotectic constant $\lambda = 0.0461\,\mathrm{kg/mol}$ reproduces the NIST solubility data within 3.9 %. The cotectic temperature then agrees with the numerical model to 0.56 K.

2.6 Regime I: the ice is all gone

If the heat brought in is more than the ice can absorb, $x = m_i$ and $W = m_w + m_i$. Eq. (2.2) then gives the temperature directly:

$$\theta_I = \frac{m_i L_0 - C_r \Delta T_r + n_d \Delta h_s + n'_e h_E - Q_x}{C_\theta + m_i \Delta c} \qquad (\theta_I < 0 \text{ means above } 0\,^\circ\mathrm{C}). \tag{2.11}$$

A bath in regime I has no latent-heat store left, so it is useless for cooling a drink.

2.7 Which regime? Boundaries in closed form

Evaluate Eq. (2.7) first. If its two checks pass, that is the answer. If the brine comes out saturated, use regime S. If $x > m_i$, use regime I. Because the bath's enthalpy rises monotonically with the ice melted, exactly one regime is consistent. Setting the checks to equalities gives the boundaries as explicit formulas.

Salt needed to reach the eutectic (salt only). Put $\theta = \theta_E$ and $W = n'_s/b_E$ into Eq. (2.2). Both C_r and C_θ are linear in m_s, so

$$m_s^* = \frac{m_w\,[L_E + c_w(\Delta T_r + \theta_E)] + m_i c_i \theta_E}{\eta_s\,(L_E + b_E \Delta h_s)/(M_s b_E) - c_s(\Delta T_r + \theta_E)}. \tag{2.12}$$

For B1 ($m_w = 1.60$, $m_i = 4.0\,\mathrm{kg}$) this is 0.963 kg, so B1's 1.00 kg is just enough and leaves 36 g of solid salt. For your original charge B4 it is 1.409 kg: the 0.95 kg you planned never reaches the eutectic and the bath sits at −14.7 °C. This is the same "mass balance is not an energy balance" point as the 24 Oct note.

Ethanol that melts all the ice. Set $x = m_i$ in Eq. (2.2) and in Eq. (2.5) or Eq. (2.9). In the unsaturated branch $\theta = N/(m_w + m_i - W_b)$ is linear in m_e, so the energy balance becomes a quadratic in m_e; the saturated branch also reduces to a quadratic. For B1's water, salt and ice the ice is gone at $m_e = 1.46\,\mathrm{kg}$ ($T_b = -25.1\,^\circ\mathrm{C}$). For B2's at 1.55 kg (−29.5 °C), and for ethanol alone at 2.57 kg (−23.6 °C).

These are limits, not recipes

> At those points the bath has no ice left, and the temperatures are below −23.6 °C, where the NaCl liquidus is extrapolated (unreliable below −27 °C). The practical floors are the regime-S values with ice to spare.

2.8 How sensitive is the bath? Closed-form derivatives

Differentiate $F(\theta) = A\theta^2 + B\theta - NL_0 = 0$ implicitly. A, B and N are all linear in each mass, so for any input m_j

$$\frac{\partial \theta}{\partial m_j} = \frac{L_0\,\partial_j N - \theta^2\,\partial_j A - \theta\,\partial_j B}{2A\theta + B}, \tag{2.13}$$

with the partial derivatives

input	$\partial_j N$	$\partial_j A$	$\partial_j B$
m_w	0	c_i	$L_0 + c_w \Delta T_r$
m_i	0	c_i	0
m_s	$\eta_s K_s / M_s$	$c_s + \Delta c\,\eta_s \omega_s / M_s$	$c_s \Delta T_r + (\eta_s/M_s)(\Delta c\,K_s - \omega_s L_0 - \Delta h_s)$
m_e	$\eta_e K_e / M_e$	$\eta_e c_{ea} + (1-\eta_e) c_e$	$c_e \Delta T_r + (\eta_e/M_e)(\Delta c\,K_e - h_E)$

For your original charge B4 (regime U, −14.7 °C) the derivatives of the depression θ (positive means colder) are, per kilogram: water −5.00, ice −0.32, salt +14.70 and ethanol +5.91 K/kg. So 100 g more salt makes it about 1.5 K colder, and 100 g more warm water 0.5 K warmer.

The surprise: more ice does not make the bath colder

> $\partial\theta/\partial m_i = -c_i\theta^2/(2A\theta + B) < 0$ always. Ice at 0 °C has to be cooled to T_b like everything else, which melts more ice and dilutes the brine. Ice is the store that soaks up the *can's* heat, not the thing that sets T_b.

In regime S the temperature is pinned, so the inputs change how much ice melts instead. From Eq. (2.8), the liquid water rises by 1.563 kg per kg of water added, 0.142 per kg of ice and 0.129 per kg of salt.

2.9 Measuring the mixing coefficient: invert the closed form

In regime U, Eq. (2.5) with $W = m_w + x$ and x from Eq. (2.2) is *linear* in the active salt n'_s, for a known θ. So a thermometer reading of the settled bath gives η_s directly:

$$\eta_s = \frac{M_s}{m_s} \cdot \frac{\theta[m_w + (C_r\Delta T_r - n'_e h_E + C_\theta\theta)/(L_0 - \Delta c\,\theta) - \omega_e n'_e] - K_e n'_e}{K_s + \omega_s\theta + \theta\,\Delta h_s/(L_0 - \Delta c\,\theta)}. \tag{2.14}$$

Example. Fully mixed, B1 should settle at −21.2 °C. If your thermometer reads −18.0 °C, Eq. (2.14) gives $\eta_s = 0.80$: a fifth of the salt is not working. Measure this once per recipe and preparation method, then keep it.

2.10 Ethanol: one correction pass

The heat of mixing per mole and the apparent heat capacity of dissolved ethanol both change with concentration: $h_E = h_\infty + g\,b_e$ and $c_{ea} = \min(c_{\text{cap}},\, c_0 + c_1 b_e)$, with $b_e = n'_e/W$. Exact substitution would turn the quadratics into cubics. Instead, evaluate the closed form once with $b_e = 4.0$ mol/kg, compute $b_e = n'_e/W$ from that result, and evaluate once more. Without this second pass the bath is off by up to 5.0 K; with it, by at most 0.73 K.

2.11 Results, and the check against the numerical model

recipe	regime	T_b (°C)	x (kg)	ice left (kg)	solid salt (g)	b_s	b_e	ρ_b (kg/m^3)	numerical T_b
B0	U	0.00	0.502	2.50	–	–	–	1000	0.00
B1	S	−21.20	1.595	2.40	36	5.16	–	1177	−21.20
B2	S	−24.38	1.934	2.07	205	4.64	2.22	1142	−23.86
B3	U	−13.24	1.775	2.23	–	–	6.26	973	−13.21
B4	U	−14.68	1.629	1.37	0	3.84	–	1140	−14.67

The last column is the earlier numerical model (water-activity liquidus from the full data tables, temperature-dependent heat capacities and H^E, root-finding). Over 116 recipes (salt:ice 0–0.35, ethanol:ice 0–0.4, water:ice 0.2–0.65) the closed form is within 0.73 K (rms 0.27 K) and the ice left within 123 g. One borderline recipe is classed U instead of S (Fig. 4). The largest single difference among the five recipes is B2 (−0.51 K), from the cotectic.

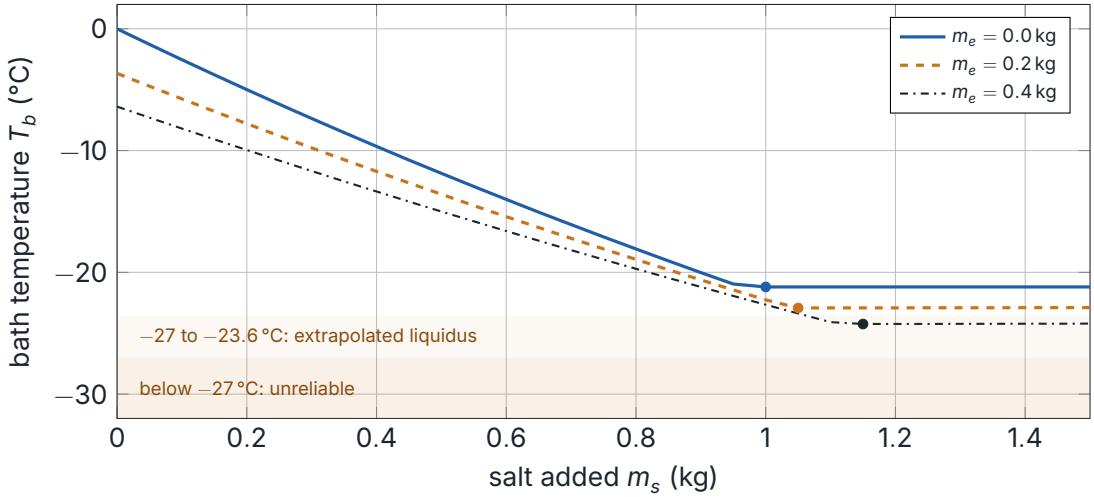


Figure 3. Bath temperature against salt for three ethanol loadings ($m_w = 1.6$, $m_i = 4.0$ kg). The dot marks where the brine saturates. With salt only, the curve then stays flat at the eutectic: it is a thermostat. With ethanol it keeps sinking slowly along the cotectic. Shading marks where the liquidus data run out.

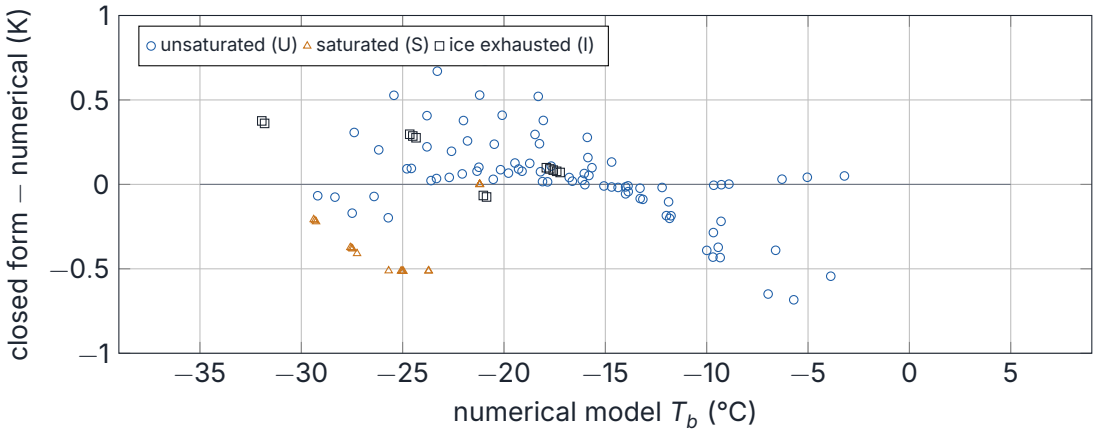


Figure 4. Closed form minus numerical model for 116 recipes, by regime.

3 Part II · The can, its air space, and where things sit in the tank

3.1 The part-filled can

Treat the can as a cylinder of outer radius R and height H, with a wall of thickness t_w. The drink of volume V_d stands at the bottom; above it is the air space (headspace gas).

$$r_i = R - t_w, \qquad h_d = \frac{V_d}{\pi r_i^2}, \qquad V_a = \pi r_i^2 H - V_d, \qquad h_a = H - h_d, \tag{3.1}$$

$$C = \rho_d V_d c_d + m_{\mathrm{Al}} c_{\mathrm{Al}}, \qquad M_c = \rho_d V_d + m_{\mathrm{Al}}, \qquad \bar{\rho}_c = \frac{M_c}{\pi R^2 H}. \tag{3.2}$$

h_d is the height of the drink column, V_a and h_a the volume and height of the air space, C the heat capacity that has to be cooled (drink plus aluminium), M_c the can's mass and $\bar{\rho}_c$ its mean density.

	Can 2 (standard)	Can 1 (slim)	
outer diameter, height	66.00, 116.6 mm	57.55, 146.2 mm	measured
drink column h_d (330 mL)	97.0 mm	127.7 mm	Eq. (3.1)
air space V_a (share of the inside), height h_a	66.5 mL (16.8 %), 19.6 mm	47.6 mL (12.6 %), 18.4 mm	Eq. (3.1)
heat capacity C	1 402 J/K	1 402 J/K	Eq. (3.2)
mass M_c, mean density $\bar{\rho}_c$	356 g, 893 kg/m^3	356 g, 937 kg/m^3	Eq. (3.2)
free-floating draft in eutectic brine / in water	88.5 / 104.1 mm	116.3 / 136.9 mm	Eq. (3.3)
force to hold it fully under eutectic brine	1.11 N	0.90 N	Eq. (3.3)

The air space does three separate things, and each enters the model in a different place:

1. it **lowers the heat capacity**: only the drink and the can need cooling (Eq. (3.2));
2. it **leaves a dry band of wall** at the top. Inside that band is gas, so the drink can only reach it by conduction along the aluminium. That band acts as a cooling fin (§3.4);
3. it **makes the can float**: $\bar{\rho}_c < \rho_b$, and here even $\bar{\rho}_c < 1000\,\mathrm{kg/m^3}$.

3.2 Buoyancy

A free can floats upright with draft d_f. To hold it at an immersion depth h_o you have to push it down with a force F:

$$d_f = \frac{M_c}{\rho_b\,\pi R^2}, \qquad F = \rho_b\, g\,\pi R^2 h_o - M_c g. \tag{3.3}$$

A 330 mL can floats even in fresh water: its draft there is 104.1 mm of its 116.6 mm. In eutectic brine its waterline sits at 88.5 mm, so the top 8.6 mm of the drink column would stick out, and it needs 1.11 N to hold it fully under. So it must be clamped, and the lid board in your rig note does exactly that. A floating can is also barely stable upright. Its centre of mass (near the middle of the drink) sits at about the height of its metacentre, so left alone it tips over.

3.3 Where the ice and the brine end up

From Part I take the ice left $m_l = m_i - x$, the brine mass $m_b = W + n_d M_s + n'_e M_e$, the brine density $\rho_b = \rho_0 + \beta_s w_s - \beta_e w_e$ (with w_s, w_e the salt and ethanol mass fractions), and any solid salt m_{ss}. Then

$$V_l = \frac{m_l}{\rho_i}, \quad V_b = \frac{m_b}{\rho_b} + \frac{m_{ss}}{\rho_{\mathrm{NaCl}}}, \quad A' = A_t - \pi R^2, \quad h_p = \frac{V_l}{\varphi_p A'}, \quad V_p = \frac{1-\varphi_p}{\varphi_p}\,V_l. \tag{3.4}$$

A' is the floor area beside the can, h_p the height the ice pack would occupy, and V_p the brine needed to fill all its pores. Ice is lighter than brine, so the layering depends on how you hold the ice (Fig. 1):

- **Ice floats freely.** By Archimedes a fraction ρ_i/ρ_b of the pack is under the surface, holding $V_f = (\rho_i/\rho_b)V_p$ of brine in its pores. The rest of the brine sits under the ice as a clear layer:

$$h_{cb} = \frac{V_b - V_f}{A'}, \qquad h_b = h_{cb} + \frac{\rho_i}{\rho_b}\,h_p, \qquad \text{ice above the surface: } \Big(1 - \frac{\rho_i}{\rho_b}\Big)h_p. \tag{3.5}$$

- **Ice pushed under a lid board.** The whole pack is submerged: $h_{cb} = (V_b - V_p)/A'$, and slush runs from h_{cb} up to $h_b = h_{cb} + h_p$.
- **Ice held down at the floor** (a mesh). Slush runs from the floor up to h_p, with clear brine above it up to $h_b = h_p + (V_b - V_p)/A'$.

If there is too little brine to float or submerge the pack ($V_b < V_f$, or $V_b < V_p$), the ice rests on the floor. Then $h_b = V_b/[(1-\varphi_p)A']$, slush runs from the floor to h_b, and dry ice sits above. Plain ice water B0 is like this. Finally, everything must fit: the top of the pack must stay below H_t.

recipe	ice held	clear brine h_{cb}	surface h_b	pack top	slush on can L_{sl}	brine on can L_{br}	fits?
B1	floats	5.0	14.8	17.6	5.7	4.0	at brim
B1	lid board	3.7	16.3	16.3	7.0	2.7	yes
B1	at floor	0	16.3	16.3	9.7	0.0	yes
B3	lid board	4.5	16.2	16.2	6.2	3.5	yes
B4	lid board	8.8	16.0	16.0	1.9	7.8	yes
B4	at floor	0	16.0	16.0	6.2	3.5	yes

All heights and lengths in cm, measured from the tank floor; the can's base is at $z_0 = 1.0$ cm.

B4 (your original charge) melts so much ice that with the lid board most of the drink column sits in clear brine: only 1.9 cm of the 9.7 cm drink column touches slush. That matters a lot in §4.

3.4 Contact zones on the can

The drink wets the wall from z_0 to $z_0 + h_d$. Overlapping that band with each layer gives the lengths of wall that have drink inside and slush, or clear brine, outside (L_{sl}, L_{br}). Wall facing ice above the brine surface, or air, is treated as insulated (A8). The bottom counts if the can is held off the floor ($z_0 > 0$).

$$A_{sl} = 2\pi R\, L_{sl}, \qquad A_{br} = 2\pi R\, L_{br}, \qquad A_{\mathrm{bot}} = \pi R^2. \tag{3.6}$$

The dry band as a fin. Above the drink line the wall has gas inside and liquid outside. It still cools the drink, because heat runs along the aluminium from the wetted wall. A thin fin of conductance $k_{\mathrm{Al}} t_w$ per unit width, losing heat to the outside with a mean coefficient $\bar{h}$, carries heat only over a length $1/m$:

$$m = \sqrt{\frac{\bar{h}}{k_{\mathrm{Al}} t_w}}, \qquad A_{\mathrm{fin}} = 2\pi R\, \frac{\tanh(mL_f)}{m}, \qquad A_{[x_a,x_b]} = 2\pi R\, \frac{\sinh m(L_f - x_a) - \sinh m(L_f - x_b)}{m \cosh mL_f}. \tag{3.7}$$

L_f is the length of dry wall in liquid. The last expression splits A_{fin} between layers when the band crosses from slush into brine. With $\bar{h} \approx 249\,\mathrm{W\,m^{-2}K^{-1}}$ (from §4), $1/m \approx 9$ mm. So only the first centimetre of the 19.6 mm dry band works, contributing $A_{\mathrm{fin}} \approx 18.2\,\mathrm{cm^2}$ of the 253.6 $\mathrm{cm^2}$ total contact area for B1 with the ice at the floor. $\bar{h}$ comes from the first pass of the cooling calculation, so the fin enters on the second pass.

4 Part III · Heat flow with no convection

The story in four steps

1. With nothing flowing, the salt next to the can cannot follow the meltwater. The brine there gets diluted, and the last ice beside the can melts at T_L, well above T_b (§4.2).
2. The slush does not melt at one temperature. Its latent heat is spread over a partly melted zone, so it behaves like a solid with a huge, temperature-dependent heat capacity (§4.3).
3. A classical similarity solution turns that zone into one number, an effective *effusivity* E (§4.4).
4. A well-mixed can in contact with such a material cools by an exact closed-form law, which comes out of a Laplace transform (§4.5).

4.1 What "no convection" does

Heat then moves only by conduction, $\partial H/\partial t = \nabla\cdot(k\nabla T)$ with H the enthalpy per unit volume, in the drink, the wall, the brine and the slush. The second consequence is easy to miss: **dissolved salt and ethanol stay put**. They could only move by diffusion, with $D \approx 1.5\times10^{-9}\,\mathrm{m^2/s}$, while heat diffuses with $\alpha \approx 1.3\times10^{-7}\,\mathrm{m^2/s}$. The ratio (the Lewis number) is about 90. In a 200 s cooling, salt moves about $\sqrt{Dt} \approx 0.5$ mm while heat reaches about 5 mm. So the water from ice melted by the can stays where it melted.

4.2 The diluted brine at the can: the coefficient you asked for

Take one cubic metre of slush next to the can, with ice volume fraction φ_p and its pores full of bath brine (mass fractions w_w, w_s, w_e):

$$W_0 = (1-\varphi_p)\rho_b w_w, \quad m_{I0} = \varphi_p \rho_i, \quad n_{s,v} = \frac{(1-\varphi_p)\rho_b w_s}{M_s}, \quad n_{e,v} = \frac{(1-\varphi_p)\rho_b w_e}{M_e}, \tag{4.1}$$

with $N_v = K_s n_{s,v} + K_e n_{e,v}$ and $W_{b,v} = \omega_s n_{s,v} + \omega_e n_{e,v}$. Because the solutes cannot leave, Eq. (2.5) applies to this cubic metre with its own fixed N_v. At depression θ it holds liquid water $W_l(\theta) = W_{b,v} + N_v/\theta$ and ice $m_l(\theta) = W_0 + m_{I0} - W_l(\theta)$. Far away it sits at θ_b with $W_l = W_0$. Heated, it melts progressively, and the last ice goes when $W_l = W_0 + m_{I0}$:

$$\theta_L = \frac{N_v}{W_0 + m_{I0} - W_{b,v}}, \qquad \boxed{\eta_c \equiv \frac{\theta_L}{\theta_b} = \frac{W_0 - W_{b,v}}{W_0 + m_{I0} - W_{b,v}}} \tag{4.2}$$

η_c is the ratio of free water in the brine alone to free water in brine plus ice. It is a "lowered coefficient" that needs no fitting: it follows from the no-convection assumption and the ice fraction. For B1 ($W_0 = 407$, $m_{I0} = 504$, $W_{b,v} = 92.5\,\mathrm{kg/m^3}$), $\eta_c = 0.384$. The last ice next to the can therefore melts at $T_L = -8.14\,°\mathrm{C}$, while the bath a few centimetres away is at $-21.2\,°\mathrm{C}$.

Two different coefficients, both lowered

> Your mixing coefficient η_s (§2.1) describes the *whole bath*: how much of the salt you poured actually dissolved. It sets T_b. The no-convection coefficient η_c describes the *few millimetres next to the can*: how much the can's own meltwater dilutes the brine there. They multiply: $\theta_L = \eta_c\,\theta_b(\eta_s)$. If undissolved salt crystals sit in the slush right against the can, η_c moves towards 1, because melting can then stay at the eutectic until the local crystals run out.

4.3 The slush is a heat sink with its latent heat spread out

Per unit volume, measuring from all-liquid at 0 °C, the enthalpy of the slush is

$$H(\theta) = -m_l(\theta)\,(L_0 - \Delta c\,\theta) - \theta\,C_v, \qquad C_v = c_w(W_0 + m_{I0}) + c_s M_s n_{s,v} + c_{ea} M_e n_{e,v}. \tag{4.3}$$

Its slope with temperature, the *apparent heat capacity*, is

$$c_{\mathrm{app}} = \frac{\mathrm{d}H}{\mathrm{d}T} = C_v - \Delta c\, m_l(\theta) + (L_0 - \Delta c\,\theta)\frac{N_v}{\theta^2} \qquad (\theta_L < \theta < \theta_b), \qquad c_{\mathrm{app}} = C_v \quad (\theta < \theta_L).$$

Three features matter.

- There is **no latent-heat jump** at T_L: the ice fraction falls continuously to zero there.
- Most of the latent heat is taken up near T_L, because N_v/θ^2 is largest there.
- The total heat needed to turn slush at T_b into liquid at T_L is

$$\Delta H = m_{I0}\,(L_0 - \Delta c\,\theta_b) + C_v\,(\theta_b - \theta_L) \qquad (\text{B1: } 196.7\,\mathrm{MJ/m^3},\ C_v = 3.92\,\mathrm{MJ\,m^{-3}K^{-1}}). \tag{4.4}$$

The numerical solution of the full conduction problem (Fig. 5) shows what this does. Behind the full-melt isotherm there is only a thin liquid layer: 3.1 mm after 240 s. Ahead of it a broad, partly melted zone reaches 1–2 cm, and it holds 37 % of the heat taken from the can after 60 s and 46 % after 240 s. A sharp-front (Stefan) model that puts all the latent heat at T_L ignores that zone, and comes out too slow: by +18 % at $\theta_b \approx 5\,\mathrm{K}$ and by +57 % at the eutectic (Appendix 6.2).

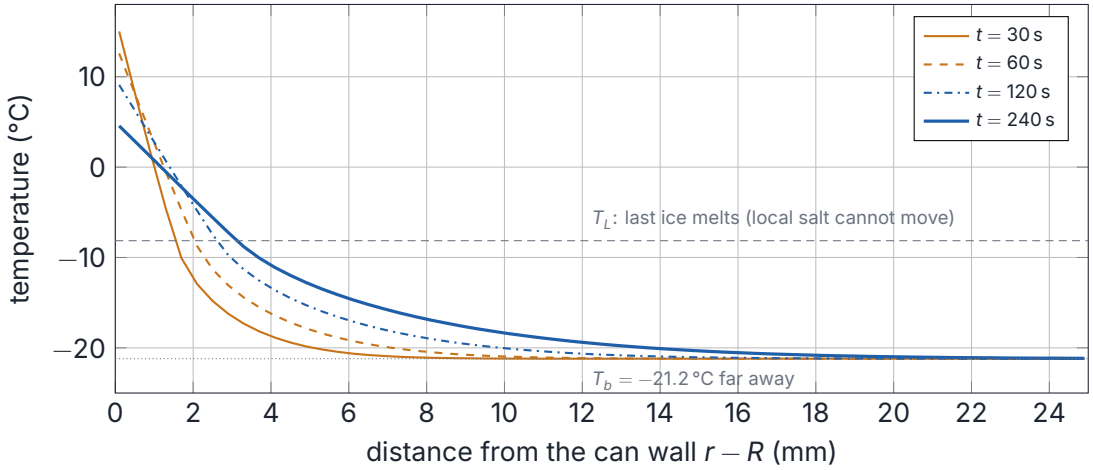


Figure 5. Temperature around a well-mixed Can 2 in B1 slush ($\varphi_p = 0.55$), from the numerical solution of the conduction problem. Left of the dashed line T_L the slush has fully melted (a few mm). To the right, a broad partly melted zone at temperatures between T_L and T_b soaks up nearly half of the heat.

4.4 One number for the slush: its effective effusivity

The classical problem. Hold a flat wall at temperature T_w against a half-space of slush at T_b. With no length scale in the problem, the solution depends only on $\xi/\sqrt{t}$ (ξ is the distance from the wall). Treat the liquid layer (conductivity k_l, heat capacity C_v, diffusivity $\alpha_l = k_l/C_v$) and the partly melted zone separately. Give the second constant effective properties: $c_m = \Delta H/(\theta_b - \theta_L)$, and k_m = the mean slush conductivity between T_b and T_L (ice fraction weighted, Maxwell–Eucken), with $\alpha_m = k_m/c_m$. The isotherm T_L then moves as $s = 2\lambda\sqrt{\alpha_l t}$ and

$$T = T_w - (T_w - T_L)\frac{\operatorname{erf}\left(\xi/2\sqrt{\alpha_l t}\right)}{\operatorname{erf}\lambda}\ (\xi < s), \qquad T = T_b + (T_L - T_b)\frac{\operatorname{erfc}\left(\xi/2\sqrt{\alpha_m t}\right)}{\operatorname{erfc}\left(\lambda\sqrt{\alpha_l/\alpha_m}\right)}\ (\xi > s).$$

There is no latent jump at T_L, so the heat flux must be continuous there. That fixes λ:

$$\frac{E_l\,(T_w - T_L)\,e^{-\lambda^2}}{\operatorname{erf}\lambda} = \frac{E_m\,(T_L - T_b)\,e^{-\lambda^2\alpha_l/\alpha_m}}{\operatorname{erfc}(\lambda\sqrt{\alpha_l/\alpha_m})}, \qquad E_l = \sqrt{k_l C_v}, \quad E_m = \sqrt{k_m c_m}. \tag{4.5}$$

This is one equation in one unknown, solved once (B1: $\lambda = 0.296$). The flux through the wall is then

$$q = \frac{E_l(T_w - T_L)}{\operatorname{erf}\lambda\,\sqrt{\pi t}} \equiv \frac{E\,(T_w - T_b)}{\sqrt{\pi t}}, \qquad \boxed{E = \frac{E_l\,(T_w - T_L)}{(T_w - T_b)\,\operatorname{erf}\lambda}} \tag{4.6}$$

E is exactly what a uniform material of effusivity E would give. So, as far as the wall can tell, the slush behaves like such a material, with an effective conductivity and heat capacity

$$k = \frac{E^2}{c_{\text{sec}}}, \qquad c_{\text{sec}} = \frac{\Delta H + C_v(T_w - T_L)}{T_w - T_b}. \tag{4.7}$$

For B1: $E_l = 1\,428$, $E_m = 3\,685$ and $E = 2\,576\,\mathrm{W\,s^{1/2}\,m^{-2}\,K^{-1}}$, with $k = 0.78\,\mathrm{W\,m^{-1}K^{-1}}$. In plain ice water ($\theta_b = 0$) the ice melts at one temperature, so Eq. (4.5) becomes the textbook one-phase Stefan condition $\lambda e^{\lambda^2}\operatorname{erf}\lambda = \mathrm{Ste}/\sqrt{\pi}$, with $\mathrm{Ste} = C_v T_w/(m_{l0}L_0)$.

Which wall temperature? The drink cools, so T_w falls from T_0 to T_1 and E changes along the way. Two limits fix the choice. When the partly melted range $\theta_b - \theta_L$ is wide compared with how far the drink ends above T_L, the slush behaves almost linearly and the time-averaged drink temperature $\bar{T}$ is right. When that range is narrow (fresh water), the slush behaves like a sharp front, the last stage dominates, and T_1 is right. Interpolating between them,

$$T_{\text{ref}} = w\,\bar{T} + (1 - w)\,T_1, \qquad w = \min(1, \psi), \qquad \psi = \frac{\theta_b - \theta_L}{T_1 - T_L}, \tag{4.8}$$

with $\bar{T}$ from Eq. (4.12) below (two or three passes converge). B1 has $\psi = 0.99$, so $T_{\text{ref}} = 10.3\,°\text{C}$. Plain ice water has $\psi = 0$, so $T_{\text{ref}} = T_1$. This rule is the only empirical step in Part III. Without it, the error against the full solution rises from 10.7 % to about 50 % for nearly fresh baths.

4.5 The cooling law

Set-up. Let $\Theta(t) = T_d(t) - T_b$ be the drink's excess over the bath. A well-mixed drink loses heat through its contact area A:

$$C\,\frac{\mathrm{d}\Theta}{\mathrm{d}t} = -A\,q(t).$$

For a medium of effusivity E and conductivity k around a cylinder of radius R, the response to a wall excess $\Theta_w(t)$ is, in Laplace variables ($\hat{f}(s) = \int_0^\infty f e^{-st}\mathrm{d}t$),

$$\hat{q}(s) = \Big(E\sqrt{s} + \frac{k}{2R}\Big)\hat{\Theta}_w(s).$$

The first term is the flat-wall response, $q = E\Theta_w/\sqrt{\pi t}$ for a step. The second is the first correction for the curved surface (for a step it adds the steady $k\Theta_w/2R$). The aluminium has no resistance and the drink is well mixed, so $\Theta_w = \Theta$. Transforming the drink equation, with $\Theta(0) = \Theta_0$:

$$C(s\hat{\Theta} - \Theta_0) = -A\Big(E\sqrt{s} + \frac{k}{2R}\Big)\hat{\Theta} \quad\Longrightarrow\quad \hat{\Theta} = \frac{\Theta_0}{s + a\sqrt{s} + b} = \frac{\Theta_0}{(\sqrt{s}+p_1)(\sqrt{s}+p_2)},$$

$$a = \frac{AE}{C}, \qquad b = \frac{A_{\text{lat}}\,k}{2RC}, \qquad p_{1,2} = \frac{a \mp \sqrt{a^2 - 4b}}{2}. \tag{4.9}$$

Split into partial fractions, $1/[(\sqrt{s}+p_1)(\sqrt{s}+p_2)] = [1/(\sqrt{s}+p_1) - 1/(\sqrt{s}+p_2)]/(p_2 - p_1)$. Then use the standard pair $\mathcal{L}^{-1}\{1/(\sqrt{s}+p)\} = 1/\sqrt{\pi t} - p\,\operatorname{erfcx}(p\sqrt{t})$, where $\operatorname{erfcx}(z) = e^{z^2}\operatorname{erfc}(z)$. The $1/\sqrt{\pi t}$ terms cancel, leaving

$$\boxed{T_d(t) = T_b + \Theta_0\,\frac{p_2\,\operatorname{erfcx}(p_2\sqrt{t}) - p_1\,\operatorname{erfcx}(p_1\sqrt{t})}{p_2 - p_1}} \tag{4.10}$$

Differentiating with $\frac{\mathrm{d}}{\mathrm{d}t}\operatorname{erfcx}(p\sqrt{t}) = p^2\operatorname{erfcx}(p\sqrt{t}) - p/\sqrt{\pi t}$ gives the cooling rate. The time average needed in Eq. (4.8) also has a closed form:

$$-\frac{\mathrm{d}T_d}{\mathrm{d}t} = \Theta_0\Big[\frac{a}{\sqrt{\pi t}} - \frac{p_2^3\operatorname{erfcx}(p_2\sqrt{t}) - p_1^3\operatorname{erfcx}(p_1\sqrt{t})}{p_2 - p_1}\Big], \tag{4.11}$$

$$\bar{\Theta} = \frac{1}{t_1}\int_0^{t_1}\Theta\,\mathrm{d}t = \frac{\Theta_0}{t_1(p_2 - p_1)}\Big[\frac{\operatorname{erfcx}(p_2\sqrt{t_1}) - 1}{p_2} - \frac{\operatorname{erfcx}(p_1\sqrt{t_1}) - 1}{p_1}\Big], \qquad \bar{T} = T_b + \bar{\Theta}. \tag{4.12}$$

The time to reach the target, t_5, is the single root of $T_d(t_5) = T_1$ in Eq. (4.10), which falls monotonically. The minimum cooling rate inside the window $[T_0, T_1]$ is Eq. (4.11) at t_5.

What the law says.

- **Short times:** $\Theta \approx \Theta_0(1 - 2a\sqrt{t/\pi})$. The rate starts infinite and falls like $1/\sqrt{t}$, because a cold medium pressed against a hot wall has no boundary layer yet. In practice the first second is limited by contact and by how well the drink mixes.
- **Long times:** $\Theta \approx \Theta_0\,a/(2\sqrt{\pi}\,b^2 t^{3/2})$, a power law rather than an exponential.
- **The effective conductance falls with time:** $q/\Theta \approx E/\sqrt{\pi t} + k/2R$, about $200\,\text{W}\,\text{m}^{-2}\text{K}^{-1}$ after one minute in B1 and about half that after four. This is why "a time constant τ" does not exist here.
- **Real or complex roots:** for a can wholly in one medium, $C \approx \pi R^2 h\rho_d c_d$ and $A_{\text{lat}} = 2\pi Rh$. Then $a^2/4b = c_{\text{sec}}/(\rho_d c_d)$, so the roots are real when the medium stores more heat per kelvin than the drink. Slush does ($c_{\text{sec}} \approx 2\rho_d c_d$). Clear brine is marginal ($a^2/4b = 0.96$): the roots are complex conjugates, and Eq. (4.10) still holds with the complex erfcx (the Faddeeva function, `scipy.special.wofz`).

4.6 Several contact zones, the bottom and the fin

Each zone of wall sees the same drink temperature, and the responses add. So the closed form keeps its shape, with

$$a = \frac{(A_{sl} + A_{\mathrm{fin},sl})E + (A_{br} + A_{\mathrm{fin},br})E_{br} + A_{\mathrm{bot}}E_{\mathrm{bot}}}{C}, \qquad b = \frac{(A_{sl} + A_{\mathrm{fin},sl})k + (A_{br} + A_{\mathrm{fin},br})k_b}{2RC}, \tag{4.13}$$

where the clear brine is an ordinary conductor, $E_{br} = \sqrt{k_b \rho_b c_b}$ (c_b is the brine's specific heat; $E_{br} = 1\,447$ for B1), and the flat bottom contributes no curvature term. The fin needs $\bar{h}$, which comes from the first pass:

$$\bar{h} = \frac{C\,(\Theta_0 - \Theta_1)}{A\,t_5\,\bar{\Theta}}. \tag{4.14}$$

Evaluation order: first $T_{\mathrm{ref}} \to E, k \to a, b \to t_5, \bar{T}, \bar{h}$; then T_{ref} and A_{fin} again; repeat once more.

4.7 Is the closed form right? Check against the full conduction solution

I solved the full problem numerically with no closed-form steps, as an independent check. It is a one-dimensional radial enthalpy method: the drink (well mixed, or conducting), the aluminium, and slush with immobile solutes, using its exact $H(T)$ from Eq. (4.3) and a Maxwell–Eucken conductivity. Halving the grid spacing and quartering the time step changes the answer by less than 0.1 s.

case	θ_b (K)	φ_p	ψ	full solution t_5 (s)	closed form t_5 (s)	error (%)
plain ice water	0.00	0.55	0.00	908	959	+5.7
weak salt 50 g	1.28	0.55	0.12	774	798	+3.1
weak salt 100 g	2.54	0.55	0.22	681	695	+2.1
salt 200 g	4.99	0.55	0.38	553	558	+0.9
salt 300 g	7.37	0.55	0.50	468	468	−0.1
salt 500 g	11.87	0.55	0.69	361	355	−1.7
salt 700 g	16.08	0.55	0.84	295	286	−3.0
B1 eutectic	21.20	0.55	0.99	239	229	−4.4
B1, loose slush	21.20	0.35	0.50	310	303	−2.3
B1, dense slush	21.20	0.65	1.35	216	192	−10.7
B2 salt+ethanol	24.38	0.55	1.09	211	199	−5.9
B3 ethanol only	13.24	0.55	0.77	336	329	−2.2
B4 rig charge	14.68	0.55	0.79	315	306	−2.6
B1, Can 1	21.20	0.55	0.99	182	174	−4.3
B4, Can 1	14.68	0.55	0.79	239	233	−2.6
plain ice water, Can 1	0.00	0.55	0.00	689	729	+5.8

Over 16 cases, from plain ice water to the cotectic, two can sizes and three ice fractions, the closed form for the lateral surface has mean error −1.4 %, rms 4.4 % and worst 10.7 % (dense slush, $\varphi_p = 0.65$). For clear brine alone it is −1.1 % (638 vs 645 s). Dropping the curvature term would be +29 % off.

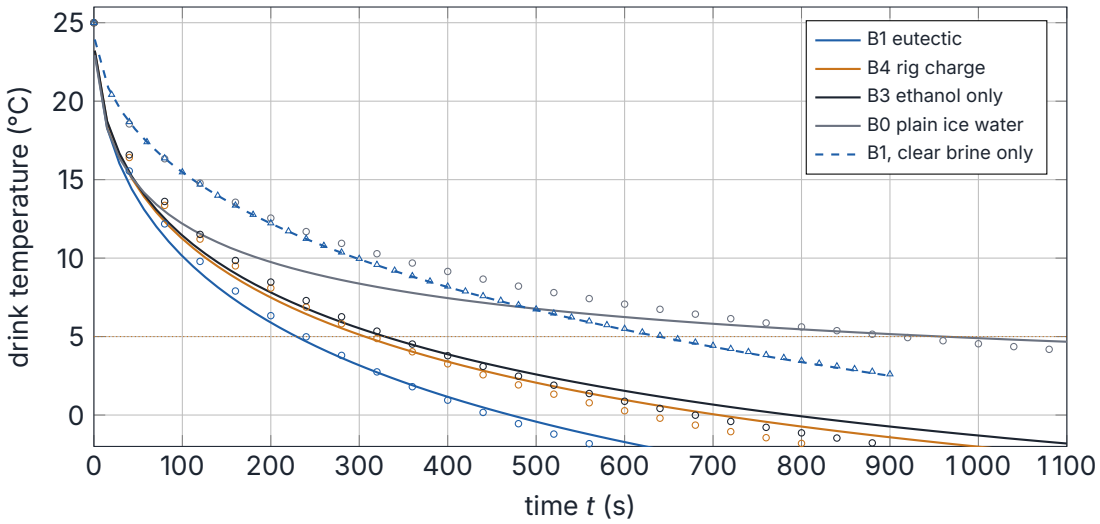


Figure 6. Drink temperature for a well-mixed Can 2 (side wall only, as in the check), from Eq. (4.10) (lines) and from the full numerical solution (symbols).

4.8 Results for your rig

Full can (side wall, bottom and fin), Can 2 with 330 mL, base 1.0 cm off the floor, $\varphi_p = 0.55$, cooling $25 \to 5$ °C:

recipe	ice held	T_b (°C)	E slush ($\mathrm{W\,s^{1/2}m^{-2}K^{-1}}$)	t_5 (s)	rate at 15 °C (K/min)	rate at 5 °C (K/min)	mean rate (K/min)
B0 plain ice	floor	0.0	5 446	640	6.9	0.22	1.87
B1 eutectic	floats	−21.2	2 574	223	7.0	1.79	5.38
	board		2 575	203	7.7	1.95	5.91
	floor		2 576	151	10.4	2.59	7.93
B2 salt + ethanol	board	−24.4	2 496	185	8.2	2.23	6.48
	floor		2 497	138	11.1	2.97	8.73
B3 ethanol only	board	−13.2	2 960	324	5.4	1.06	3.71
	floor		2 963	222	8.1	1.52	5.41
B4 original charge	board	−14.7	2 867	447	3.8	0.81	2.68
	floor		2 870	287	6.0	1.23	4.18

What the table says:

- **Salt pays**, even with no convection. Going from plain ice water to the eutectic cuts the time by 4.2 × with the ice at the floor.
- **Ice placement pays almost as much.** Slush around the whole drink column beats a floating pack by 1.47 ×. Clear brine has no latent store, so its effusivity is only 1 447.
- **Ethanol does not pay here.** B2 is only slightly faster than B1 despite being 3 K colder. B3 (ethanol only) is much slower because it is warmer and has melted more of the ice.
- **B4 is slow for a geometric reason as much as a thermal one:** with a lid board almost all of its drink column sits in clear brine.

A check against published data. SINTEF timed a 0.33 L can from 21 to 5 °C: about 600 s in ice water and 150 s in a −16 °C salt bath. The same calculation (21 °C start, Can 2, ice at the floor, $m_s = 0.696$ kg for −16 °C) gives 445 s and 128 s (154 s with a lid board). With no convection inside the drink either, the full numerical solution gives 2 197 s and 752 s. Both measurements sit near the well-mixed limit and far from the no-convection-inside limit: a real drink mixes itself well by natural convection. The ice-water measurement is about a third slower than the well-mixed prediction, which is what you would expect if water rather than packed slush surrounded part of the can.

4.9 No convection inside the drink either

Taken literally, “disregard convection” also stops the drink from mixing, so heat has to diffuse through it too. This is the **true minimum cooling rate**: any convection, inside or outside, can only speed things up.

The contact temperature. Two materials of effusivity $E_d = \sqrt{k_d \rho_d c_d}$ (drink, 1536) and E (slush) brought into contact at once jump to a common surface temperature. To leading order it stays there:

$$\boxed{T_c = \frac{E_d\, T_0 + E\, T_b}{E_d + E}} \tag{4.15}$$

For B1, $T_c = -7.38\,°\mathrm{C}$; the full numerical solution’s wall temperature after 300 s is $-7.37\,°\mathrm{C}$.

The drink at the wall must supercool or freeze

> T_c is far below the drink’s freezing point ($-0.77\,°\mathrm{C}$). In a no-convection world the layer of drink against the wall either supercools, which canned drinks often do, or freezes onto the wall while the centre is still near 25 °C. The numbers below assume it supercools; freezing would add a second moving front.

Closed form for the mean drink temperature. Inside a solid cylinder the response, to the same order, is $E_d\sqrt{s} - k_d/2r_i$. The curvature now *reduces* the flux, because the inside runs out of heat. Matching it to the outside response $E\sqrt{s} + k/2R$ at the wall and inverting the transform (Appendix 6.1) gives

$$\boxed{\frac{\bar{\Theta}(t)}{\Theta_0} = 1 - \frac{A}{C\,D}\left[E_d E f_2 + \beta f_3 - \gamma f_4\right]} \tag{4.16}$$

$$D = E_d + E, \qquad g_d = \frac{k_d}{2r_i}, \qquad g_s = \frac{k}{2R}, \qquad c = \frac{g_s - g_d}{D}, \qquad \beta = E_d g_s - E g_d, \qquad \gamma = g_d g_s,$$

$$f_1 = \operatorname{erfcx}(c\sqrt{t}), \qquad f_2 = \frac{1 - f_1}{c}, \qquad f_3 = \frac{1}{c}\left(2\sqrt{\frac{t}{\pi}} - f_2\right), \qquad f_4 = \frac{1}{c}\,(t - f_3).$$

It needs a single erfcx and holds while the drink’s Fourier number $\alpha_d t / r_i^2 \lesssim 0.15$. For a linear outside medium it agrees with the full solution to 4 %. In slush it is −13 % (B1) and −9 % (B4) against the full solution, because the wall temperature drifts down through the partly melted range. Use the full-solution numbers as the bound:

	T_c (°C)	t_5 mean, full solution	t_5 mean, Eq. (4.16)	t_5 centre, full solution	rate at 5 °C (full)
B1	−7.38	813 s	711 s	1940 s	0.66 K/min
B4	−3.57	1052 s	954 s	2193 s	0.49 K/min

So for B1 the drink’s mean temperature takes about 813 s, and its centre 1940 s, compared with 151–223 s for a well-mixed drink. These are side-wall-only numbers, which is conservative, as a bound should be.

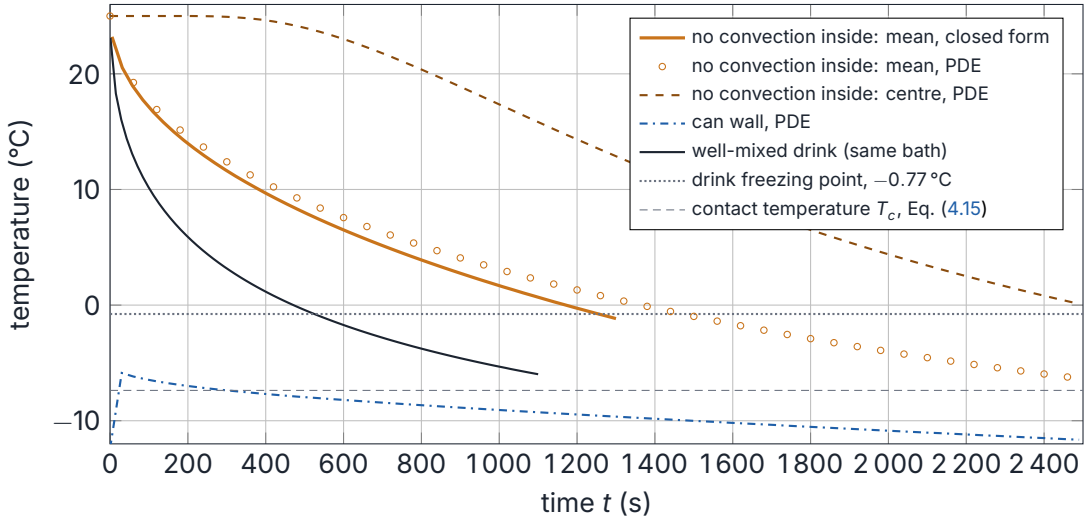


Figure 7. B1, Can 2, side wall only: with no convection inside the drink, the wall drops at once to about T_c and stays below the drink's freezing point. The mean temperature falls like $\sqrt{t}$ and the centre lags far behind. The well-mixed drink in the same bath is shown for comparison.

5 The whole chain, design and what to measure

5.1 From your four masses to the drink temperature

Evaluation order: each line uses only quantities from the lines above and the measured constants

1.	Active solute n'_s, n'_e	Eq. (2.1)
2.	Heat capacities C_r, C_θ	Eq. (2.3)
3.	Depression strength N and bound water W_b	Eq. (2.5)
4.	A, B, Q_0; bath depression θ_U; then W, x	Eqs. (2.6), (2.7)
5.	If saturated: W_S and θ_S. If $x > m_i$: θ_l	Eqs. (2.8)–(2.11)
6.	With ethanol: update h_E, c_{ea} at $b_e = n'_e/W$ and repeat 1–5 once	§2.10
7.	Brine: $b_s = n_d/W$, $b_e = n'_e/W$, m_b, w_s, w_e, ρ_b	§3.3
8.	Can: r_i, h_d, V_a, C, M_c	Eqs. (3.1), (3.2)
9.	Tank: V_l, V_b, h_p, V_p, then the layers for your way of holding the ice	Eqs. (3.4), (3.5)
10.	Contact lengths and areas A_{sl}, A_{br}, A_{bot}	Eq. (3.6)
11.	Slush at the can: W_0, m_{l0}, N_v, $W_{b,v}$; then θ_L, η_c, C_v, ΔH, c_m, k_m	Eqs. (4.1)–(4.4)
12.	Effusivity: T_{ref}, then λ, E, k; clear brine E_{br}	Eqs. (4.8), (4.5)–(4.7)
13.	Cooling: a, b, $p_{1,2}$; $T_d(t)$; t_5; rate; $\bar{T}$; $\bar{h}$; fin A_{fin}	Eqs. (4.9)–(4.14), (3.7)
14.	Repeat 12–13 twice (the reference temperature and the fin settle)	
15.	Strict lower bound, no convection inside: T_c and $\bar{\Theta}(t)$	Eqs. (4.15), (4.16)

The only numerical steps are two scalar roots, λ in Eq. (4.5) and t_5 in Eq. (4.10); both functions are monotonic. Everything else is explicit.

5.2 What depends on what

derived quantity	is a function of	equation
D n_s', n_e'	m_s, m_e; η_s, η_e	(2.1)
D C_r, C_θ	m_w, m_s, m_e, m_i; c's, η_e	(2.3)
D N, W_b	n_s', n_e'; $K_s, K_e, \omega_s, \omega_e$	(2.5)
D θ_b $(= -T_b)$	m_w, m_s, m_e, m_i; T_r; $L_0, \Delta c, \Delta h_s, h_E, K$'s, ω's, b_E, θ_E, λ; η_s, η_e	(2.7), (2.8), (2.9), (2.11)
D W, x, n_d	as θ_b	same
D ρ_b, m_b, V_b, V_l	W, n_d, n_e', x; $\rho_0, \beta_s, \beta_e, \rho_i$	(3.4)
D h_d, V_a, C, M_c	$V_d, R, H, t_w, m_{\mathrm{Al}}$; $\rho_d, c_d, c_{\mathrm{Al}}$	(3.1), (3.2)
D layers h_{cb}, h_b, h_p	V_b, V_l, ρ_b; A_t, R, φ_p; how the ice is held	(3.5)
D L_{sl}, L_{br}, A's	layers; z_0, h_d, H, R	(3.6)
D θ_L, η_c	θ_b, brine composition, ρ_b; φ_p, ρ_i	(4.2)
D ΔH, C_v, c_m, k_m	θ_b, θ_L, slush composition; $L_0, \Delta c$, c's, k_i, k_b	(4.3), (4.4)
D λ, E, k	$T_{\mathrm{ref}}, T_b, T_L$; k_l, C_v, k_m, c_m	(4.5)–(4.7)
D a, b, $p_{1,2}$	A's, E, k, E_{br}, k_b, C, R	(4.9), (4.13)
D $T_d(t)$, t_5, rate	a, b, T_b, T_0; T_1	(4.10), (4.11)
D T_c, $\bar{\Theta}(t)$ (no convection inside)	$E_d, E, k, k_d, r_i, R, A/C, T_0, T_b$	(4.15), (4.16)

5.3 Design: what each mass does to the time

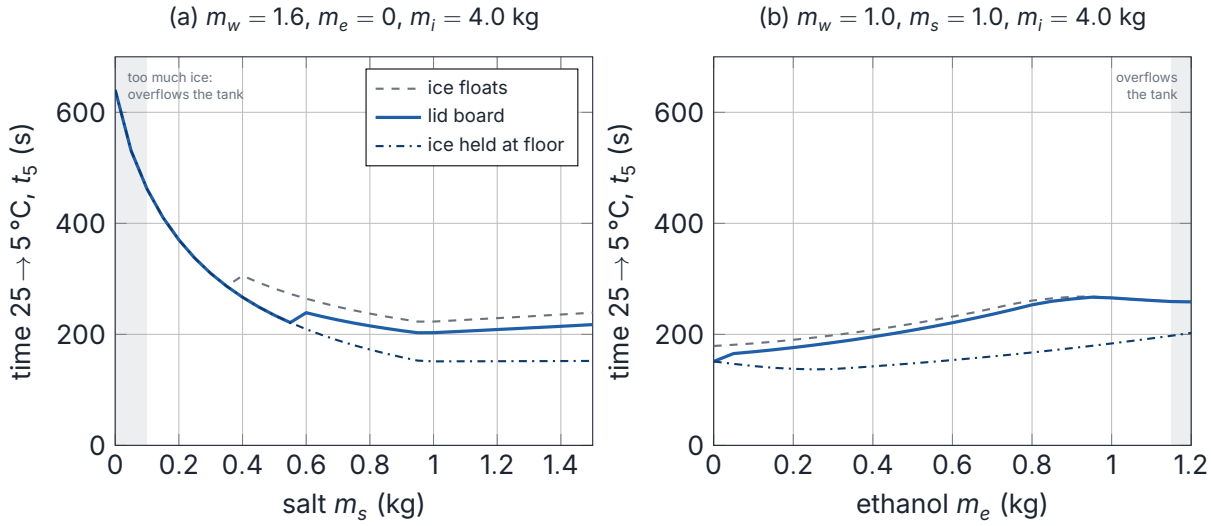


Figure 8. Time to cool Can 2 from 25 to 5 °C against (a) salt and (b) ethanol, for the three ways of holding the ice. Grey bands: charges that do not fit your 17.5 cm tank with the can in. The small steps are where clear brine first reaches the can's base, which then sits in brine instead of slush.

- **Salt (a).** The time falls steeply until the brine saturates, near m_s^* (Eq. (2.12)). Past that point the bath is a thermostat, and extra salt only brings in warm mass that melts a little more ice. With the ice floating or under a board, that ice comes off the pack around the can.
- **Ethanol (b).** Each 100 g lowers T_b, but each 100 g also melts ice through its warmth and its heat of mixing. With no convection the second effect wins sooner than you would guess, because the can ends up in clear brine.
- **Ice.** Below saturation, more ice at 0 °C makes the bath very slightly *warmer* (§2.8). What it really changes is how much slush there is around the can.
- **Water.** In regime U, each kilogram of warm water raises T_b by 5.0 K. In regime S the temperature stays put, and each kilogram of water instead adds 1.563 kg of liquid water (itself plus the ice it melts).

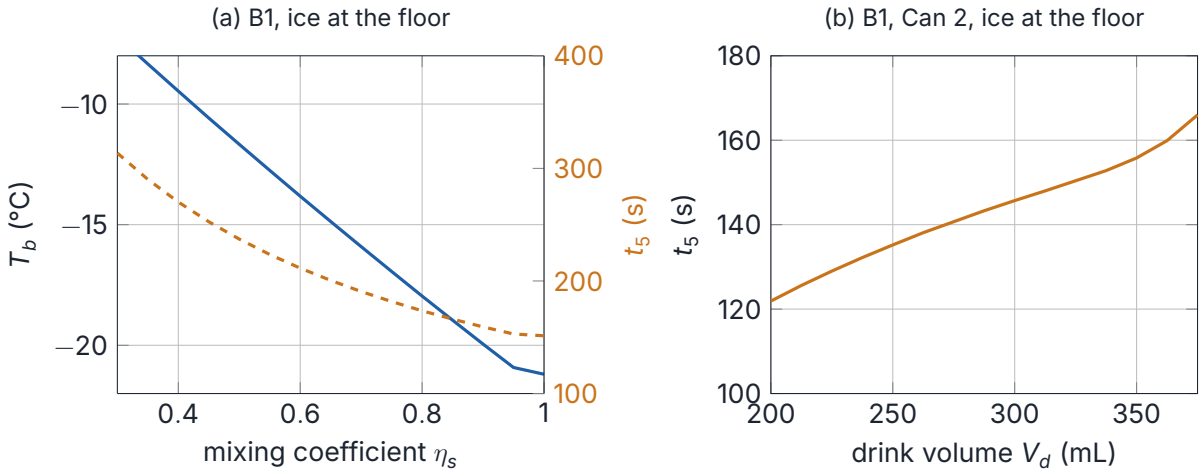


Figure 9. (a) B1 with the ice at the floor: bath temperature (blue, left axis) and time to 5 °C (amber dashed, right axis) against the salt mixing coefficient η_s. (b) Time to 5 °C against the volume of drink in Can 2: nearly doubling the drink changes the time by about a third.

With $\eta_s = 0.3$ (badly mixed salt) B1 only reaches −7.2 °C and needs 314 s. Changing the fill from 200 to 375 mL moves t_5 only from 122 to 166 s, because the drink's heat capacity and its wetted wall both scale with the fill. What remains comes from the fixed bottom, the aluminium and the shrinking fin.

5.4 What to measure, and how

1. φ_p**, the ice packing.** Fill a jug with crushed ice to a mark, top it up with ice water to the same mark, and record the water added: $\varphi_p = 1 - V_{\text{water}}/V_{\text{mark}}$. It enters η_c, the layering and E.
2. η_s**, the salt mixing.** Let the bath settle (no stirring) and read T_b at least 5 cm from the can, then apply Eq. (2.14). Do it for each way of adding the salt (all at once, in layers, pre-dissolved).
3. **The layering.** Mark h_{cb} and h_b on the tank wall and compare with Eq. (3.5). That tests φ_p and ρ_b in one go.
4. T_L**, the diluted brine.** A fine thermocouple bead 2–3 mm from the can wall should read between T_L and T_b while the can cools (Fig. 5). If it reads close to T_b, convection is carrying the meltwater away and η_c is closer to 1 than Eq. (4.2) says.
5. **The shape of the cooling curve.** Fit Eq. (4.10) with E as the only free parameter, and also fit an exponential. The ratio $E_{\text{fit}}/E_{\text{predicted}}$ is a direct measure of how much convection actually helps in your tank: 1 means none.

5.5 Limits of this model, bluntly

- **Real baths convect.** Meltwater is fresh and light and rises; brine is heavy and sinks. In reality the diluted layer is partly swept away, so the local coefficient lies between η_c and 1. Treat this model as the no-flow limit, not a forecast of a stirred tank.
- **The well-mixed drink** is an assumption. A real drink sits between the two limits given here, near the well-mixed one according to the SINTEF data.
- **The slush as a linear medium** is an approximation: within ±10.7 % of the full conduction solution in 16 cases. The reference-temperature rule, Eq. (4.8), is the one empirical step. The bottom and the fin are added by superposition and were not checked against a 2-D solution.
- **Bath thermodynamics.** The liquidus is good to 1.31 K, and the cotectic below −23.6 °C is extrapolated (B2). Δh_s, c_s and the ethanol mixing table are placeholders carried over from the earlier model; the H^E table came from a scrambled extraction and needs checking against the paper. Hydrohalite formation and dissolution kinetics are not modelled; η_s absorbs them empirically.
- **Geometry.** The can is a cylinder (dome and neck ignored) in perfect contact with the slush. The ice pack is uniform, although real packs are looser near walls. The tank height, 17.5 cm, was inferred,

not measured.

6 Appendix

6.1 Laplace pairs and the two inversions

With $\hat{f}(s) = \int_0^\infty f(t)e^{-st}\,\mathrm{d}t$, the pairs used are

$$\mathcal{L}\{1\} = \frac{1}{s}, \quad \mathcal{L}\{t\} = \frac{1}{s^2}, \quad \mathcal{L}\Big\{\frac{1}{\sqrt{\pi t}}\Big\} = \frac{1}{\sqrt{s}}, \quad \mathcal{L}\Big\{2\sqrt{\tfrac{t}{\pi}}\Big\} = \frac{1}{s^{3/2}}, \quad \mathcal{L}\{\mathrm{erfcx}(p\sqrt{t})\} = \frac{1}{\sqrt{s}\,(\sqrt{s}+p)}.$$

Writing $1/(\sqrt{s}+p) = 1/\sqrt{s} - p/[\sqrt{s}(\sqrt{s}+p)]$ gives $\mathcal{L}^{-1}\{1/(\sqrt{s}+p)\} = 1/\sqrt{\pi t} - p\ \mathrm{erfcx}(p\sqrt{t})$, the pair used for Eq. (4.10).

Well-mixed can (§4.5). $\hat{\Theta}/\Theta_0 = 1/[(\sqrt{s}+p_1)(\sqrt{s}+p_2)]$. Partial fractions and the pair above give Eq. (4.10), valid for real or complex-conjugate $p_{1,2}$.

Drink with no convection inside (§4.9). Let $\sigma = \sqrt{s}$. The drink, initially uniform at excess Θ_0, delivers $\hat{q} = Y_d(\Theta_0/s - \hat{\Theta}_w)$ to the wall, with $Y_d = E_d\sigma - g_d$. The outside takes $\hat{q} = Y_s\hat{\Theta}_w$, with $Y_s = E\sigma + g_s$. Equating the two:

$$\hat{\Theta}_w = \frac{Y_d\Theta_0}{s(Y_d+Y_s)}, \qquad \hat{q} = \frac{Y_dY_s\,\Theta_0}{s\,(Y_d+Y_s)}, \qquad \frac{\hat{\bar{\Theta}}}{\Theta_0} = \frac{1}{s} - \frac{A}{C}\frac{\hat{q}}{s\,\Theta_0} = \frac{1}{s} - \frac{A}{CD}\,\frac{E_dE\sigma^2 + \beta\sigma - \gamma}{\sigma^4(\sigma+c)},$$

using $Y_d + Y_s = D(\sigma + c)$ and $Y_dY_s = E_dE\sigma^2 + \beta\sigma - \gamma$. The inverse of $1/[\sigma^k(\sigma+c)]$, call it f_k, follows from $1/[\sigma^k(\sigma+c)] = c^{-1}\{\sigma^{-k} - 1/[\sigma^{k-1}(\sigma+c)]\}$. Starting from $f_1 = \mathrm{erfcx}(c\sqrt{t})$, this gives the recursion in Eq. (4.16). Numerically, $c\sqrt{t}$ is small, so evaluate the recursion with extra precision (or by its power series); the subtractions cancel otherwise.

6.2 The classical Stefan limit (sharp melting front)

If all the latent heat were absorbed at a sharp front at temperature T_f (true for plain ice water, with $T_f = 0\,°\mathrm{C}$), the liquid layer of thickness s around the can would carry heat quasi-steadily: $q = k_l(T_w - T_f)/[R\ln(1 + s/R)]$. Conservation of energy relates s to how far the drink has cooled. Including the sensible heat stored in the layer (mean excess about $\Theta/2$), with $\Theta = T_d - T_f$:

$$\Big(1 + \frac{s}{R}\Big)^2 = 1 + \frac{\beta\,(\Theta_0 - \Theta)}{1 + \gamma\Theta}, \qquad \beta = \frac{C}{\pi R^2 h\,\Delta H}, \qquad \gamma = \frac{C_v}{2\Delta H}.$$

For a well-mixed drink, $\mathrm{d}t = -(CR/Ak_l)\ln(1 + s/R)\,\mathrm{d}\Theta/\Theta$. The integral is two logarithms of linear functions divided by Θ, and each gives a dilogarithm Li_2:

$$t(\Theta) = \frac{CR}{2Ak_l}\Big[\ln(1+\beta\Theta_0)\ln\frac{\Theta_0}{\Theta} - \mathrm{Li}_2\frac{(\beta-\gamma)\Theta_0}{1+\beta\Theta_0} + \mathrm{Li}_2\frac{(\beta-\gamma)\Theta}{1+\beta\Theta_0} + \mathrm{Li}_2(-\gamma\Theta_0) - \mathrm{Li}_2(-\gamma\Theta)\Big].$$

(Verified by differentiating: sympy and a numerical quadrature agree to 10^{-10}.) Against the full solution this law is good for plain ice water (−6 %). Applied to salty slush with $T_f = T_L$, it is +18 % off at $\theta_b \approx 5\,\mathrm{K}$ and +57 % off at the eutectic, because it ignores the partly melted zone of §4.3. That failure is what led to the effusivity treatment.

6.3 Verification log

- **Algebra** (sympy): the regime-U quadratic (2.6), the eutectic and cotectic results (2.8)–(2.10), regime I (2.11), the eutectic boundary (2.12) and the dilogarithm antiderivative were each re-derived and checked by substitution.
- **Derivatives** (2.13): agree with finite differences of the closed form to four significant figures. **Inversion** (2.14): round trip exact. **Ethanol limit**: within 0.6 % of a brute-force root.

- **Bath**: 116 recipes against the earlier numerical model, worst 0.73 K, rms 0.27 K, one borderline regime call.
- **Full conduction solution**: halving the grid changes t_5 by under 0.1 s; energy is conserved to 0.3 %; the textbook Stefan problem is reproduced; a linear medium matches Eq. (4.10) within 1.1 %; a conducting drink in a linear medium matches Eq. (4.16) to 4 %.
- **Laplace inversions**: Eqs. (4.10) and (4.16) agree with a numerical inverse Laplace transform (Talbot's method, 30 digits) to 10^{-15}, for real roots, complex roots and the conducting drink. Eqs. (4.11) and (4.12) agree with numerical differentiation and quadrature of Eq. (4.10).
- **Closed form against the full solution**: 16 cases, rms 4.4 %, worst 10.7 %.

6.4 Sources

- R. J. Bodnar (1993), "Revised equation and table for determining the freezing point depression of H_2O–NaCl solutions", *Geochim. Cosmochim. Acta* 57, 683–684. NaCl liquidus to the eutectic.
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- H. S. Carslaw, J. C. Jaeger (1959), *Conduction of Heat in Solids*, 2nd ed., Oxford: contact of two semi-infinite bodies; Neumann's two-region solution; small-time expansions for a cylinder.
- SINTEF can-cooling times (0.33 L can, 21 → 5 °C) and your measured can and tank dimensions: project notes `b8-web-sweep` and `b8-cooling-rig`.
- The earlier numerical bath model used for the comparison: `b8_core.py` (29 Oct 2026).